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Felipe Julian Miotto

EVALUATION OF COLLABORATIVE ROBOTS APPLICATION IN AIRCRAFT FINAL ASSEMBLY

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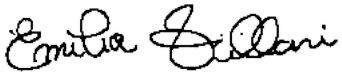

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EVALUATION OF COLLABORATIVE ROBOTS APPLICATION IN AIRCRAFT FINAL ASSEMBLY

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Para minha família.

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Resumo

Este trabalho apresenta estudos para a aplicação de robôs colaborativos na etapa de montagem final de aeronaves. Inicialmente, uma compreensiva análise bibliográfica foi realizada visando o entendimento dos processos e atividades envolvidos nesta etapa do processo produtivo aeronáutico. Além disso, caracterizou-se o cenário atual da robótica colaborativa, através da identificação de aplicações e estudos para implantação de Cobots em processos de montagem, incluindo cases da indústria aeronáutica e de outros setores como automotivo, eletroeletrônico e metal mecânico. Diferentes frameworks para avaliação da aplicação de robôs colaborativos foram identificados na literatura, que permitiram o desenvolvimento de uma metodologia própria que permitisse o mapeamento, identificação e seleção da aplicação mais adequada. Tal metodologia foi definida de modo a permitir a tomada de decisão levando em consideração o balanço entre os benefícios esperados e a complexidade no desenvolvimento da solução. Baseado nesta metodologia um aprofundado mapeamento dos processos da montagem final de uma aeronave comercial de corredor único foi realizado, permitindo a identificação de oito potenciais aplicações para robôs colaborativos. Tais aplicações foram então avaliadas de acordo com os critérios definidos na metodologia, permitindo a identificação da instalação de painéis do teto da fuselagem como aplicação mais indicada à ser implementada. A partir da definição da aplicação, estudos foram realizados para o desenvolvimento de um projeto conceitual da solução robótica à ser utilizada, seguindo a metodologia do *system design process*, parte do conceito do *systems engineering*. Partindo da identificação e avaliação dos *stakeholders*, levantamento das necessidades e restrições, foi possível o mapeamento do cenário atual e a definição do cenário futuro no escopo do CONOPS, levando à definição dos requisitos para a aplicação. Finalmente, a partir dos requisitos mapeados, uma arquitetura de sistemas foi definida, e diferentes soluções foram levantadas para cada componente. Tais soluções foram então avaliadas através de estudos de *trade-off* e simulações em ferramenta computacional, permitindo a definição da solução conceitual mais adequada para a aplicação estudada.

Abstract

This work presents studies for the application of Collaborative robots in the final assembly stage of aircraft manufacturing. Initially, a comprehensive bibliographical analysis was carried out in order to understand the processes and activities involved in this stage of the aircraft production process. In addition, the current scenario of Collaborative robotics was characterized, through the identification of applications and studies for the implementation of Cobots in assembly processes, including cases from the aeronautical industry and from other sectors such as automotive, electrical / electronics, and metal mechanics. Different frameworks for evaluating the application of collaborative robots were identified in the literature, which allowed the development of a methodology that would allow the mapping, identification, and selection of the most appropriate application. This methodology was defined in order to allow decision-making taking into account the balance between the expected benefits and the complexity in the development of the solution. Based on this methodology, an in-depth mapping of the final assembly processes of a single-aisle commercial aircraft was carried out, allowing the identification of eight potential applications for Collaborative robots. Such applications were then evaluated according to the criteria defined in the methodology, allowing the identification of the installation of fuselage ceiling panels as the most suitable application to be implemented. From the definition of the application, studies were carried out for the development of a conceptual project of the robotic solution to be used, following the methodology of the system design process, part of the concept of systems engineering. Starting from the mapping and assessment of stakeholders, and needs and restrictions identification, it was possible to map the current scenario and define the future scenario within the scope of CONOPS, leading to the definition of application requirements. Finally, from the mapped requirements, a systems architecture was defined, and different solutions were identified for each component. Such solutions were evaluated through trade-off studies and simulations in a computational tool, allowing the definition of the most adequate conceptual solution for the studied application.

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List of Abbreviations and Acronyms

Cobots	Collaborative Robots
AHP	Analytic Hierarchy Process
CONOPS	Concept of Operations
DfX	Design for Excellence Guidelines
HMI	Human-Machine Interface
AMS	Air Management System
EV	Aircraft's Vertical Tail
EH	Aircraft's Horizontal Tail
PSU	Passenger Service Unit
LRU	Line-replaceable Unit
SSM	Speed and Separation Monitoring Cobot operating mode
SMS	Safe Monitored Stop Cobot operating mode
HG	Hand Guiding Cobot operating mode
PFL	Power and Force Limiting Cobot operating mode
OCRA	Occupational Repetitive Action
NIOSH	National Institute for Occupational Safety and Health
RULA	Rapid Upper Limb Assessment
HRAA	Human-Robot Activity Allocation
PDCA	Plan, Do, Check, and Act
DMAIC	Define, Measure, Analyze, Improve, and Control

List of Symbols

H_D	Height from the ground of the entrance door for the Cobot
h_D	Height dimension of the cabin entrance door
w_D	Width dimension of the cabin entrance door
W_F	Width of the fuselage cabin floor
H_F	Height of the fuselage cabin - floor to ceiling in the fuselage center-line
W_OB	Free space between overhead bins - aisle
l_p	Ceiling panel length
l_p2	Side panel length
w_p	Ceiling panel width
w_p2	Side panel width
t_p	Ceiling panel thickness
W_p	Ceiling panel weight

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1 Introduction

The manufacturing industry has evolved significantly through the evolution of processes, techniques, and technology, marked by the different industrial revolutions (PÉREZ *et al.*, 2020). Recently, in order to keep competitiveness, companies are seeing the need to become increasingly efficient, reduce costs, and increase productivity, with production systems that are connected to all the value chain and can be rapidly reconfigured to answer the variable demand from the market, (VILLANI *et al.*, 2018), (SALUNKHE *et al.*, 2019).

The deployment of Cobots is a key technology to allow companies for more flexible manufacturing systems, (SALUNKHE *et al.*, 2019). Compared to conventional industrial robots, their lower cost, ease to set up and program, and the possibility to work safely side by side with humans can lead to an increase in productivity, (BLOSS, 2016). Furthermore, this characteristic allows expanding automation to new applications, as is the case of assembly in the aeronautical industry on which this work is based.

1.1 Motivation

The Aeronautics manufacturing industry is historically high dependent on a manual workforce. Even with the introduction of automated systems, special characteristics of this industry, such as close tolerances, short or medium series production, and the geometrically complex and large components are the main factor limiting the increase of automation, (PÉREZ *et al.*, 2020).

It is important to highlight that, considering the structural assembly process, the aerospace industry has already successfully implemented automated solutions using industrial robots for drilling, deburring and riveting activities. However, since the aircraft manufacturing process is complex and comprehends several stages, the level of automation varies significantly between the phases.

The final assembly stage, performed after the aircraft's structural assembly, encompasses the installation and test of all systems, interior components, and the most expensive

parts of the aircraft as the engines and power units. As a result, this stage is characterized by a wide range of activities, with a high level of customization and complexity, most of them being performed inside the aircraft fuselage and sometimes simultaneously in the same workspace.

These characteristics make the final assembly one of the stages with the lowest level of automation in aircraft manufacturing. According to Salunkhe *et al.* (2019), more than 90% of the tasks are performed by humans, and generally are performed in non-ergonomic conditions, an important aspect related to safety in the workplace (PÉREZ *et al.*, 2020).

In this context, Cobots are seen as a great opportunity to increase the level of automation in the aeronautic industry, mainly because they can be deployed to work in the same workspace as the human operator. Combining the advantages of automation with the cognitive and flexibility of the human, (VILLANI *et al.*, 2018), Cobots can reduce humans' physical workload, while increasing productivity, process and product quality.

Therefore, the application of Collaborative Robots in the aircraft's final assembly stage is one of the goals of the aeronautics industry, once they have the technology necessary to overcome the challenges brought by the characteristics of the activities performed in this phase. Furthermore, as OEMs are increasingly looking for higher manufacturing efficiency, improving the level of automation of the final assembly is a key factor, once these activities are performed internally, without subcontracting or outsourcing, even for programs with a high level of risk partnerships, such as the Boeing 787 and the Airbus A380, (HORNG, 2006).

The benefits achieved through the deployment of Cobots, however, vary significantly depending on the application characteristics, requiring the correct addressing of aspects such as interaction with humans, safety, end effector design, operational concept, and integration with the current process, to achieve the full potential of this technology, (FERREIRA; FLETCHER, 2022). Therefore, is crucial to develop studies to identify the applications with high improvement potential and to address the main challenges related to their deployment, allowing companies to successfully implement them in the real assembly environment.

1.2 Objectives

Considering the potential benefit of deploying Cobots in the aeronautics industry, this dissertation aims to evaluate the application of Collaborative Robots in aircraft's final assembly process, to increase the level of automation in this manufacturing stage.

In order to achieve the main objective, the following specific objectives were defined:

- Mapping the macro processes and activities of aircraft's final assembly and identifying potential applications for Cobots.
- Evaluate and select the application to be developed considering expected benefits and development complexity.
- Define the main requirements, restrictions, and functions for the application.
- Identify potential conceptual solutions to address the required functions defined.
- Evaluate and select a final conceptual solution for the application.

1.3 Methodology

To achieve the objectives outlined, a set of resources and methods were defined as presented below and resumed in Figure 1.1.

First, following the literature review and the information collected with specialists through guided interviews, the aircraft's final assembly process was mapped, and potential applications for the deployment of Cobots were identified. After that, the identified potential applications were evaluated through the developed methodologies, allowing the selection of the application to be further developed.

The next steps are related to the definition of requirements and restrictions of the application, captured through the definition of a Concept of Operation and translated into technical requirements and functional decomposition. This allows the identification of potential solutions, which are evaluated and compared for the selection of the best conceptual solution for the application.

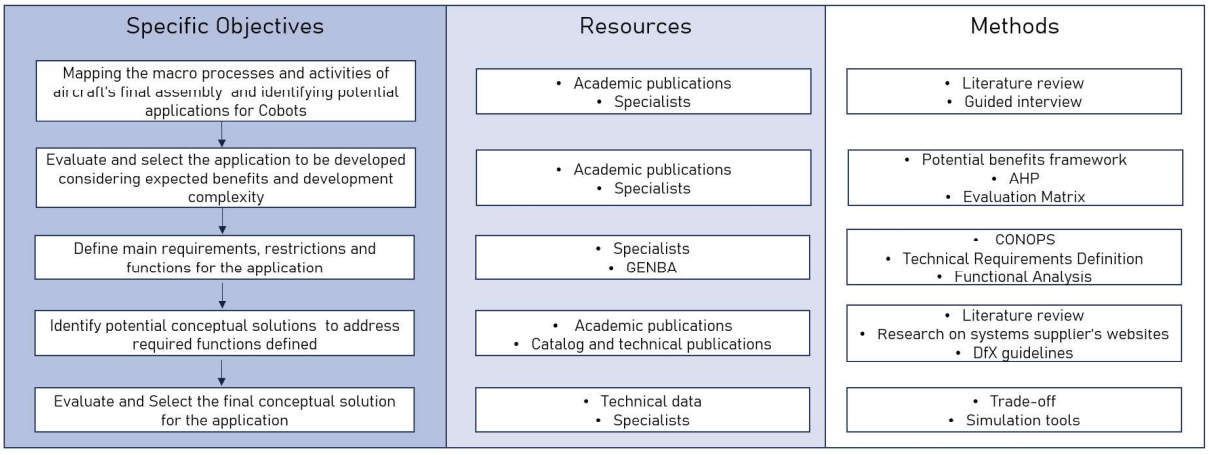


FIGURE 1.1 – Methodology schematic.

1.4 Research Delimitation

The present work focuses on studying the application of Collaborative robots in the aircraft final assembly process. The scope of the work is limited to process mapping, potential application identification, and the definition of a methodology for the selection of the application most indicated to be developed. Furthermore, for the application selected, a conceptual solution is also developed.

The development of the work was based on data available in published works and studies, and on information collected information with specialists. No measurements or data were collected by experimental tests.

The development of the conceptual solution was also based on available information on previous works, catalogs, and equipment suppliers' websites. The scope was limited to a conceptual phase solution, with no detailing, testing, or manufacturing of the solution. The conceptual solution was developed considering the nominal operating condition, while, off-nominal scenarios were not evaluated.

During the conceptual solution development, economic aspects related to the application, such as ROI (Return on investment), payback time, and implementation costs were not derived through requirements and must be addressed in a detailed development phase, which is out of the scope of this work.

1.5 Work Structure

Additionally to the introduction, this work is divided into five more chapters, structured according to the objectives, and described below:

- Chapter 2 is a literature review that discusses the importance and benefits of Collaborative Robots in the context of Industry 4.0, followed by main applications in the context of the manufacturing and aeronautics industry. Moreover, a discussion regarding the aircraft's final assembly process and Cobots deployment is performed, including main frameworks to properly evaluate the application of Collaborative Robots. Finally, a set of methodologies regarding the conceptual design process are presented.

- In Chapter 3, the methodology defined for the process mapping and potential application identification is presented. Furthermore, the chapter presents the methodology for the evaluation of the potential applications to allow the selection of the one to be developed in this work, considering the balance between the potential benefits expected and the development complexity.

- In Chapter 4, the methodology presented in Chapter 3 is applied in the context of the application of Collaborative Robots to aircraft's final assembly, according to the specific objectives defined. First, the final assembly process mapping is presented, followed by the identification of potential applications. In the sequence, the obtained results on evaluating the potential applications are presented, finishing with the selected application.

- Chapter 5 presents the conceptual design process for the application selected. First, the development of the CONOPS is performed through the definition of the current scenario and the envisioned operation. Based on the system design process, the high-level requirements are defined as well as the definition of the functions required for the system, Based on these definitions, different conceptual solutions are identified and evaluated, leading to the final conceptual solution. The chapter ends with the requirements verification, concluding the fulfillment of the specific objectives defined.

- In Chapter 6 the conclusions of the work are discussed, including a brief review of the objectives and main results, as well as a recommendation for future works.

2 Literature Review

This chapter presents an overview of the theoretical foundations used in this work, based on a comprehensive bibliographic survey. Initially, an overview of Collaborative Robots application in the context of Industry 4.0 is presented, followed by current applications including in the aeronautics industry, then focusing on aircraft final assembly. In the sequence, the process of deploying Collaborative Robots in current manufacturing environments is discussed, including the main methodologies used to evaluate the deployment of Cobots and to design a conceptual application.

2.1 Collaborative Robots in the Context of Industry 4.0

In recent years, the manufacturing industries are facing a challenge where globalization and the demand for customized products from customers are requiring companies to adapt their processes with agility (SALUNKHE *et al.*, 2019),(SCHUH *et al.*, 2017). In this context, Industry 4.0 plays an important role in changing the way companies plan, operate, and conceive their production systems.

The term Industry 4.0 was first introduced in 2011 by the German group Industry-Science Research Alliance to describe the trend of widespread integration between data from all company levels, to create a digitally connected industrial production. The term "4.0" was adopted as a reference to a 4th industrial revolution, considering the possible effects of this trend that are beyond the industrial context, which may lead to cultural and social changes, as experienced in the previous industrial revolutions (SCHUH *et al.*, 2017), (KAGERMANN *et al.*, 2013).

As presented in Figure 2.1, industrialization began at the end of the 18th century when steam-powered equipment revolutionized the production of goods, moving from a fully manual way to an application of mechanical machines. At the beginning of the 20th century, the industries faced an important change in their production systems through the application of electrically-powered equipment the division of labor, and mass production, which characterized the second industrial revolution (KAGERMANN *et al.*, 2013).

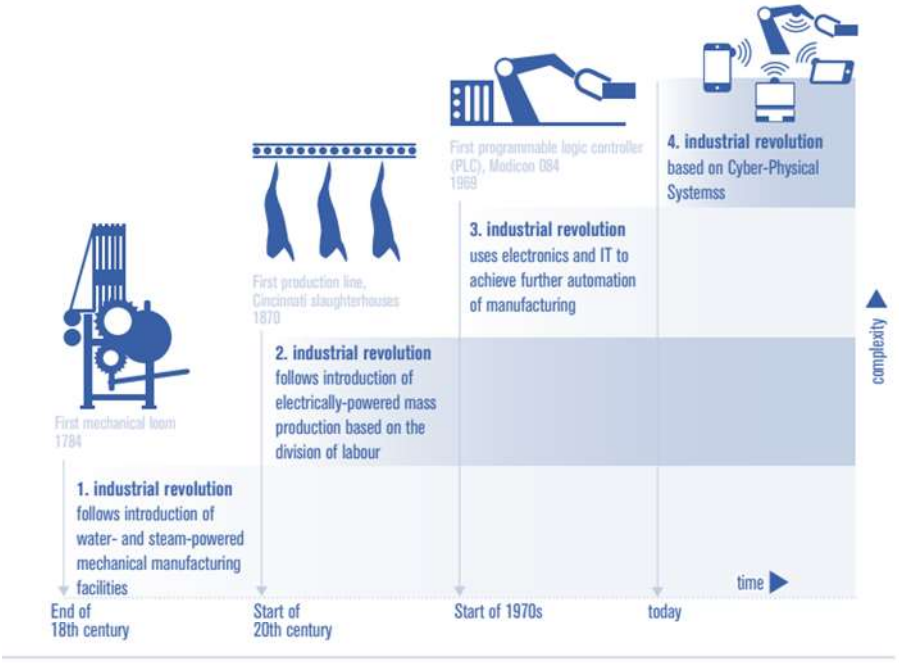


FIGURE 2.1 – Industrial revolutions(KAGERMANN *et al.*, 2013).

As a continuous evolution, during the 1970s the world witnessed the beginning of the third industrial revolution, which continued up to the present day. This revolution is marked by information technology (IT) and electronics application through process automation.

In this context, the so-called 4th industrial revolution is based on the evolution of manufacturing systems to cyber-physical systems in which physical production plants, warehouses, and all production environments are digitally connected through standard Internet-based protocols, allowing data exchange and analysis, and self-configuration to adapt to external changes, creating an automated and integrated production flow, (COLOTLA *et al.*, 2016), (KAGERMANN *et al.*, 2013).

Differing from previous industrial movements, Industry 4.0 isn't based on a single major invention as electricity but is based on nine enabler technologies, including Augmented Reality, Additive Manufacturing, the Internet of Things, Big Data & Analytics, Simulation, Horizontal & Vertical Integration, Cloud, Cyber-security and Advanced Robotics.

Advanced robotics refers to a new paradigm of the application of robots in industrial environments. Since the beginning of the 1960s, when General Motors installed the Unimate - what is considered the first industrial robot – (GASPARETTO; SCALERA, 2019), robots have been widely applied to different purposes, mainly in Metal and machinery, Electrical/Electronics, and automotive sectors, (IFR, 2021). Despite the robustness and high accuracy speed and quality of movements, industrial robots must work separated from humans, given that no contact with humans is allowed according to safety require-

ments.

For this reason, the application of robots in manufacturing processes, especially in components assembly, is limited, once generally the presence of human workers is required taking into account the variety of components sizes, formats, and weights, as well as possible restricted working spaces like insides car or aircraft fuselages.

With these points in mind, a new era of advanced robots is emerging, called Collaborative robots. This new generation of robots has safety systems that allow them to share the same workspace with humans without any safety barrier, (CHOWDHURY *et al.*, 2020). Furthermore, according to Salunkhe *et al.* (2019), Cobots are intended to have direct contact with humans, which potentially is changing the way people perceive and interact with robotic technologies.

Since there is no need for separation between humans and Cobots, new applications now can be addressed by these robots, as final assembly operations, in which this work is focused to study. The application of Collaborative Robots brings several benefits to assembly operations as presented in the following section.

2.2 Collaborative Robots Advantages for Manufacturing

One of the main differences between industrial robots and collaborative robots is the purpose of their application. Conventional robots are robust, work at high speeds, can carry high payloads, and operate separately from humans, and in many cases, they replace factory workers (CHOWDHURY *et al.*, 2020). On the other hand, Cobots are conceived to work in the same workspace as human workers, to collaborate and reduce their strain, rather than replacing human labor.

According to Villani *et al.* (2018) the main benefit brought by Cobots is the possibility to combine the advantages of robots with the flexibility and cognitive skills of human workers. This combination brings the opportunity to increase the level of automation of some processes that until then were not possible to be automated. According to Bloss (2016), logistics and assembly tasks are the most benefited areas, with a forecast of almost 100 million new possible tasks to be automated.

One of the most important improvements that can be achieved through the application of a collaborative robot is the worker's stress and fatigue reduction, which is possible considering that the Cobot, may be used to perform non-ergonomic tasks, including carrying heavy loads, repetitive tasks, and awkward positions (VILLANI *et al.*, 2018), (PALOMBA *et al.*, 2021). Furthermore, considering the characteristics of the robot, like repetitiveness, an improvement in process and product quality can be achieved, (ZANELLA *et al.*, 2017).

Even though the ergonomics benefits already justify the application of Collaborative robots, the correct task allocation between humans and robots may lead to a productivity enhancement, allowing the cycle times reduction, as presented by Palomba *et al.* (2021), Tsarouchi *et al.* (2017) and Salunkhe *et al.* (2019), for example.

Another important characteristic of Cobots is the ability to be rapidly reprogrammed to respond to changes in the environment, an aspect that in combination with other enabling technologies like the Internet of things, can bring the flexibility and the agility to manufacturing environment expected by the Industry 4.0, (CHOWDHURY *et al.*, 2020).

2.3 Safety Aspects of Collaborative Robots

The great potential benefits that can be achieved by the application of Cobots come with a number of challenges to be addressed. According to Villani *et al.* (2018), the main challenges for the introduction of collaborative robots in industrial applications are the design methods applied (including task planning and allocation, sensors, and control laws), intuitive interfaces (related to programming approaches, input modes, and reality enhancement), and above all, Safe interaction.

Safety issues are the primary challenges to be addressed in any application of collaborative robots once the human operator can work alongside the robot. Studies regarding this aspect over the years led to safety standards and collaborative operating modes that unified requirements and guidelines to help the development of new safety applications.

Safety standards for Cobots can be divided into three levels according Villani *et al.* (2018). The first is related to basic safety standards that can be applied to machinery, in which ISO 12100 and IEC 61508 define the basic terminology and methodology used in achieving the safety of machinery. The second level refers to generic safety standards, such as emergency stops devices and two-hand control devices for example. The third level specifies individual safety standards for specific machines. This level of standards superimposes the previous in case are applied to the machinery of the scope of evaluation. For industrial robots, ISO 10218-1/2 dictates the standards that regulate safety aspects. With the introduction of Cobots, it was verified that the established standard ISO 10218 would not be sufficient to achieve maximum safety while allowing a flexible level of collaboration with human operators. For this reason, the International Organization for Standardization specified the ISO/TS 15066, which is a technical standard that complements the ISO 10218 with the specific aspects for collaborative robots ((VILLANI *et al.*, 2018), (ROSENSTRAUCH; KRÜGER, 2017)).

The specific standards for Cobots introduce the concept of operating modes, specifying how collaborative robots can be implemented in industrial applications. Four operating

modes were established for Cobots:

- Safe monitored stop (SMS): The Cobot only moves if the Human operator is outside of the collaborative workspace. The workspace generally is marked by laser sensors that detect objects, and when the human operator enters to perform any tasks, the Cobot's motion stops until the operator leaves.
- Speed and separation monitoring (SSM): Human operator is allowed to enter the collaborative workspace. In this case, the robot motion speed decreases / increases according to the distance between the human and the robot, with a limiting distance where the Cobot stops.
- Hand guiding (HG): This mode goes a step further in collaboration. In this case, contact between human and robot is allowed, once the operator is able to guide the robot by moving it by hand.
- Power and force limiting (PFL): This mode follows a complete collaborative workspace, dealing with the possibility of unpredicted collisions between humans and robots. In this case, the maximum forces allowed to contact each part of the human body are defined, and this information enters as a restriction for the robot, limiting the torque allowed in each of the robot joints.

These operating modes defined by the specific standards were considered as part of the development of the conceptual design solution for the scope of this work, discussed in Chapter 5.

2.4 Collaborative Robots Applications

As presented before, Collaborative Robots present several benefits that are leading companies to move towards their application, fundamentally intending to improve ergonomics aspects, productivity, flexibility, and quality, (SIMÕES *et al.*, 2020). Also, considering their characteristics, the main applications that are emerging are related to product packing, production line loading and unloading, parts testing, workplace materials logistics, and assembly operations, (BLOSS, 2016).

The following sections will bring some applications of Collaborative Robots to exemplify the cited motivation and benefits expected by companies. First, general industrial applications are presented, and then focusing on aeronautical case studies and research.

2.4.1 Industrial Applications

The integration of Cobots in the industry has been growing. As presented by Cohen *et al.* (2022), in 2016 the US market share of Cobots represented $100 million and reached $710 million in 2018, with a forecast to reach $12.3 billion in 2025, mainly due to the fast growth of using Collaborative robots in assembly applications.

In this context, several industrial segments are applying Cobots to increase the level of automation of their process. According to Ferreira and Fletcher (2022) several automotive manufacturers already have implemented Cobots. BMW Group, for example, has been applying collaborative robots since 2013, when the company introduced Cobots from Universal Robots, to equip the insides of the doors of an X3 model with sound and moisture insulation, working side by side with humans. The main motivation was based on ergonomic considerations, (SCHILLMOELLER, 2013). Another application from BMW is the Dingolfing plant, where the KUKA LBR iiwa Cobot was introduced, basically to improve ergonomics. The application is related to bevel gears assembly process, where the Cobot is responsible to take, handle and position the heaviest part of the assembly, removing this ergonomically demanding job from the human worker, (HRC, 2020a).

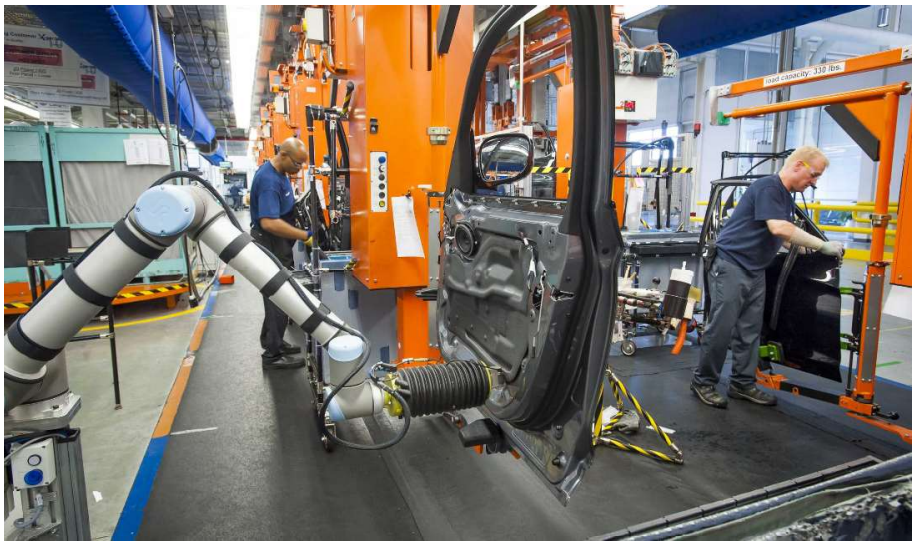


FIGURE 2.2 – Automotive application of Cobot - BMW Group - (SCHILLMOELLER, 2013)

Ford as well, applied the KUKA LBR iiwa Cobot to perform the unergonomic task of adjusting fog lights, working in parallel with a human operator who adjusts the main headlights. The motivation was to reduce worker strain while improving the quality of the adjustment, (HRC, 2020b).

Another example is the CSVW company, which applied a SIASUN collaborative robot (SCR5) to create an HRC work cell, where the Cobot completes the screw tightening task of the automobile engine assembly line, (HEER, 2018).

Villani *et al.* (2018) presents other application examples, including the case of Volkswa-

gen, where Cobots handle delicate glow plugs into cylinder heads, allowing the operator complete the assembly task in an upright healthy posture, (A/S, 2013). Other examples are Audi, which introduced Cobots to assist human workers ergonomically in pick and place activities and tightening underfloor car components,(TEC, 2021), and Skoda, where Cobots assist delicate gear-boxes assembly, both strongly motivated by ergonomics improvements for human workers, (IFR, 2015).

Although the main industrial applications are related to the automotive sector, other important industrial applications of Cobots can be found. The electrical/electronic segment has been experiencing an increase in the level of automation (IFR, 2021), which is supported by the application of collaborative robots. Some examples presented by Ferreira and Fletcher (2022) include the case of ASM Assembly Systems, which implemented the Rethink Robotics Sawyer to increase productivity and relieve employees from performing repetitive tasks. In this application the Cobot is used to insert circuit boards into a test adapter, check for defects, and position the boards according to the test results, allowing the human workers to perform more complex tasks while the Cobot carries out monotonous activities, (ROBOTICS, 2018).

A further example of the application of Cobots in the electric/electronic segment is the case of Qisda, where Cobots were introduced to perform assembling, testing, and packaging activities. The main motivation was the fact that the company has a large portfolio of products that are very different in sizes and specifications, requiring frequent adjustments on the production lines. With the application of Cobots, this adjustment time has been shortened from 30 minutes to less than three minutes, leading the company to improve its production efficiency by more than 70%, (HWANG, 2019).

FIGURE 2.3 – Cobot application in metal mechanics industry, (SHAWN, 2019).

In addition to the aforementioned applications, the plastics and metal mechanics sec-

tors are other important fields of application of Collaborative solutions. This is the case of Trelleborg Company, where collaborative robots were applied to perform machine tending tasks on CNC machines of a plastic and polymer manufacturer, bringing the desired flexibility for production, (SHAWN, 2019). Another example came from Scott Fetzer Electrical Group (SFEG), where Cobots were introduced in the sheet metal department, being applied in the entire production cycle, optimizing the production efficiency by 20% and removing the monotonous and potentially hazardous activities from human workers, (METALWORKIGN, 2016).

To support the increasing number of industrial applications, research studies on Cobots deployment are fundamental and have been increasing over the last years. As presented by Villani *et al.* (2018), research has been conducted focusing on solving the main challenges in Human-Robot Collaboration. The first is to guarantee a safe interaction between humans and robots since this is one of the main advantages of collaborative robots. Furthermore, to access the full potential of a collaborative application, the interaction between robot and human must be efficient and assertive, which brings significant difficulties when it comes to an industrial application, considering aspects such as noise, illumination, and transit of people and vehicles. Finally, the design of the application, which includes the task planning and allocation, as well as sensors, and control laws are important aspects to guarantee that the potential benefits of the application of Cobots can be accessed.

Following the same tendency of the industrial applications, the automotive industry represents most of the research publications related to Cobots, (SEGURA *et al.*, 2021). Makrini *et al.* (2018) presented the results of the project called ClaXon, which studied the interaction between humans and Cobots in factories, resulting in the integration of Cobot in the Brussels Audi manufacturing plant, for applying glue to reinforcement plates used on cars' roofs. The study focused on the development of gesture and facial recognition solutions and performed interviews to assess the social acceptance of the cobot by human workers.

Other interesting examples of studies related to the automotive industry came from Volvo. Salunkhe *et al.* (2019) presents research focused on testing the feasibility and assessing the potential of the introduction of Cobots in real industrial operations. Following the results, the use of Cobots in nut assembly on a wheel hub showed real benefits in ergonomics and quality improvement. Furthermore, the company studied the challenges involved with Cobot deployment, related to production planning and preparation. In this case, the collaborative application was related to engine assembly, as shown in Figure 2.4, and the results showed that the planning and preparation process of current assembly systems need to be reviewed to enable the implementation of re-configurable modern factories, mainly including virtual commissioning as a part of the preparation process, (HANNA *et al.*, 2019).

FIGURE 2.4 – Study of application of Cobot in Engine assembly, (BENGTSSON, 2019)

Considering other industrial applications studies, Gualtieri *et al.* (2020) presented a complete study on designing a human-centered collaborative workstation, with a focus on physical ergonomic sand production efficiency improvements. The study focused on the application of Collaborative Robots to perform taping tasks on wire harness assembly. The study applied different methodologies to assess the potential and to evaluate the real improvements in human operator health.

A resume of the discussed applications and studies of Collaborative Robots in the context of Industrial Applications is presented in Table 2.1, where can be observed the main industrial segments benefited from the application of this technology. Furthermore, most of the applications identified are related to handling/positioning, screwing, and picking-placing, in accordance with the information presented by Segura *et al.* (2021) and Bloss (2016).

Reference	Environment	Sector	Company	Task	Expected Benefit/Objective
(SCHILLMOELLER, 2013)	Industrial Application	Automotive	BMW	Installation of insulation adhesive foils in the doors inside equipping.	Improve ergonomics conditions for human workers and increase product quality. .
(HRC, 2020a)	Industrial Application	Automotive	BMW	Positioning of bevel gears components of car gearbox.	Reduce non-ergonomic conditions regarding heavy components handling.
(HRC, 2020b)	Industrial Application	Automotive	Ford	Adjusting car fog lights.	Reduce worker's strain. through reduction of non-ergonomic tasks.
(HEER, 2018)	Industrial Application	Automotive	CSVW	Tightening of car engine screws.	Increase production flexibility.
(A/S, 2013)	Industrial Application	Automotive	Volkswagen	Handling delicate components for cylinder heads assembly.	Optimizing ergonomic working processes.
(TEC, 2021)	Industrial Application	Automotive	Audi	Tightening of underfloor car components.	Improve working postures for underfloor assembly operations.
(IFR, 2015)	Industrial Application	Automotive	Skoda	Insertion of gear actuator pistons into transmission assembly.	Increase product quality considering tight assembly tolerances.
(ROBOTICS, 2018)	Industrial Application	Electrical/Electronic	ASM	Insertion of circuit boards into a test adapter.	Increase productivity and relieve employees from performing repetitive tasks.
(HWANG, 2019)	Industrial Application	Electrical/Electronic	Qisda	Assembling, testing, and packaging electronic components.	Improve the production efficiency.
(SHAWN, 2019)	Industrial Application	Metal Mechanics	Trelleborg	Executing machine tending tasks on CNC machines.	Optimize manufacturing time and improve agility.
(METALWORKIGN, 2016)	Industrial Application	Metal Mechanics	SFEG	Positioning, picking and placing, and testing activities.	Solve ever-changing tasks with a high mix – low volume optimizing production.
(MAKRINI *et al.*, 2018)	Research Study	Automotive	Audi	Glue application in reinforcement plates for car's roof.	Study of human-robot interaction, including social studies with human workers.
(SALUNKHE *et al.*, 2019)	Research Study	Automotive	Volvo	Insert and apply torque into weels nut.	Testing feasibility and assessing Cobots potential for nut assembly process.
(HANNA *et al.*, 2019).	Research Study	Automotive	Volvo	Ladder frame on top of the engine assembly, tightening bolts.	Cobot impact study in engine production planning and preparation.
(GUALTIERI *et al.*, 2020)	Research Study	Electrical/Electronic	ELVEZ d.o.o.	Designing a human-centered collaborative workstation for wire harness assembly.	Designing a human-centered collaborative work cell to improve Human ergonomics conditions.

TABLE 2.1 – Cobots industrial applications literature's summary.

Finally, considering the objective of the present work, the next section will focus on the applications and research related to the aeronautical manufacturing industry.

2.4.2 Aeronautical Manufacturing Applications

According to Barattini *et al.* (2019), the application of robotics technology in aerospace manufacturing has a slow rate of adoption compared to other industrial sectors. This is motivated by some special challenges faced by this sector, mainly related to space restrictions, tight levels of tolerance, and the size of components, as presented by Laudante *et al.* (2020) and Pérez *et al.* (2020). However, the introduction of Cobots as presented

in section 2.2, has turned possible new applications that have broadened the potential of robotic applications in complex environments like the aeronautical.

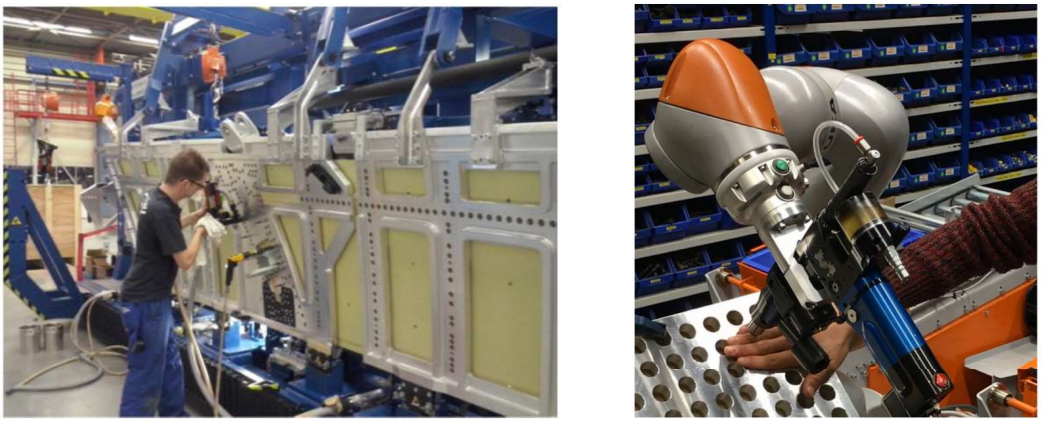
The Airbus company, for example, presented in 2014 its concept called "Factory of the Future", an research and development program to accelerate the application of Industry 4.0 technologies, which included the application of collaborative solutions in their manufacturing plants, (GREENFIELD, 2014), and (SLAMA, 2014). Since then, the company studied the implementation of collaborative solutions mainly focused on screwing, tightening, and measurement activities, as presented by Mueller *et al.* (2015). One example is the application of a two-axis collaborative robot from Kawada Industries, to perform riveting tasks on the rudder spar assembly station of the Airbus A380 model, sharing the same workspace with human workers, (WEBER, 2015).

FIGURE 2.5 – Dual arm Cobot to perform riveting tasks on Airbus A380 aircraft, (BOEING, 2017).

Boeing company implemented a Universal Robot UR10 to their production facility in Melbourn, Australia, to perform machining activities on several 737 model components. The main motivation was to reduce ergonomically difficult labor for human workers, and as presented by the study, with the implementation of the collaborative solution, a reduction of machining cutter consumption was achieved,(BOEING, 2017).

Other examples are presented by Muijs and Snijders (2017) related to Fokker company. The first is the application of Cobot in the production of the Airbus A350 outboard flap, where the Cobot task was to pick up an Automatic Drilling Unit from a table and insert it in a drill jig. The main goal was to free up well-trained workers for more complex tasks. Another application presented by Muijs and Snijders (2017) is the implementation of a Cobot on the shop floor to assist the human operator in the application of sealant for landing gear bushings, focusing on reducing operating time and creating a repeatable quality process.

FIGURE 2.6 – Application of collaborative robots at Boeing, (BOEING, 2017).



(a) Initial process, performed by human worker.

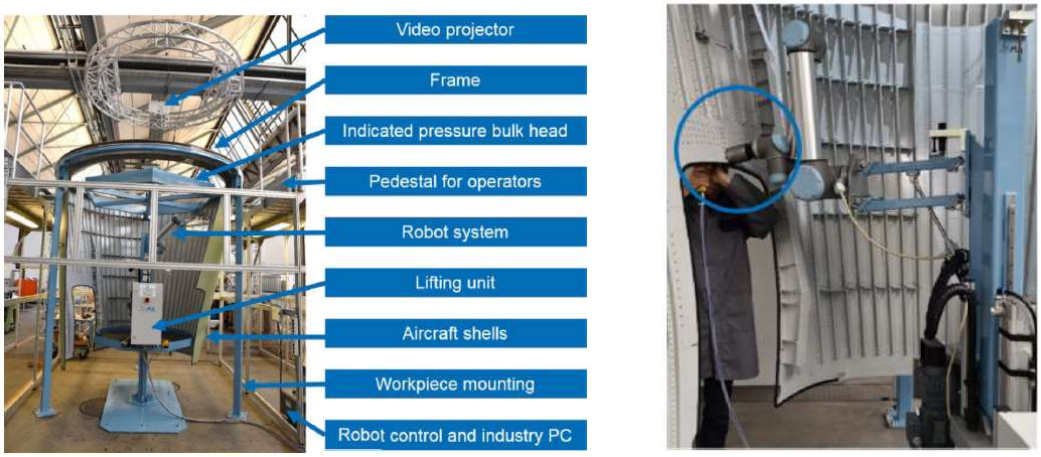
(b) Cobot solution for drilling unit insertion.

FIGURE 2.7 – Fokker Cobot application for Airbus A350 outboard flap manufacturing, (MUIJS; SNIJDERS, 2017).

As presented for other industrial segments, the implementation of Collaborative robots in aeronautical manufacturing is based on several research initiatives. One example presented in the literature is the project VALERI - Validation of Advanced, Collaborative Robotics for Industrial Applications -, an initiative coordinated by the Fraunhofer Institute for Factory Operation and Automation (IFF) including companies such as KUKA Robotics GmbH and Airbus. The project goal was to develop and study the technologies necessary to deploy Cobots to work side by side next to humans, focusing on three different use cases: Sealant application, optical quality control of the applied sealant, and optical inspection of braided carbon fiber parts, (SAENZ *et al.*, 2017a), (SAENZ *et al.*, 2017b).

Studies related to structural assembly are presented by Serebrenny *et al.* (2019) and Li *et al.* (2021). A concept of a collaborative workspace for drilling and riveting activities is presented in the first one, where the Cobot performs the most monotonous operations, making it possible to reduce the operational time with minimal interference with the existing process. Li *et al.* (2021) by the other hand focused on the human-robot interaction challenge, through the development of a transfer learning-enabled action recognition approach for the installation of brackets on structural assembly of aircraft fuselage panels. A similar study is presented by Costanzo *et al.* (2021), where human detection and tracking are addressed through a multimodal approach supported by the treatment of images acquired from different sources, to guarantee a safe operation during a cooperative assembly of fuselage panels.

Additionally, to the previous studies focused on implementing Cobots for structural assembly activities, Mueller *et al.* (2017) presents a project to deploy Cobots for riveting application. The study is part of the European Union's Horizon 2020 and focuses on the study of a Human-Robot collaborative solution for riveting the pressure bulkhead of an aircraft, which is motivated by the non-ergonomics and onerous characteristics of the activity. The work presents a complete development of a demonstrator, including process planning, virtual modeling, the definition of the human-robot interface, and the construction of the demonstrator for an Airbus A320 aircraft. As presented, the integration of the Cobot allowed the improvement of working conditions, while setting free a second operator, previously required, for 3-4 hours, also, showing potential for process quality improvement.



(a) Demonstrator schematics.

(b) Cobot and human worker performing riveting activity.

FIGURE 2.8 – Cobot for riveting application demonstrator, (MUELLER *et al.*, 2017).

Another more recent project, the SHERLOCK EU presented by Dimitropoulos *et al.* (2021), aims to develop a high payload collaborative robot solution, including safety

modes of operation and human-robot interface studies, to allow the seamless integration of the Cobot and operators in a common workspace. One of the future visions of the project is to introduce a collaborative mobile manipulator to assist human operators in the co-manipulation and positioning of large composite aeronautics parts, aiming to remove non-value-added activities from human operators.

Table 2.2 presents the consolidated applications and studies discussed above. As can be verified, most of the current applications of Cobots in the aeronautics industry are related to structural assembly processes. However, the manufacturing of an aircraft comprehends several stages that can benefit from the deployment of collaborative solutions, such as the final assembly stage, which will be discussed in the next section, and where this work is based.

Reference	**Environment**	**Sector**	**Company**	**Task**	**Expected Benefit/Objective**
(WEBER, 2015)	Industrial Application	Aeronautics	Airbus	Riveting on the rudder spar assembly.	Reduce operator's ergonomics issues.
(BOEING, 2017)	Industrial Application	Aeronautics	Boeing	Machining activities on aircraft components.	Reduce ergonomically difficult labor for humans.
(MUIJS; SNIJDERS, 2017)	Industrial Application	Aeronautics	Fokker	Inserting Automatic Drilling unit in a drill jig for outboard flap manufacturing.	Free up humans for more complex tasks.
(MUIJS; SNIJDERS, 2017)	Industrial Application	Aeronautics	Fokker	Applying sealant on aircraft's landing gear bushings.	Reducing operating time and improve process quality.
(SAENZ *et al.*, 2017a)	Research Study	Aeronautics	Airbus	Sealant application on fuselage pannels.	Safety and risk assessment Study for increasing technology maturity.
(SAENZ *et al.*, 2017b)	Research Study	Aeronautics	Airbus	Sealant application on fuselage pannels.	Monitoring system study for safeguarding development and application, to increase technology maturity level.
(SEREBRENNY *et al.*, 2019)	Research Study	Aeronautics	-	Drilling and riveting activities on aircraft fuselage.	Concept design of a collaborative solution.
(LI *et al.*, 2021)	Research Study	Aeronautics	-	Aircraft fusealge pannel bracket assembly.	Develop human action recognition system.
(COSTANZO *et al.*, 2021)	Research Study	Aeronautics	-	Quality inspection and deburring tasks on fuselage structural assembly.	Human tracking and detection study through multimodal images acquisition for human action recognition system.
(DIMITROPOULOS *et al.*, 2021)	Research Study	Aeronautics	COMAU, others.	Aeronautics components holding and positioning.	Safety modes and human-robot interface studies for a high payload Cobot solution.
(MUELLER *et al.*, 2017)	Research Study	Aeronautics	Airbus	Collaborative riveting.	Development and test of a Collaborative riveting application demonstrator.

TABLE 2.2 – Cobots aeronautics applications literature's summary.

2.5 Aircraft Final Assembly and Collaborative Robots

Aircraft manufacturing is a complex process composed of several stages, from the transformation of raw materials into primary components to customer delivery. According to Junior *et al.* (2019), the aircraft manufacturing process can be resumed in three main stages, which comprehend the manufacturing of primary parts, the structural assembly, and the final assembly, as presented in Figure 2.9.

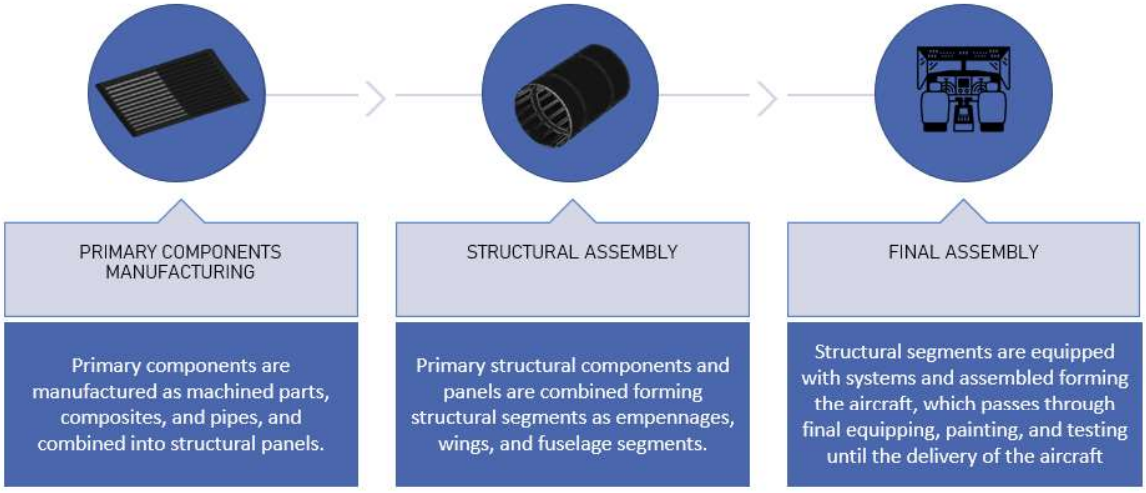


FIGURE 2.9 – Aircraft general manufacturing sequence.

The final assembly process encompasses all activities developed after the structural assembly until the delivery of the aircraft to the customer, (BARBOSA, 2007). The sequence of the final assembly varies from each aircraft OEM since it depends on the industrial strategy, production rates, and product design restrictions, however, a general sequence is presented by Barbosa (2007), shown in Figure 2.10 and described in the following paragraphs.

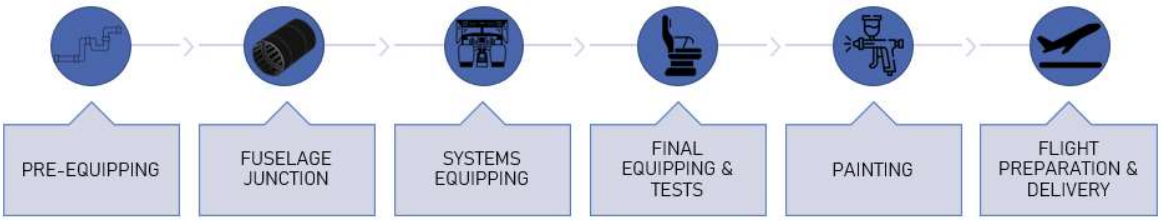


FIGURE 2.10 – Aircraft general final assembly process.

In the pre-equipping phase, mechanical, electrical, and electronic systems start to be installed in the big structural segments such as wings, empennages, pylons, and fuselage sections. Some examples of installed systems are brackets, supports, pipes, valves, cables, and insulation materials.

After the pre-equipping phase, the fuselage segments are joined, forming the fuselage body. This stage requires special tooling to guarantee the correct alignment between the fuselage segments. Figure 2.11 shows the fuselage junction process of the Boeing 777 aircraft.

FIGURE 2.11 – Fuselage junction of a Boeing 777, (BOEING, 2011).

With the fuselage body assembled, the equipping phase begins when most of the systems are installed. In this phase, big cables, power cables, and insulation blankets are installed, as well as, fuel, hydraulic, environmental, and other significant systems. Furthermore, the main cockpit components and systems are installed. Figure 2.12 illustrates one of the assembly activities performed during this phase.

FIGURE 2.12 – Insulation blankets installation, one of the activities performed during the systems equipping phase of the Boeing 737 assembly, (RYANAIR, 2014).

Generally, after the systems equipping, wings and empennages are assembled to the fuselage body and the previously installed systems in the structural components are connected. Then, the final equipping phase starts which is characterized by the installation of the higher added value items to the aircraft. During this phase, interior components such as overhead bins, ceiling, side panels, seats, floor mats and other interior monuments, as well as the aircraft avionics and Engines. During the final equipping, several functional tests are performed before the aircraft follows the painting process and flight preparation phase, which represents the last stage of the final assembly. Figure 2.13 presents one of the cited activities performed during the final equipping phase.

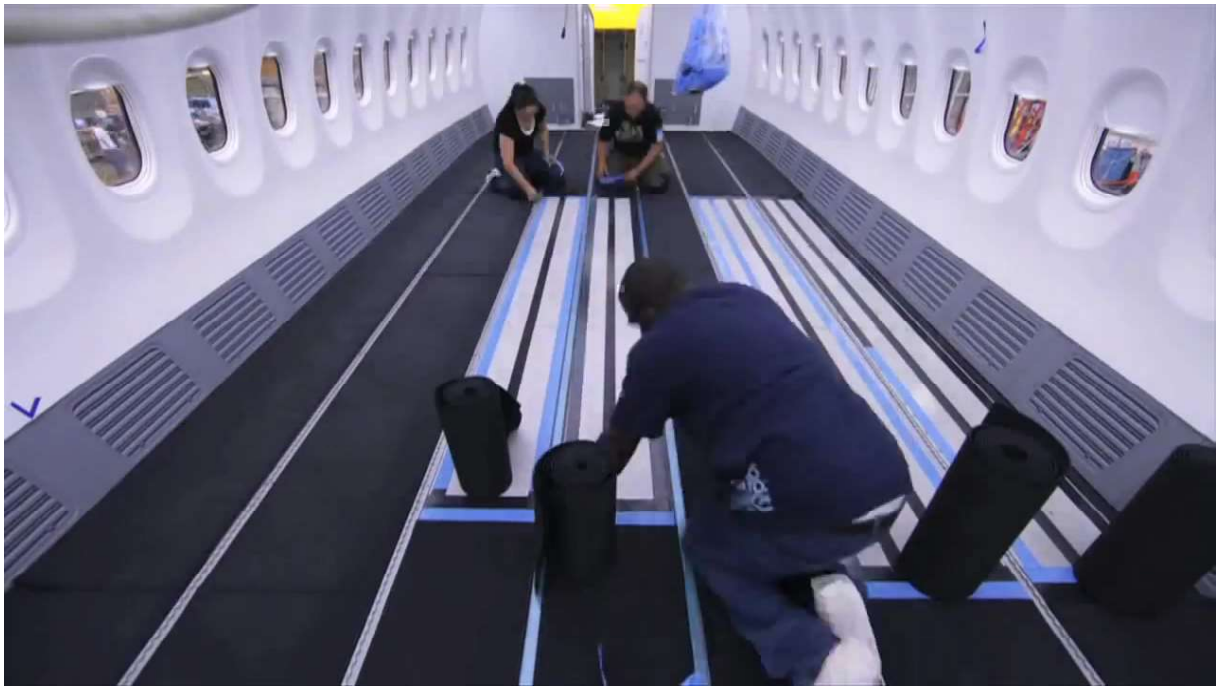


FIGURE 2.13 – Floor mat installation during the final equipping phase of the Boeing 737 assembly, (RYANAIR, 2014)

It is important to highlight that, the level of automation of each stage of the aircraft's general manufacturing process presented is different. Structural assembly stages in the aeronautical industry already present a high level of automation, mainly in the drilling and riveting activities, making use of large C-frame machines or industrial robots, (SARH *et al.*, 2009). Furthermore, as presented in the last section, structural assembly operations are one of the most common applications for Cobots in the aeronautical industry.

In the final assembly stage of aircraft production, however, the level of automation is very limited, and according to Salunkhe *et al.* (2019) humans do over 90% of the activities. The main reasons are based on the fact that this assembly stage, as presented, evolves activities of high diversity, complexity, and customization, (JUNIOR *et al.*, 2019), that are scheduled in a very complex process chain, (ROVEDA *et al.*, 2017).

In addition, as most of the final assembly activities are performed manually, human operators must use hand tools or advanced assembly systems to perform activities that combined with space restrictions of the aircraft cabin or cargo areas, lead to non-ergonomic conditions, (OTTOGALLI *et al.*, 2021), (ROVEDA *et al.*, 2017). For these reasons, one of the goals of the aeronautics industry is to increase the level of automation of the final assembly stage to reduce manufacturing times and costs, and improve ergonomics conditions, (OTTOGALLI *et al.*, 2021).

Considering the characteristics of the final assembly process, and the aspects presented in Section 2.2, Collaborative Robots are one of the main technologies that can play an important role in this transition to a more automated process. Even though most of Cobot's

aeronautics applications are related to structural assembly operations, some studies have been published in the direction of the application of collaborative robots in aircraft final assembly activities.

The ASIMOV project, (VERNE, 2017), for example, has the objective to develop a prototype for an industrial collaborative robot (Cobot) with the capability to carry out assembly operations inside an aircraft structure in order to assist the other workers with their low-added-value tasks. As shown in Figure 2.14, the prototype consists of a Collaborative robot mounted on a mobile robot, which performs floor fixing activities, improving human working conditions.

FIGURE 2.14 – ASIMOV project prototype, (VERNE, 2017).

Another example is the H2020 CleanSky 2 project, which studied the application of collaborative solutions to improve assembly tasks in the aircraft cabin. As part of the project, Roveda *et al.* (2017) presented a conceptual project of a lightweight mobile arm while Ottogalli *et al.* (2021) presented a process evaluation and ergonomics assessment of a Collaborative final assembly line, through virtual reality simulation.

(a) VR environment for assembly, and the detail of semi-automatic assembly of overhead bin.

(b) Ergonomics analysis for collaborative assembly (sidewall panel on the left, overheard bin on the middle, and lining on the right).

FIGURE 2.15 – Simulation of interiors assembly, (OTTOGALLI *et al.*, 2021).

Although these cited studies, the application of collaborative solutions in the real final assembly process is still a challenge, considering all the process characteristics presented before. As indicated by Salunkhe *et al.* (2019), to increase the use of Cobots on final assembly, more successful cases are needed, expanding the range of possible solutions. Therefore, the present work intends to evaluate the application of Collaborative robots in the final assembly process of an aircraft, mapping the main activities, identifying potential applications, and developing a conceptual solution for the most suitable application, then contributing to the increase of maturity of collaborative solutions applied in this scenario.

2.6 Frameworks for Cobots Application Evaluation

Collaborative robots, as presented, have the potential to transform the aeronautical industry, increasing the level of automation of aircraft assembly and as a consequence, improving productivity, quality, and operators' ergonomics, (SIMÕES *et al.*, 2020). However, as presented in section 2.4, the aircraft's final assembly encompasses several activities with different characteristics, restrictions, and requirements. As a consequence, the potential benefits that can be achieved by the introduction of Cobots vary significantly depending on the application, and for this reason, an evaluation is needed to identify stations with high improvement potential, (COHEN *et al.*, 2022).

Furthermore, the application of Cobots in a company has an impact not only on the

production indicators but also, involves social implications, and for this reason, must be taken at a strategic level of the organization, (COHEN *et al.*, 2019). Consequently, in the last years, a considerable number of works have been published related to the frameworks for Cobot deployment in manufacturing environments, considering the evaluation of the expected benefits, helping the identification of the best application for the Cobot's implementation.

Gualtieri *et al.* (2020) proposed a methodology to evaluate the deployment of Collaborative robots, that consists basically of two main steps: Analysis of the current situation in terms of physical ergonomics; and the evaluation of the potential for collaborative robotics. To conduct the first step, the author presented a framework shown in Figure 2.16, which consists of an algorithm that orientates the identification of non-ergonomic tasks that could be improved by a collaborative operation. Depending on the answers, the algorithm leads to the application of specific methodologies to quantify the level of ergonomics issues, such as the NIOSH (National Institute for Occupational Safety and Health) lifting index; OCRA (Occupational Repetitive Action) checklist, or the RULA (Rapid Upper Limb Assessment) for working postures.

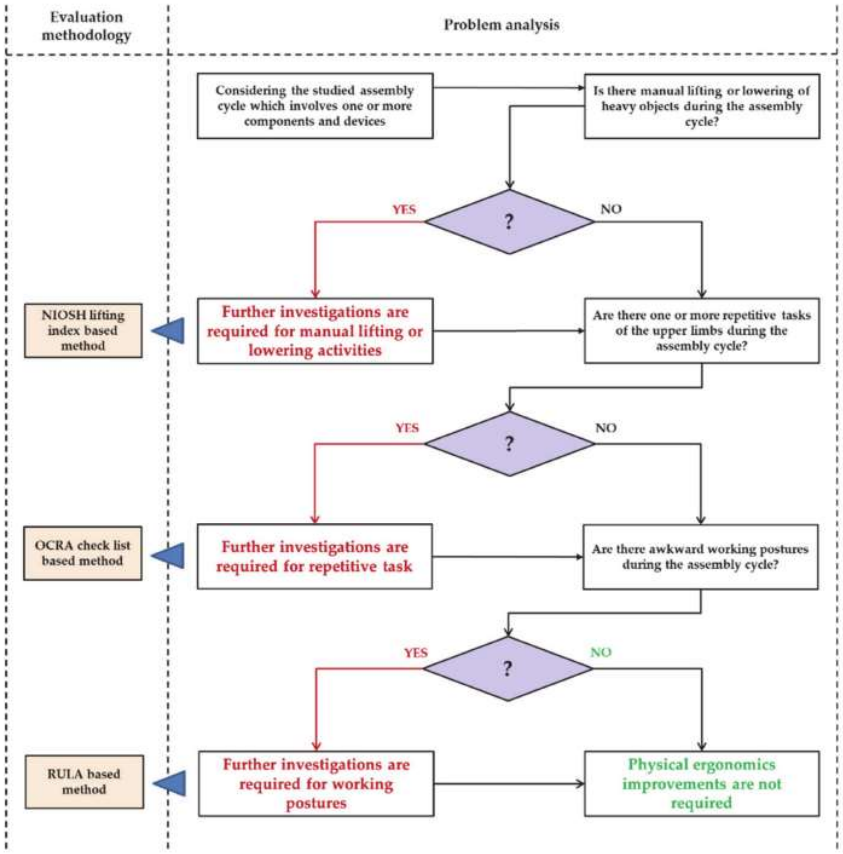


FIGURE 2.16 – Identification of non-ergonomic tasks framework, (GUALTIERI *et al.*, 2020).

The second step of the methodology presented by the author consists of the application

of a so-called Quick Assessment algorithm, shown in Figure 2.17, that allows evaluating the potential of the application of a collaborative robot to perform certain tasks. The algorithm is based on the evaluation of three main aspects: Safety and ergonomics, product and process quality, and economics, where the sum of the product between the score evaluated for each task and the weight of each aspect represent the potential index for a collaborative application. The obtained value is then classified, and for a potential index higher than 18, the activity is considered with a good potential for the application of a collaborative robot.

Category	Process Critical Issue (PCI)	Weight (W_i)	Score (S_{ij})
Safety and ergonomics	PCI 1: Are there physical ergonomic problems related to: —lifting/lowering or carrying of objects OR —repetitive tasks of the upper limbs characterized by repeated work cycles OR —static or awkward working postures?	3	0–3
	PCI 2: Are there occupational risks for the operator's safety which are not properly managed OR occupational risk for operator's health which are not properly managed? (not yet considered in the previous point)	3	0–3
	PCI 3: Is there high work monotonousness OR very low requirements in terms of task qualification of the manual work?	3	0–3
Product and Process Quality	PCI 4: Are there not-constant/satisfactory product quality OR unsuitable process quality levels according to the nominal values (i.e., characterized by high variability/low standardization)?	2	0–3
Economics	PCI 5: Is there an inefficient use of time and/or resources without a real advancement of production, which means tasks which cannot generate value to the final costumers (not value-added activities) OR Is there low/not satisfactory process productivity and/or production efficiency?	1	0–3
	(*) Score meaning: 0—Improvements are not required for that activity / the problem does not exists 1—The achievable improvements could be moderate for that activity 2—The achievable improvements could be good for that activity 3—Improvements could be very significant for that activity		

FIGURE 2.17 – Quick assessment algorithm, (GUALTIERI *et al.*, 2020).

The presented methodology was applied in a real case study of a redesign of a wire harnesses assembly cell. According to the author, the new proposed collaborative assembly station reduced the operator's work condition in terms of manual handlings by more than 12% and 50% in terms of postural improvements. Moreover, it was verified a production efficiency improvement by reducing the cycle times by more than 12%, concluding that both the methodology used and the proposed solution were effective.

A similar framework is presented by Gualtieri *et al.* (2019), where the conversion process of a manual assembly process is analyzed to verify its feasibility. The proposed methodology called HRAA (Human-Robot Activity Allocation) is a procedure that, starting from an existing process, allows the separation of tasks and activities between human and robot in a new collaborative workspace, considering aspects such as safety and ergonomics, process quality, economics, and technical feasibility.

As shown in Figure 2.18 the framework is based on four evaluation steps that allow the verification that an assembly task or process is suitable for the application of a collaborative robot. Furthermore, if the collaboration is feasible, the framework allows the definition of the best task assignment combination for the activities, identifying the task performed only by the robot or by the human, and those performed in collaboration.

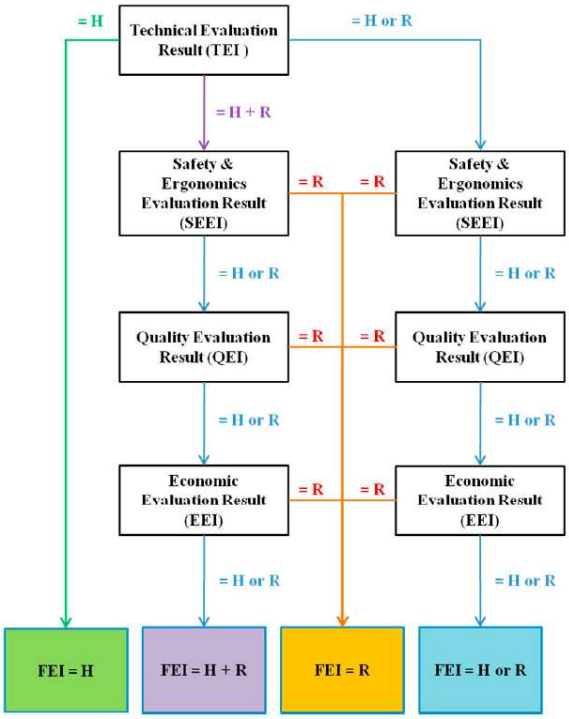


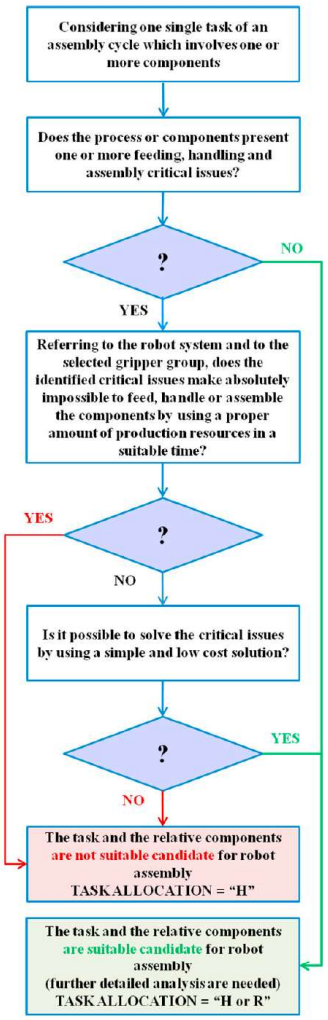
FIGURE 2.18 – HRAA methodology, (GUALTIERI *et al.*, 2019).

According to the proposed framework, the first stage is the Technical evaluation analysis (TEI), which aims to evaluate if an activity is efficiently executable by the robot considering technical reasons. This evaluation is performed through the application of the algorithm shown in Figure 2.19(a), which considers the influence of "technical critical issues" on the capability of the Cobot to execute the task. As presented in Figure 2.19(b), in general, the main complexities arise from product geometry, dimensions, materials, assembly location, and sequence.

Following the proposed methodology, the next stage is the Safety and Ergonomics evaluation (SEEI), which is performed using the pre-cited RULA method. Then, the Quality evaluation (QEI) is performed based on the evaluation of the Coefficient of Variation (CV), which is a largely used KPI to evaluate the quality of a manufacturing process. In this case, depending on the value of the CV, the algorithm identifies the best task assignment. Finally, the Economics evaluation (EEI) is performed considering the classification of the activity in a Value Added (VA) or a Not Value Added (NVA).

The previous methodology was applied to a real case study, presented by Gualtieri

et al. (2020), related to the assembly process of pneumatic cylinders. According to the author, the methodology is suitable for preliminary feasibility analysis and can help the proper use of collaborative robots in assembly cells to improve performance, operators working conditions, and production quality.



(a) Technical Evaluation Index (TEI).

Feeding critical issues
– The component is magnetic or sticky
– The component is nest or tangle
Handling critical issues
– The component has none symmetry axis
– The component is fragile or delicate
– The component is flexible
– The component is very small or very big (referring to a human hand)
– The component is light so that air resistance would create conveying problems
– The component is slippery
Assembly critical issues
– Components do not have a "datum surface" (reference surface) which simplify a precise positioning during the assembly
– Components cannot be easily orientate
– Components do not include features which allow a self-aligning during the assembly
– Components cannot be located before they are released
– Components provide resistance to insertion
– Components do not provide chamfers or tapers that help to guide and position the parts in the correct position
– Components have not a suitable base part on which to build the assembly
– Components cannot be assembled in layer fashion from directly above (z-axis assembly)
– The assembly is overconstrained
– It is difficult to reach the assembly area / the components access for assembly operations is restricted or not easy to reach
– The component and/or the assembly sequence requires high physical dexterity
– The assembly requires high accuracy and/or demanding insertion tolerances
– The assembly needs to reposition the partially completed sub-assembly, other components or fixtures
– The assembly requires to reorient the partial assembly or to manipulate previously assembled parts
– Components requires to be compress during the assembly
– The component and/or the assembly sequence requires two hands for handling
– The component and/or the assembly sequence require typical human skills (for example touch perception, haring, ability to interpret situations…)

(b) Critical Issues.

FIGURE 2.19 – Framework of Cobot deployment evaluation, (GUALTIERI *et al.*, 2019)

Borges *et al.* (2021), presented an alternative framework to help the decision-making of the deployment of a collaborative robot, drastically reducing the risks involved. As shown in Figure 2.20, the proposed framework is based on four stages: (i) assessment of the current situation, including the ergonomics evaluation and collection of production data; (ii) modeling of the key factors of the system; (iii) simulation of different scenarios when inserting a Cobot to the system; and (iv) technical, ergonomic and economic evaluation.

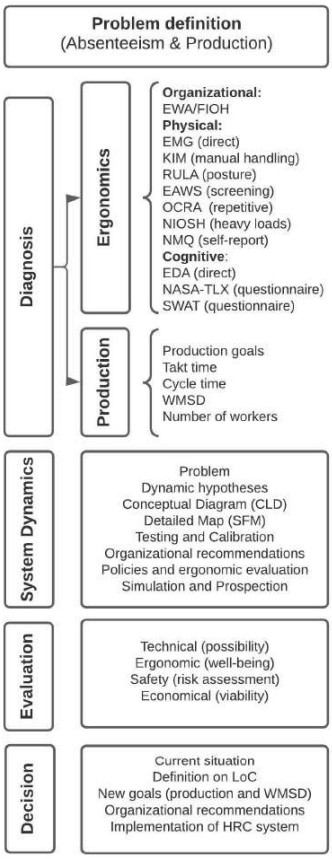


FIGURE 2.20 – Decision making of Cobot deployment, (BORGES *et al.*, 2021).

The first stage defined in the framework is based on several tools to evaluate the current situation of the process, considering the organizational, physical, and cognitive aspects related to ergonomics, while productivity is evaluated through well-known KPIs such as takt time and cycle time. Results obtained in this first stage are used for the second and third stages, which consist of computational modeling and simulation of several cases to understand the best Level of Collaboration (LoC) for each task of the process.

Finally, the results obtained from the simulation indicate the best task assignment between humans and robots, identifying those performed by the robot. Thereby, it is possible to perform a technical evaluation, based on the critical issues presented in Figure 2.19(b), (GUALTIERI *et al.*, 2019). The proposed collaborative workstation is then evaluated through ergonomics, safety, and economics aspects to verify the possible gains that can be achieved, allowing the decision to be made.

The presented framework was applied in a real case study related to the deployment of a Collaborative robot to assist human workers in the manual assembly of electrical components. According to Borges *et al.* (2021), the main objectives were to reduce absenteeism due to Work-related Musculoskeletal Disorders (WMSD) by 25% while increasing productivity by 20%. After the study, the authors concluded that considering the objectives, assessments, evaluations, and investments, the workstation studied justifies the

application of collaborative robots. Furthermore, the presented framework well addressed the difficulties related to the decision-making of deploying Cobots mainly by the fact that is based on reliable data and process, reducing the risk of failure of the project

Other frameworks to evaluate the deployment of Cobots have been discussed in the literature, as presented by Zanella *et al.* (2017). Although this methodology is focused on automotive use cases, the framework in a general way, is similar to the previous presented, as shown in Figure 2.21.

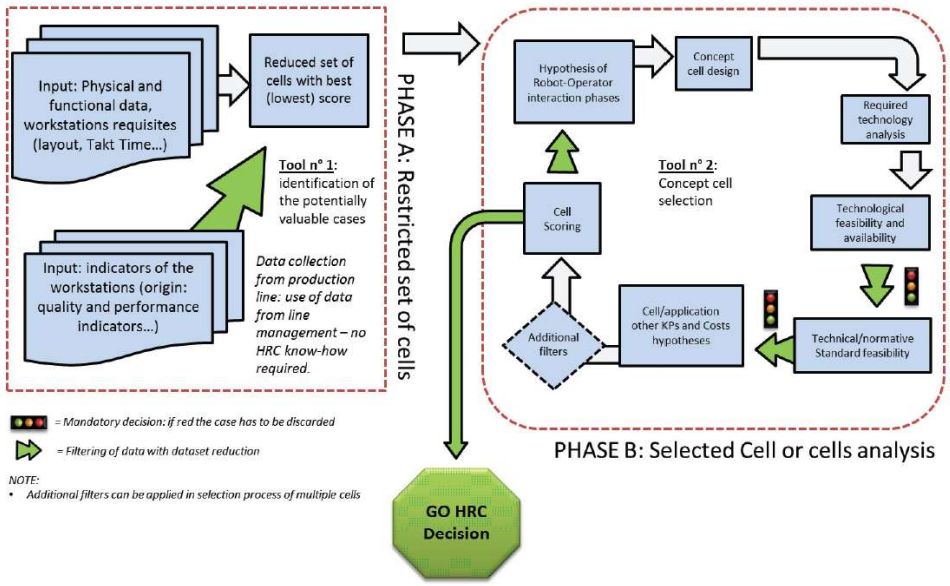


FIGURE 2.21 – Framework focused on automotive evaluation, (ZANELLA *et al.*, 2017).

The presented methodology is applied to evaluate and score a set of candidate applications for Cobot deployment, based on two phases. The first phase, similar to the previously presented frameworks, consists of the data collection of the current situation, based on the productivity and ergonomics indexes, as well as the first filtering of the possible applications. This filter guarantees that, for example, a current profitable automatic application is eliminated from the possibilities while a very non-ergonomic takes a higher classification on the application possibilities pool. As a base for this phase, takt time, Non-Value Added Activities (NVAA), ergonomics classification, and logistics analysis are carried out to support the understanding of the current situation for each potential application.

The second phase of the framework consists of the evaluation of feasibility and benefit analysis, which is based on five key parameters: Technological complexity; HRC relevance; Benefits/Costs indicator; Ergonomics & Safety; and Logistic Interface. To allow the evaluation of the cited parameters, in this phase, is required to know how the application will work, including a draft of the main working steps and a concept design of the cell in terms of functionalities, systems, and hardware. The final overall score is based on the

sum of the above-listed parameters.

Malik and Bilberg (2019) and Cohen *et al.* (2022) are other examples of frameworks developed to help the decision-making process of deploying Cobots on manufacturing systems. The first one is similar to the one presented by Gualtieri *et al.* (2019), which is based on the evaluation of the best task assignment strategy between the human and the robot for an application. This methodology was applied to a real case study of a linear actuator assembly and implemented through a tablet application, and the results showed that the methodology allowed a rapid evaluation of the potential application for Cobots.

Cohen *et al.* (2022) by the other hand, presented a methodology that is tightly based on productivity indexes. The author limited the analysis to the addition of a Cobot in a current manual workstation and proposed a model based on the comparison of productivity before and after the cobot deployment. The main contribution of this work is a quantitative methodology that allows the evaluation of the suitability of the deployment of a Cobot for both a single workstation and a complete assembly line.

Finally, Table 2.3 presents the consolidated methodologies discussed above.

Reference	Methodology Steps	Considered Aspects	Case Study	Results
(GUALTIERI *et al.*, 2020)	- Analysis of current ergonomics; - Evaluation of potentials for Cobots; - Workstation Re-design .	- Safety & Ergonomics; - Product & Process Quality - Economics.	Wire harnesses assembly cell	- 12% reduction on manual handling work; - 50% postural improvements; - 12% cycle times reduction.
(GUALTIERI *et al.*, 2019)	- Technical Feasibility Evaluation; - Safety & Ergonomics Evaluation; - Quality Evaluation; - Economic Evaluation.	- Objects characteristics; - Safety & Ergonomics; - Product & Process Quality - Economics.	Pneumatic cylinders assembly.	Methodology considered feasible and suitable for preliminary evaluation.
(BORGES *et al.*, 2021)	- Diagnosis; - Task assignment evaluation; - Evaluation solution; - Decision,	- Production; - Safety & Ergonomics; - Economics.	Electrical components assembly.	Methodology well addressed the deployment, reducing decision-making risks.
(ZANELLA *et al.*, 2017)	- Current state evaluation & first filter; - Feasibility and benefits Evaluation.	- Production (takt times); - Non-value Added Activities - Ergonomics; - Logistics;	Automotive general assembly example.	Cyclic methodology analysis brings incremental definition detailing helping decision-making.
(MALIK; BILBERG, 2019)	- Current tasks evaluation; - Task assignment evaluation.	- Objects characteristics; - Logistics (feeding); - Operation characteristics; - Safety;	Electronic linear actuator assembly.	Rapid evaluation of the potential for Cobot application.

TABLE 2.3 – Cobots deployment evaluation frameworks summary.

As can be seen, several methodologies related to the Cobots deployment evaluation for manual assembly lines are available in the literature. Despite the specific characteristics of each one, all the previous presented methodologies have in common the general sequence of: (i) well understanding of the current situation in terms of ergonomics, productivity, and quality; (ii) identifying potential applications considering aspects such as technological feasibility; (iii) identifying the most suitable applications; (iv) define the application and evaluate the expected benefits compared to the original situation. These frameworks were

used as a base for this work, helping the definition of the methodology used for selecting the application to be studied, which is presented in the next chapter.

2.7 Conceptual Design Methods and Tools

This section presents an overview of the main methods and tools used in this work related to the conceptual design and decision-making processes.

2.7.1 System Design Process

The development of a new system that successfully achieves all the expected goals generally is complex, nonlinear, and includes several moments of uncertainty. For these reasons, it is essential for the development process to be performed following a methodology that helps to organize all the inputs, define alternatives, evaluate, and select the most adequate solution.

In this context, the Systems Engineering approach was introduced, which according to Kapurch (2010), is defined as a methodical, multi-disciplinary approach to the design, realization, technical management, operations, and retirement of a system. The system can be understood as any the combination of elements that function together to produce the capability required to meet a need.

The main motivation for the use of the systems engineering approach is that it seeks a safe and balanced design in the face of opposing interests and multiple, sometimes conflicting constraints. Systems engineering is composed of three main elements that are connected, as presented in Figure 2.22. The System Design process is the element responsible for defining the design solution from stakeholders' expectations, while the Product Realization Process is the phase where the solutions are created and validated. Between these elements, Technical Management is responsible to manage communication, assessing progress against the plans and requirements, and controlling the technical execution of the project (KAPURCH, 2010).

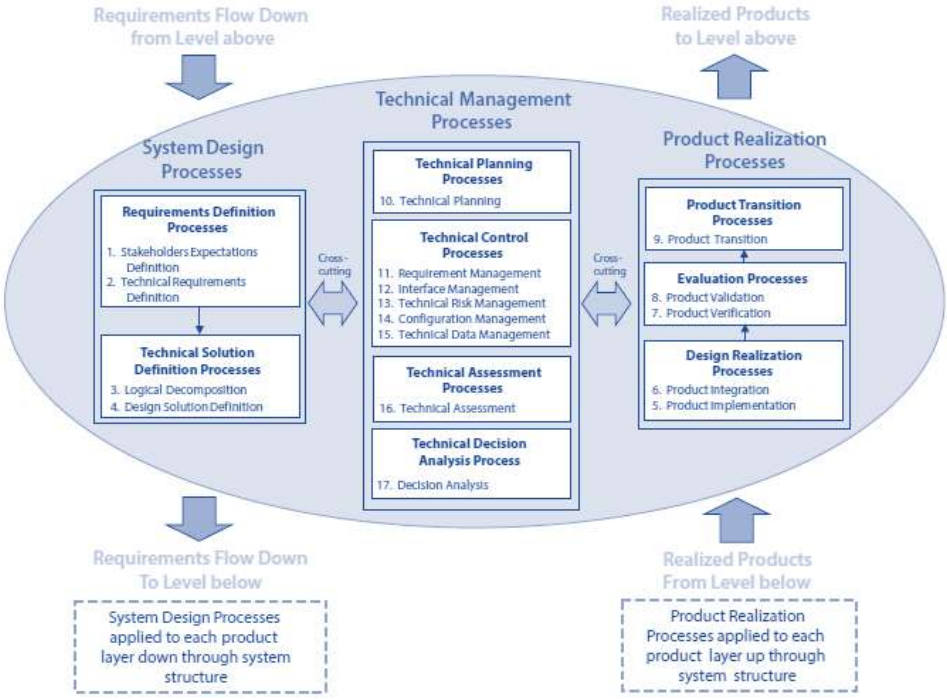


FIGURE 2.22 – The Systems Engineering Engine (NPR 7123.1). Adapted from (KAPURCH, 2010).

Considering the scope of this work, the Systems Design Process was the main element considered in the systems engineering approach. According to Kapurch (2010), the system design processes are interdependent, highly iterative, and recursive processes resulting in a validated set of requirements and a design solution that satisfies a set of stakeholder expectations. To comply with this objective four system design processes must be executed, starting with the stakeholder expectations, technical requirements, logical decomposition, and design solutions. Figure 2.23 shows the reference System Design process.

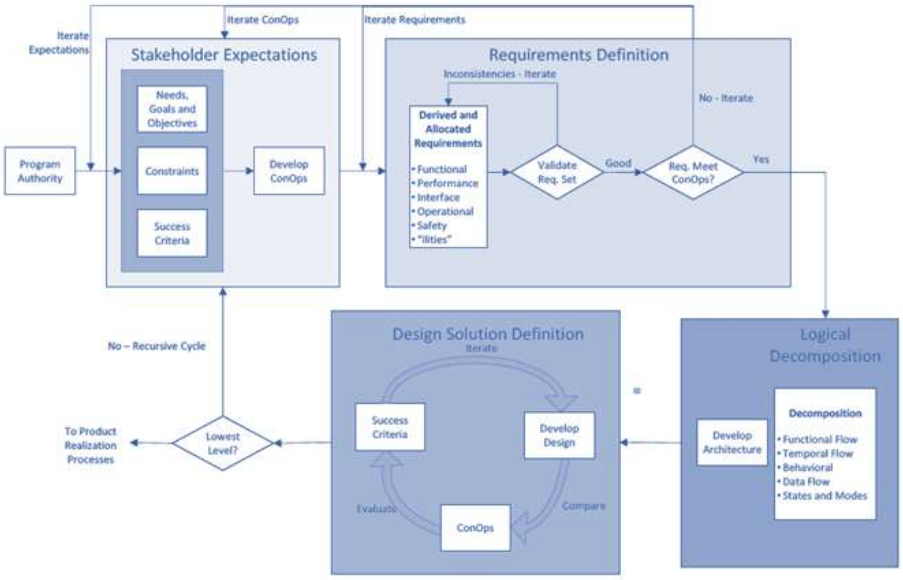


FIGURE 2.23 – Systems Desing process. Adapted from (KAPURCH, 2010).

The presented approach was applied to the definition of the solution design in the scope of this work and will be presented in Chapter 5. Special details for some methods and tools of the system design process are highlighted in the next sections, including the CONOPS, the technical requirements definition, and the functional analysis.

Finally, it is important to highlight that adding to the previously mentioned benefits of this approach, the systems design process was considered for this work because the correct understanding and definition of the concept of operations mean that the stakeholder's expectations will be correctly captured. As a consequence, the requirements and the solution design will be aligned with the real needs, leading to a higher chance of a successful operation over the product life cycle.

2.7.2 Concept of Operations

The development of a new project generally starts with the identification of a problem or an opportunity from a current situation, including the expectations of the people involved. In this sense, according to the System Design Process, the first step is regarding the Stakeholder Expectations Definition, which establishes the foundation for the design of the solution, (KAPURCH, 2010).

The main purpose of the Stakeholder analysis is to identify the most interested parts and how will interact with the solution. To guarantee that the stakeholders expectations are correctly captured, one important design tool is used: The concept of operations, or CONOPS. According to Fairley and Thayer (1997), the CONOPS development process is based on analyzing a problem domain and an operational environment to specify the

characteristics of a proposed system from the users' perspective.

The CONOPS describes the overall high-level concept of how the system will be used to meet stakeholder expectations, usually in a time-sequenced manner. Typical information contained in the CONOPS includes a description of the major phases, operation timelines, and operational scenarios, being generally visually represented through diagrams.

In the context of this work, the CONOPS methodology is applied to capture all important aspects of the application environment allowing the identification of the main functions required as well as the definition of the design requirements. The use of CONOPS and use cases are highly indicated once compose the System Design methodology, and often reveal requirements and design functions that might be overlooked,(KAPURCH, 2010).

2.7.3 Technical Requirements Definition

The Technical Requirements Definition process aims to transform mapped stakeholder's expectations into a technical view of a solution that complies with the operational needs of the user, represented through the CONOPS, (INCOSE..., 2015).

The process of requirements definition is recursive and interactive once it evolves with the maturity of the development, however, must be properly defined from the beginning, once a change in requirements later in the development cycle can lead to a significant cost impact and even the cancellation, (KAPURCH, 2010), (INCOSE..., 2015). Furthermore, the requirements should enable the description of all the inputs, outputs, and their relation, as well as constraints and the interaction of the system with the environment.

According to Carson (2015), there are four types of requirements. The first one is the Functional/Performance type, which defines functional behavior, performance, and the condition of performance of the system. Secondly, Design requirements delimit the design space of the project by limiting the range of acceptable design, fabrication, and materials solutions. The third type is the Environmental, which aims to guarantee the proper consideration of the operational scenario in the solution. Finally, there are requirements of suitability, which include other factors that may influence the solution's design, including safety and regulation, human factors, and logistics, for example.

The Requirements Definition process, according to Kapurch (2010), encompasses some stages presented in Figure 5.12. First, with the stakeholder's needs and the identified Concept of Operations, design and product constraints are defined jointly with the definition of the functional expectation in technical terms. This functional definition leads to the definition of performance requirements that, with constraints and other aspects are translated into technical requirements written in "shall statements". Finally, the requirements are verified to guarantee that they are clear, correct, complete, and achievable, and metrics

and measurements are defined to be further verified, forming the baseline of the product structure.

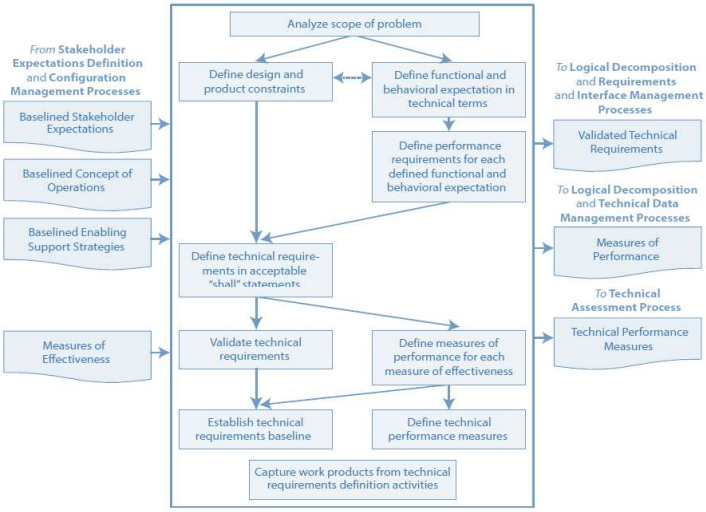


FIGURE 2.24 – Technical Requirements Definition process framework. Figure adapted from (KAPURCH, 2010).

In the scope of this work, the Technical Requirements Definition process will be applied during the conceptual design of the solution, for the application of Cobots in the aircraft's final assembly.

2.7.4 Functional Analysis

After the definition of the top-level requirements, the execution of functional analysis is required to start formulating a conceptual system architecture, which organizes the functional elements of the system, their relationship, and dependencies. As a consequence, this analysis allows the correct understanding of the system composition and the contribution of each element to achieve the defined requirement, (KAPURCH, 2010).

The functional analysis is performed in three key steps. First, it is necessary to translate the top-level requirements (defined through the process discussed in the last section) into functions that should be performed by the system to accomplish the requirements. In the sequence, the defined functions must be decomposed and allocated to lower levels of the system breakdown structure. Finally, it is necessary to identify and describe functional interfaces, (KAPURCH, 2010).

For this work, the application of the Functional Analysis is a key to assure that all functions required for the collaborative solution will be addressed during the design of the application. Furthermore, this analysis allow the definition of potential solutions at a lower functional level, which are further combined to form the complete concept, according to the work objectives.

2.7.5 Design For Excellence Guidelines

The product development process is based on the translation of the high-level requirements and the outputs of Functional Analysis into a design solution, (KAPURCH, 2010). This process involves the generation of design alternatives to address the required functions and functionalities identified.

During the solutions alternative's generation, aspects from the entire product life cycle must be considered to guarantee that the solution not only addresses the functional requirements, but also considers all the interactions with the product from the conception, manufacturing, use, and disposal, (EASTMAN, 2012).

In this context, from the 1980s several guidelines were introduced to help the proper consideration of these aspects during product development. The first guideline introduced and implanted was the DFA - Design for Assembly – once the mass production required the manufacturing process to be increasingly efficient, leading to the need of considering assembly aspects such as inserting, gripping, orientating, etc, during the development stage.

The DFA revolutionized the design solution process, bringing significant economic results, and resulted in an expansion of the concept of considering life-cycle aspects during the solution design. This led to the emergence of new guidelines such as Design for Manufacturing (DFM), Design to Cost (DtC), Design for Life Cycle (DfLC), Design for Reliability, Design for Maintainability, and most recently, the Design for Environment (DfE). Moreover, due to the similarity of the terms, the guidelines were grouped through the definition of the term DfX, defined as Design for "X", where "X" represents the different aspects considered, such as Assembly, Manufacturing, Quality, Cost, etc, (EASTMAN, 2012), (ULRICH; EPPINGER, 2011).

DfX analyses are characterized by a systematic procedure, which according to Eastman (2012), can be divided into seven steps. First is the Product Analysis, which comprehends the system breakdown allowing a clear understanding of what elements and their specifications, compose the solution. In the sequence, a clear understanding of the processes-product interactions is recommended, through the operational process and flow process charts. Once the product and process information are collected, their interactions can be measured in terms of the relevant performance indicators, being related to the performance requirements defined.

The fourth stage is the center of the application of DfX tools, which encompasses the comparison of the information gathered in the previous stages with benchmarking or reference information to verify what are the required modifications to the current solution. To guarantee that the proposed design changes will properly address identified gaps, there is the fifth stage, which is characterized by the identification of the causes of the lower

performance that is requiring an evolution of the design.

Different alternative solutions are studied in the further step, focusing on addressing the identified problems regarding the different aspects of the design (assembly, cost, quality, etc). Finally, if necessary, the solution design modifications are prioritized and implemented, finalizing one cycle of DfX evaluation.

The presented guidelines were considered in the context of this work during the conceptual design phase, mainly during the combination of the identified solutions for each functionality, aiming to guarantee that the concept addresses a broader set of life-cycle aspects.

2.7.6 Multi-criteria Decision Methods

The decision-making process is one of the main challenges faced by organizations since it generally involves the selection of adequate parameters and their influences on the final decision, (VARGAS; IPMA-B, 2010).

In the context of this work, the decision-making process is a key factor, being present in two important stages: The selection of the application most suitable for the deployment of Cobot; and the definition of the conceptual design of the application. Furthermore, as presented in section 2.5 the decision of implementing Cobots is a complex decision requiring the balancing of different variables, leading to the use of a multi-criteria decision approach.

There are several different multi-criteria methods available in the literature, including, for example, KT (Kepner-Tregoe), ELECTRE (Elimination and Choice Translating Algorithm), PROMETHEE (Preference Ranking for Organization Method for Enrichment Evaluation), as presented by Guarnieri (2015). However considering the possibility of utilizing both qualitative and quantitative criteria, defining different weights for criteria, and presenting a hierarchical structure, the AHP (Analytic Hierarchy Process) was selected for the scope of this work.

According to Saaty *et al.* (2008), the AHP is performed in four stages, beginning with the definition of the analysis objective. In the sequence, the problem is organized in a hierarchical structure, with the objective on the top, the criteria on the intermediate level, and the alternatives to be evaluated at the lower level, as presented in Figure 2.25.

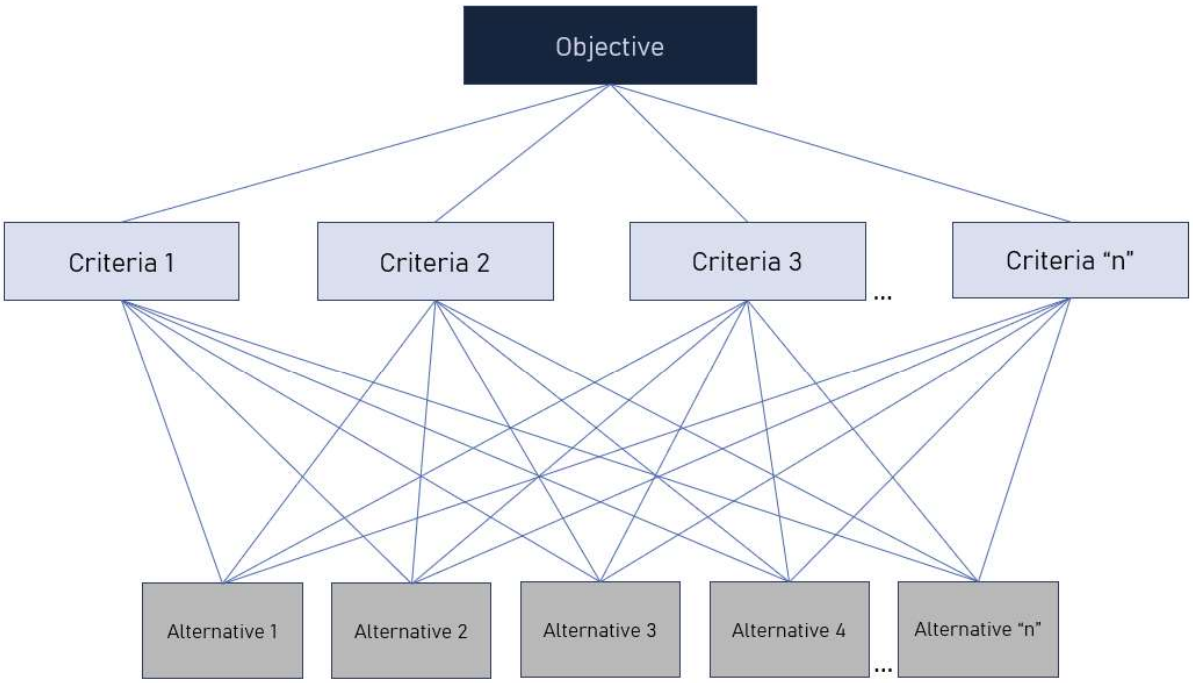


FIGURE 2.25 – Hierarchical structure of the problem, with criteria and alternatives,(VARGAS; IPMA-B, 2010).

The third stage of the method is based on the construction of a set of matrices for pairwise comparison. The first matrix is applied to the elements that compose the set of criteria, which are pairwise compared using a numerical scale proposed by Saaty *et al.* (2008), (Figure 2.4), to set the level of dominance between the elements for each pairwise comparison.

SCALE	NUMERICAL EVALUATION	RECIPROCAL
Extremely prefered	9	1/9
Very strong to extreme	8	1/8
Very strongly preferred	7	1/7
Strong to very strong	6	1/6
Strongly prefered	5	1/5
Moderate to strong	4	1/4
Moderatly prefered	3	1/3
Equal to moderate	2	1/2
Equally prefered	1	1

TABLE 2.4 – SAATY Scale (SAATY *et al.*, 2008).

After the comparison, the evaluation matrices must be normalized, through the sum of the elements of each column, followed by the division of each element by the sum of column values, (PAMPLONA, 1999), and the priority of each element is obtained through

the mean of the normalized values of their line.

Finally, according to Saaty *et al.* (2008), an evaluation is required to verify if there is a consistency between pairwise comparisons performed. Therefore, the author defined a condition that CR (Consistency Ratio) must be lower than 10%, where, the CR is defined by Equation 2.1.

$$CR = \frac{CI}{R} \tag{2.1}$$

CI is the consistency index, which is defined by Equation 2.2, where "λ_{max}" is the highest eigenvalue of the evaluation matrix and "n" is the matrix order. R is the random consistency index, a fixed value that depends on the number of evaluated criteria, as shown in Table 2.5.

$$CI = \frac{\lambda_{max} - N}{N - 1} \tag{2.2}$$

N	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
R	0	0	0.58	0.9	1.12	1.24	1.32	1.41	1.45	1.49

TABLE 2.5 – Random consistency indexes, (SAATY *et al.*, 2008).

3 Methodology Definition for Application Selection

This chapter presents the methodology developed for the process of identifying and selecting the collaborative application to be studied in this work. The methodology was defined considering the previous frameworks, methods, and tools described in Chapter 2, in which a new framework was proposed to comply with the characteristics of the aircraft's final assembly environment. The methodology defined is based on four macro stages, where the first one is related to performing current process mapping, followed by the identification of the potential applications for Collaborative robots, the evaluation of the potential applications, and finally the selection of the most promising application to be explored in the scope of the work. The four steps are represented in Figure 3.1, and are further detailed in the next sections.

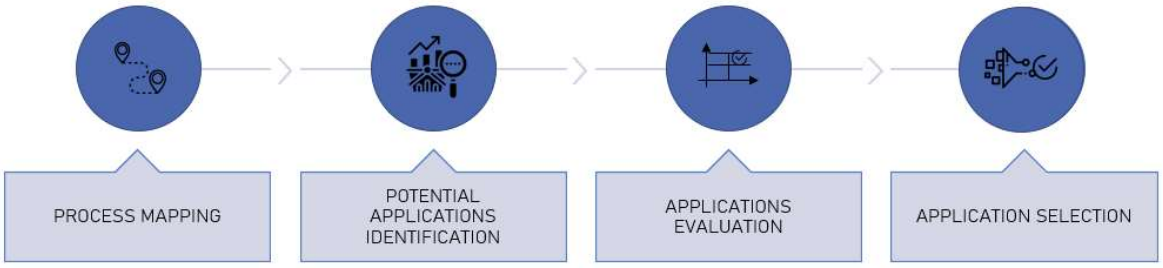


FIGURE 3.1 – Application selection framework.

3.1 Process Mapping Methodology

The first step defined for the methodology consists of mapping the current scenario, setting up the reference process, and identifying potential issues and opportunities. A good understanding of the current scenario is the base of almost all methodologies used

for process improvement, such as PDCA, DMAIC, etc. Furthermore, in the context of increasing the level of automation through the application of Collaborative robots, the mapping of the current process is also considered in most of the frameworks verified in the literature review. As presented in Chapter 2, the frameworks (PALOMBA *et al.*, 2021) and (BORGES *et al.*, 2021) consider this as the first stage of the evaluation for the Cobots application. Different tools and methods can be applied to map a current process. This step was based on two main elements for the framework to be used in this work. First, the mapping process considered the research on academic publications to create the baseline process of aircraft assembly, which was presented in section 2.4. In the sequence, the mapping process considered the execution of interviews with specialists to refine the process flow, and to collect deeper information regarding the real aircraft manufacturing process of a single-aisle commercial jet. The collected information will generate a process flow chart, where the final assembly activities and their predecessors and successors can be verified. Figure 3.2 represents the process defined for the current process mapping.

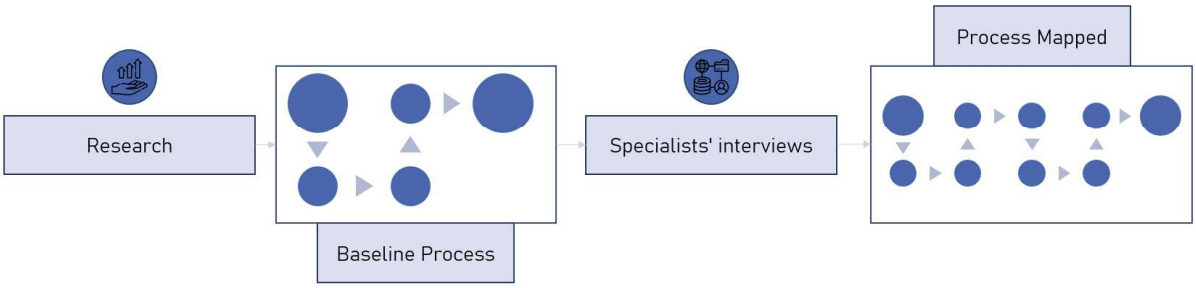


FIGURE 3.2 – Process mapping workflow.

The mapping of the current process allows the identification of the main phases and steps of the aircraft assembly and is the base for the next stage of the methodology.

3.2 Potential Applications Identification Methodology

The second step of the methodology defined consists of identifying the most promising applications for Collaborative robots in the aircraft assembly process. The basis for the identification of potential applications is the correct process mapping from the previous phase.

As presented in Chapter 2.2, the implementation of Collaborative robots has three main motivations: Ergonomics improvements, quality improvements, and productivity increase. As detailed, one of the most important improvements that can be achieved through the application of Collaborative robots is the worker's stress and fatigue reduction, which is possible considering that the Cobot may be used to perform non-ergonomic tasks,

including carrying heavy loads, repetitive tasks, and awkward positions (VILLANI *et al.*, 2018), (PALOMBA *et al.*, 2021).

Additionally, considering the characteristics of robots, like repetitiveness, an improvement in process and product quality can be achieved, (ZANELLA *et al.*, 2017). Even though the ergonomics benefits already justify the application of collaborative robots, the correct task allocation between humans and robots may lead to a productivity enhancement, allowing the cycle times reduction, as presented by (PALOMBA *et al.*, 2021), (TSAROUCHI *et al.*, 2017) and (SALUNKHE *et al.*, 2019), for example.

In this context, the collaborative potential applications identification methodology was defined to capture those activities mapped with the potential to improve at least one of the pillars cited above. The identification of the potential applications was based on data collection with specialists, through guided interviews, using a questionnaire developed to allow an analysis of the current situation in terms of Process & Productivity, Ergonomics & Safety, and Quality, finishing with general questions to collect the specialist's suggestions and opinions. The questionnaire was developed based on the methodology of GUALTIERI *et al.* 2020, and is presented in Table 3.1.

TABLE 3.1 – Process mapping questionnaire

CLASS	QUESTIONS	
PROCESS AND PRODUCTIVITY	Which step of the process?	Record the stage at which we are collecting information.
	What is the overall objective of this stage of the process?	Confirm the purpose of the process step, from the expert's point of view.
	What activities were carried out and the sequence?	Confirm the steps currently mapped and collect possible unmapped activities.
	Is this sequence always the same? Is there any factor that changes the sequence of these activities?	Evaluate whether there is a change in the sequence of activities. Influences Cobot reprogramming.
	Are these activities always the same for any aircraft?	Complementary question to the previous one.
	What are the tools used for each activity?	Understand the resources used, the type of tools that operators use. Can serve as a basis for identifying requirements for an effector.
	What information is needed in carrying out the activities? Do they need to be checked every time?	Understanding if he needs to follow a step-by-step, can influence the way he operates with Cobot.
	How is the position of the items to be installed defined? Is there any reference for placement of these items?	Understand which guides are used for positioning and installing items.
	Are the installed items rigid or flexible?	Understand the difficulty of handling items.
	Is there any activity that needs more than one operator to be performed? What is the role of each one?	Understand if there is collaboration at any stage and if there is any activity that requires more than one operator to be performed.
	Do any of the activities listed require high movement from the assembler? What is the reason?	Important to check the movement of the Cobot .
	How do the components to be installed get to the assembly space?	Understand, in the case of the Cobot application, if there will be a need for a power supply system from the components to the robot.
	How many HH are needed for each activity?	Collect the amount of effort required to perform the activity.
	What is the cycle time for this step?	understanding if he needs to follow a step by step, can influence the way he operates with Cobot.
	Number of cycles for this activity? (repeatability)	Understand if the complete activity cycle must be repeated more than one time per aircraft.
ERGONOMICS AND SAFETY	Is there heavy objects moving during any activity in this stage? If yes, what weight?	Understand if there are ergonomic problems in activities.
	What is the frequency of this movement?	Understand if weight movement is constant throughout the work period (Cobot)
	How is the level of repetitiveness?	Understand the level of repetitiveness of tasks.
	In any of the listed activities does the fitter need to be in an uncomfortable or awkward position?	Understand if there are possible ergonomic problems in the operation.
	For how long ?	Understand if this awkward position is present in only a portion of the operator's working time.
	In any of the activities listed is there access difficulty to perform it?	Understand if the operator has difficult access to perform the activity, which may even impact the possibility of placing a Cobot.
	Is it common for workers who work in the activities listed to be on leave (for health reasons)? Does any stand out?	Evaluate whether there is any indication of the occurrence of occupational problems at this stage.
	In any of the activities, are there occupational risks related to the operator that are not yet well addressed?	Evaluate whether there are occupational hazards, which may be or cause accidents to operators.
QUALITY	What are the main types of nonconformities that can be observed at this stage?	Understand if there are nonconformities, and the reason that leads to their occurrence, aiming to identify potential for improvement.
	What are the main types of rework that may be required at this stage?	Understand if there are reworks, and the reason for their occurrence, in order to identify potential for improvement.
	How is the inspection done after carrying out the activity?	Map the activities that require inspection, as additional possibility use for the Cobot.
TRAINING	How is the training carried out to carry out the activities?	Identify the process of preparing operators to carry out activities.
GENERAL	In your view, could any of the activities benefit from automation?	Collect the specialist's view of the possibility of applying these technologies in the process.
	Have potential automation solutions been studied? If so, why were they not implemented?	Collect with the specialist if such technologies have already been thought of, and indicate what the difficulties of implementing them may be.

The method consisted of the definition of a set of questions to be used during the interviews to collect the required information and allow the applications' selection. The first part of the questionnaire, related to Process and productivity, was defined to capture deeper information about the process step in the analysis, including activities sequence and restrictions, tools used, characteristics of the component assembled, as well as, process indicators such as cycle and man-hour required. Once one of the pillars of the implementation of Cobots is related to productivity, this set of questions may capture those activities that require a higher workload and influences more the aircraft production cycle. Furthermore, the characteristics of the product assembled, and the tools required may allow

identify aspects with more prone to automation.

The second set of questions, related to ergonomics and safety, aims to collect aspects such as repetitive work, handling of heavy loads, and working postures, for example, that allow the creation of the current scenario regarding ergonomics and safety issues. Based on the information collected by this section, it is possible to identify those activities with higher ergonomics issues, which combined with the characteristics of the process mapped in the questions before, identify applications with more potential for reducing human workload or operational risks, another pillar of the Cobots' application.

The third set of questions presented in Table 3.1 was defined to capture quality aspects and eventually, quality escapes verified in each assembly step. Also, with the defined questions it is possible to identify applications that require quality inspections, that may indicate a potential function to be performed by a Collaborative robot. Additionally, the questionnaire collects information regarding the training aspects for professionals' preparation to perform the activity. This aspect was defined to capture activities with lower or higher human qualifications required, indicating eventually activities with a higher possibility to be automated to free human operators to perform more specialized activities that can't be automated.

Finally, the questionnaire finishes collecting the specialist's opinions regarding the application of automation. This section was introduced to allow the capture of aspects related to the expert previous experiences that can't be collected by previous questions, however, may bring important insights to help the identification of activities most promising for the application of Cobots.

Based on the collected information, the first set of applications with the potential for the application of Collaborative robots can be obtained. At this stage, the recommendation is to select all the applications that presented an opportunity in any of the fields by the deployment of a Collaborative robot. At the end of this step of the methodology, a number of potential applications can be identified, considering a high-level scenario identified for the cobot operation, that is set to be evaluated in the next stage.

3.3 Potential Applications Evaluation and Selection Methodology

Most of the methodologies used to evaluate the application of Collaborative Robots consider aspects such as safety and ergonomics, process quality, economics, and technical feasibility, as presented in section 2.5 related to (GUALTIERI *et al.*, 2019), (BORGES *et al.*, 2021), and (ZANELLA *et al.*, 2017). The evaluation of both, improvements and feasibility

aspects are necessary to allow a broader view of the application and its consequences, helping the decision-making. As part of the methodology developed, the potential applications evaluation stage aims to evaluate the identified applications to have an indication of those applications with the most potential to be further analyzed. This is an important step of the methodology once as well as any other type of initiative, the prioritization of projects is required to comply with the restricted resources of the companies. The methodology developed was based on balancing two main pillars: the benefits expected and the technical development complexity. The selection process is illustrated in Figure 3.3, and as a result, the potential applications can be positioned into a matrix that allows decision-making, according to the company's directives. For example, based on the matrix a company can choose an application with a lower benefit expected compared to others, but that requires less development effort and can be implemented in the short term. Other companies may look to develop the solutions with the highest potential benefit, even if this solution would take a longer development time to be implemented in the assembly line.

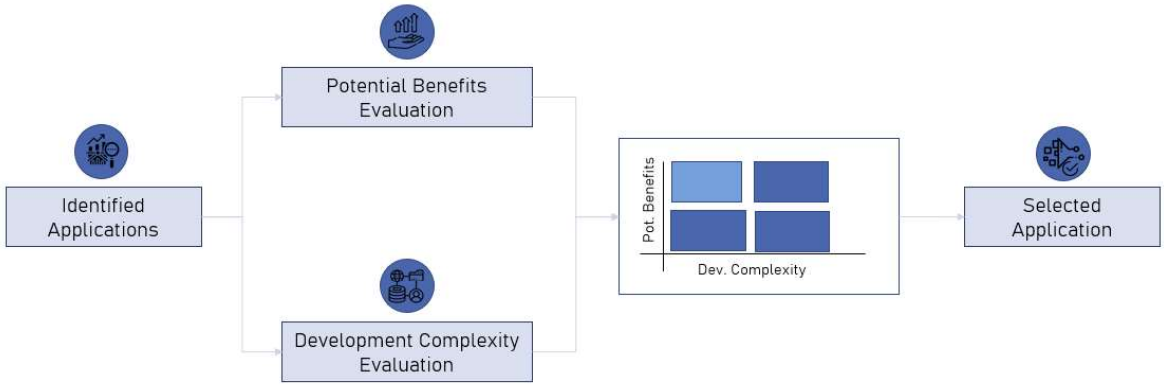


FIGURE 3.3 – Potential applications evaluation workflow.

Based on the framework of Figure 3.3, the next sections will detail the methodology defined for the evaluation of both potential benefits and development complexity for the evaluation of the potential applications.

3.3.1 Potential Benefits Evaluation Methodology

As part of the methodology developed, the potential benefits evaluation aims to evaluate the identified applications regarding their potential improvements for the current process. The developed framework was based on a simplification of GUALTIERI *et al.* (2020), GUALTIERI *et al.* (2019) and BORGES *et al.* (2021), presented in section 2.5, focusing in two main pillars: Productivity and Ergonomics enhancements.

In order to quantitatively evaluate the potential applications concerning the two pillars cited above, a numerical scale was defined and used to evaluate the applications, considering the information collected in previous stages of the methodology (Process Mapping and Potential applications identification). Figure 3.4 presents the general view of the methodology used to compose the score related to potential benefits evaluation.

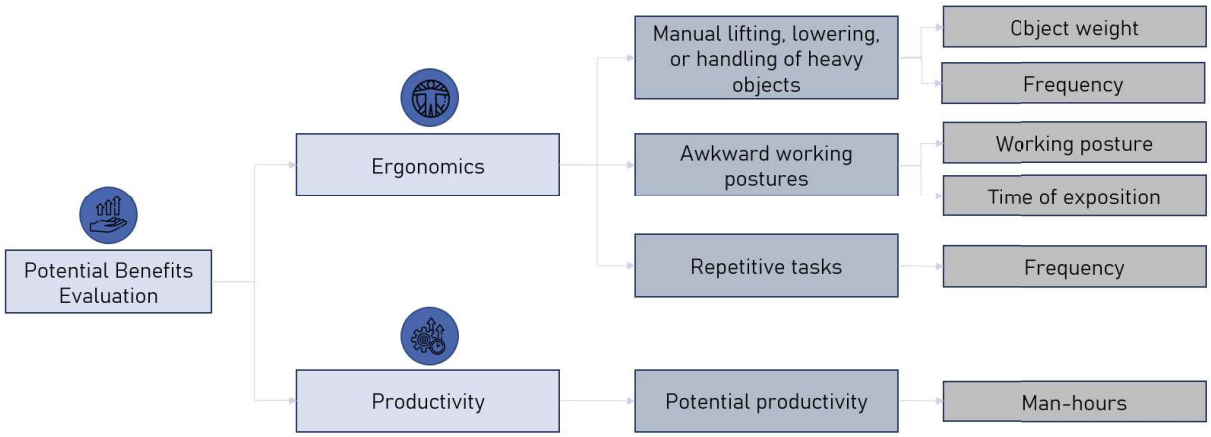


FIGURE 3.4 – Potential benefits evaluation methodology.

As can be seen, the evaluation of the ergonomics aspects was broken into three main categories: heavy object manipulation, awkward working postures, and repetitive tasks. For the first element, two aspects are required to compose the evaluation score: the weight of the object and the frequency that the object is handled during a working cycle, being the score obtained as a product of these aspects. For both, a scale was determined, where the weight scale was defined according to McAtamney and Corlett (1993):

1. For objects weighing less than 2kg;
2. For weights between 2 kg and 10 kg with intermittent movements;
3. For weights between 2 kg and 10 kg static (held);
4. For objects held weighing more than 10 kg.

For the frequency, a Likert-type scale was created, according to Nemoto and Beglar (2014):

1. Never;
2. Small portion of working time;
3. Moderate portion of working time;
4. Large portion of working time;

5. All working time.

The second element of the ergonomics evaluation, similarly to the first, depends on two components. Taking into account that it refers to awkward positions, working postures were classified according to McAtamney and Corlett (1993), allowing the definition of a scale:

1. Arms in a low position, neck, and torso straightened;
2. Arms up to shoulder height or inclined neck or inclined trunk;
3. Arms up to shoulder height and neck inclined down, or arms up to height shoulder height and trunk inclined, or trunk and neck inclined;
4. Arms above 90^{o} and/or neck extended upwards or trunk very inclined or crouched.

Composing the score, a frequency scale related to the amount of time spent in awkward postures was adopted identically that used for the handling of the heavy object element, and the final score is obtained as a product of these scales.

The last element related to ergonomics considerations refers to repetitive tasks, which aim to contemplate activities that may not include awkward postures or handling heavy objects, however, are performed repeatedly and also can lead to human worker injuries. For this element, a scale was defined based on considering the number of times that the activity is performed during a working cycle:

1. Up to 10 times;
2. Up to 20 times;
3. Up to 50 times;
4. Up to 100 times;
5. More than 100 times.

Additionally to the considerations for the ergonomics pillar, a scale was created to evaluate the potential improvement of productivity by the increase in the level of automation in the current processes. For this analysis, it was considered the use of the man-hour as the quantitative metric, since it represents a measurement of the amount of effort required to perform a task. Based on the characteristics of the potential applications identified, for the scale, it was considered that the higher the man-hour is, the higher the opportunity to improve productivity:

1. Up to 10 man-hours;
2. Up to 20 man-hours;
3. Up to 30 man-hours;
4. Up to 50 man-hours;
5. More than 50 man-hours.

Based on the defined parameters and methodological approach, each potential application can be evaluated regarding the potential benefit, receiving a final score to be used to position it in the final evaluation matrix. For the complete potential benefits evaluation, the ergonomics benefits and productivity benefits are calculated separately composing in the end the complete score. Considering the evaluation of potential applications regarding ergonomics aspects, Table 3.2 presents the defined arrangement used to compose the score, where:

- HOV – represents the score of the Heavy objects handling, which is composed of the multiplication of the object weight score (1 to 4) and the frequency of the handling (1 to 5).
- RT – represents the score of the repetitive tasks (1 to 5).
- AWP – represents the score of the awkward working position, which is composed of the multiplication of the working posture (1 to 4) and the time of exposition.

TABLE 3.2 – Evaluation of Ergonomic Score

	HOV (Wheight x Frequency)	RT (Frequency)	AWP (Posture x Exposit. Time)	TES Ergonomics Score
Potential application 1	HOV (1)	RT (1)	AWP (1)	**TES (1)**
Potential application 2	HOV (2)	RT (2)	AWP (2)	**TES (2)**
Potential application 3	HOV (3)	RT (3)	AWP (3)	**TES (3)**
...	...	...	...	...
...	...	...	...	...
...	...	...	...	...
...	...	...	...	...
Potential application n	HOV (n)	RT (n)	AWP (n)	**TES (n)**

For each potential application, the evaluation is performed for the three elements, resulting in the values HOV (1 to n), RT (1 to n), and AWP (1 to n). Finally, based on the scores of HOV, RT, and AWP, the Total Ergonomic Score (TES) can be calculated, by the Equation 3.1:

$$TES = \frac{\frac{HOV}{MHOV} + \frac{RT}{MRT} + \frac{AWP}{MAWP}}{n} \tag{3.1}$$

Where "MOHOV", "MRT", and "MAWP" represents the maximum possible value for each parameter, and "n" is the number of the parameters, which in this case is three.

For the productivity score, a similar procedure was defined and is represented in Table 3.3, where the "PP" is the score for the productivity parameter (1 to 5).

TABLE 3.3 – Evaluation of Productivity Score

	Productivity (Man-hour scale)	TPS Productivity Score
Potential application 1	PP (1)	**TPS (1)**
Potential application 2	PP (2)	**TPS (2)**
Potential application 3	PP (3)	**TPS (3)**
...	...	...
...	...	...
...	...	...
...	...	...
Potential application n	PP (n)	**TPS (n)**

The "TPS" is the total Productivity score, equivalent to the "TES", and is calculated by the Equation 3.2:

$$TES = \frac{\frac{PP}{MPP}}{n} \tag{3.2}$$

Where, "MPP" is the maximum possible value for the parameter "PP", and "n" is also the number of parameters, that in this case is equal to one.

Finally, the total potential benefit score (TPBS) can be calculated based on the "TES" and the "TPS". Table 3.4 illustrates the result expected, where the "TPBS" is calculated

based on the Equation 3.3.

TABLE 3.4 – Total Potential Benefits Score

	TES	TPS	TPBS
Potential application 1	TES (1)	TPS (1)	**TPBS (1)**
Potential application 2	TES (2)	TPS (2)	**TPBS (2)**
Potential application 3	TES (3)	TPS (3)	**TPBS (3)**
...	...	...	**...**
...	...	...	**...**
...	...	...	**...**
...	...	...	**...**
Potential application n	TES (n)	TPS (n)	**TPBS (n)**

$$TPBS = \frac{w1 * TES + w2 * TPS}{w1 + w2} \tag{3.3}$$

Where w1 and w2 are variables that represent the relative weight between the two elements that composes the "TPBS" score.

It is important to mention that, according to the presented by Gualtieri *et al.* (2020), the ergonomic aspect was considered with a weight three times higher than the productivity, following the Industry 4.0 principles which aim to promote sustainable production systems concerned with operators' occupational safety and well-being.

3.3.2 Development Complexity Evaluation Methodology

As mentioned before, despite the benefit expected by the application of Cobots, the complexity of the development of the concept can make it impractical. As a consequence, considering that the work aims to contribute to the increase in the level of maturity of this technology and allow the future insertion in the aircraft's final assembly, it was necessary to define a methodology to evaluate the complexity of the development of the potential applications identified.

Several factors can increase the complexity of a collaborative application, including the

characteristics of the object manipulated (as weight, shape, size, symmetry, and others), the sequence of the task performed, the characteristics of the environment where the Cobot will be deployed and many other characteristics, as presented by Gualtieri *et al.* (2019) in Figure 2.19 of section 2.5. Taking it into account, four main elements were defined to serve as a base for the evaluation of the identified potential applications, presented in Figure 3.5.

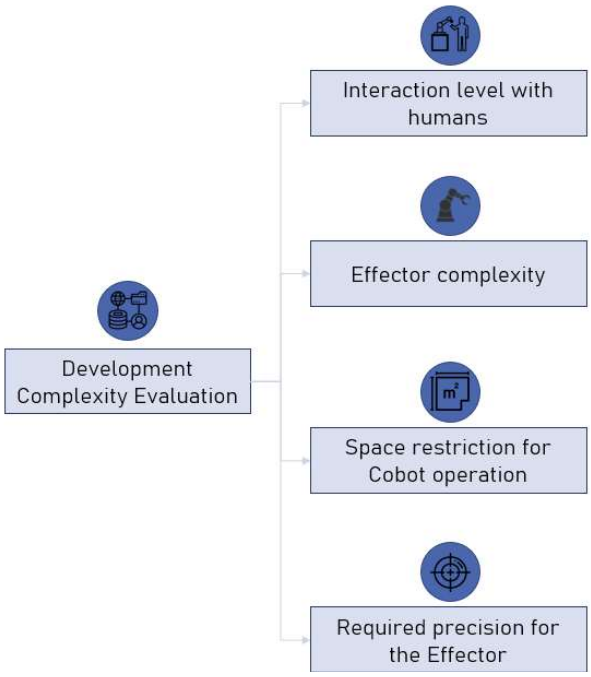


FIGURE 3.5 – Composition of the Development Complexity evaluation.

The first element aims to capture the challenges regarding the level of interaction between the human and the Cobot to perform the activity. For this element, it was necessary to define and differentiate the possible levels of interaction, which were based on Zaatari *et al.* (2019) and Segura *et al.* (2021), where the contact is classified into four possible levels, presented below.

- Independent: Human operator and robot work on separate workpieces and are detached from each other but in the same workspace.
- Sequential: Human operator and robot work on consecutive processes on the same workpiece at a separate time.
- Simultaneous: Human operator and robot work on separate processes on the same work-piece at the same time.
- Supportive: Human operator and robot work synchronously to complete a common process on the same workpiece.

In this case, the higher the dependence between humans and Cobot to perform the activity, the higher the complexity of the development. This derives from the fact that it is necessary to properly address the communication between humans and robots, which is challenging in industrial environments.

The second element is related to the end effector complexity, which encompasses three main aspects of the application. The first is related to the characteristics of the manipulated object, once it affects directly the design of the effector, for example, considering a fragile or flexible component to be manipulated. The second is related to the operation performed by the Cobot, once it also implies different functionalities required that influence the complexity. The last is related to the number of different operations required for the Cobot to perform and in this case, the higher the number of different functions required to the effector, higher the complexity.

Space restriction for the Cobot operation is the third element used to evaluate the complexity of the development. This aspect was defined to guarantee that the different characteristics related to the environment of each potential application are properly considered in the evaluation. In this case, higher the restriction for the Cobot operation, the higher the complexity of the solution, once it increases the complexity of programming aspects, and may require special effectors to allow its operation, besides the fact that the interaction with humans also is hampered.

Finally, the required precision for the effector is the last aspect defined for this evaluation. This aspect contemplates both the precision of Cobot positioning and the precision of task execution, as process parameters and their tolerances, for example. In this case, the higher the precision required for both, the higher the complexity of the solution, once it is necessary higher Cobot stiffness, sensing, and control technologies.

It is important to highlight that, for this last parameter, the precision aspect could be substituted for the concept of dexterity. Dexterity is a common metric used to compare applications and solutions in the robotic's world. Among the elements that may compose this concept, the most common are related to object size, object shape, gripping strategy, reachability, and speed. The first three elements are most related to the effector characteristics, while the last two parameters are mainly related to the Cobot arm. However, all cited elements are connected and interrelated, composing together this definition. As a result, the dexterity could be an interesting parameter to be evaluated, and further be included in this evaluation methodology.

Once the main elements necessary to evaluate the potential applications were selected, the process of evaluating the applications against the elements was defined. For this analysis, it was considered the application of an adapted model of the AHP (Analytic Hierarchy Process) methodology presented in section 2.6.5, in which Figure 3.6 shows the

hierarchical structure of the problem for this application.

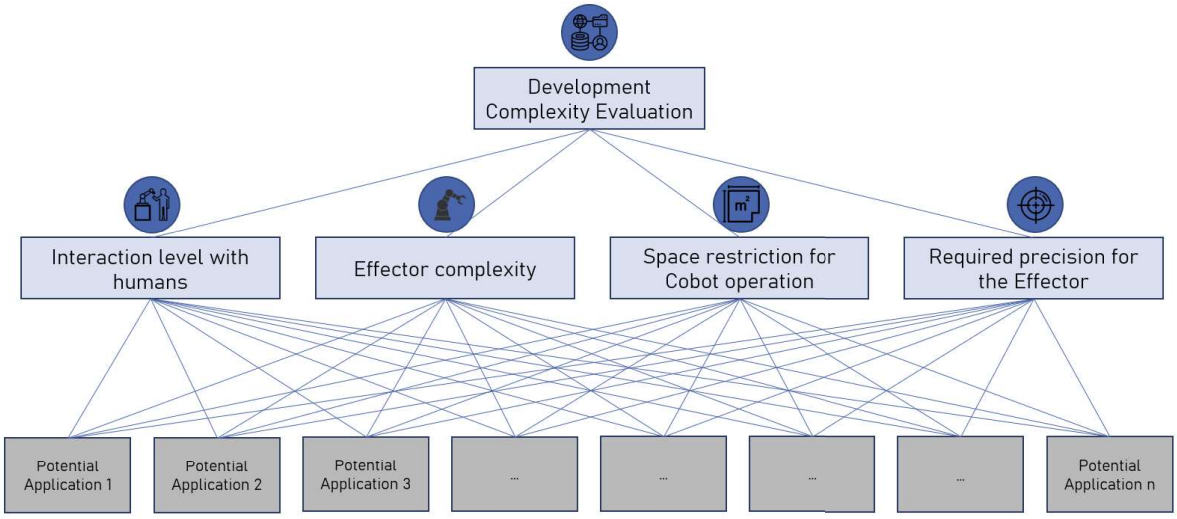


FIGURE 3.6 – Hierarchical structure of the evaluation.

Considering the hierarchical structure defined, the following steps must be executed to obtain the final score for each application:

1. Define the level of importance of each of the criteria, by pairwise comparison;
2. Applications are then pairwise compared for each of the pillars that compose the complexity of development evaluation;
3. Compose the final score for each application, considering their score for each pillar multiplied by the importance of each criterion.

Considering this sequence, the first stage is to define the level of importance of each of the criteria, in this case, the four pillars of the development complexity evaluation. This evaluation is performed using the AHP comparative matrix, where a pairwise comparison for each criterion is performed, considering the Saaty scale presented in section 2.7.6. Table 3.5 shows the comparison matrix for the criteria, in which the cells in blue are the value and the cells in white are the reciprocal values.

TABLE 3.5 – Comparative matrix of criteria.

	Interaction level with humans	Effector complexity	Space restriction for Cobot operation	Required precision for the effector
Interaction level with humans	M (1,1)	M (1,2)	M (1,3)	M (1,4)
Effector complexity	M (2,1)	M (2,2)	M (2,3)	M (2,4)
Space restriction for Cobot operation	M (3,1)	M (3,2)	M (3,3)	M (3,4)
Required precision for the effector	M (4,1)	M (4,2)	M (4,3)	M (4,4)

As the next step, the comparison matrix must be normalized to allow the definition of the local priority of the elements. For this normalization, each element of the matrix is divided by the sum of the column, resulting in the MN (i,j) elements, as indicated by Equation 3.4. Table 3.6 shows the representation of the normalized criteria matrix.

$$MN_{(i,j)} = \frac{MN_{(i,j)}}{\sum_{k=1}^{n} MN_{(n,j)}} \tag{3.4}$$

TABLE 3.6 – Normalized comparative matrix of criteria.

	Interaction level with humans	Effector complexity	Space restriction for Cobot operation	Required precision for the effector	LOCAL PRIORITY
Interaction level with humans	MN (1,1)	MN (1,2)	MN (1,3)	MN (1,4)	LP 1
Effector complexity	MN (2,1)	MN (2,2)	MN (2,3)	MN (2,4)	LP 2
Space restriction for Cobot operation	MN (3,1)	MN (3,2)	MN (3,3)	MN (3,4)	LP 3
Required precision for the effector	MN (4,1)	MN (4,2)	MN (4,3)	MN (4,4)	LP 4

As a result of the normalized matrix, it is possible to obtain the Local priority of each criterion, represented by the ”LP” elements. The local priority is obtained through the mean of elements in each column, a procedure exemplified for the element ”LP1” through the Equation .

$$LP1 = \frac{\sum_{k=1}^{m} MN_{(1,m)}}{m} \tag{3.5}$$

Where ”m” is the number of the columns, in this case, four.

The obtained results for the criteria will be used to compose the final score of the potential applications according to the AHP methodology. Additionally, the consistency of

the results obtained for the comparison between the criteria must be performed, according to the AHP methodology presented in Chapter 2.

The next step to be performed is the pairwise comparison between the potential applications. In this case, for each criterion defined, the potential applications are compared to each other, which in this case will result in four normalized matrices (once four criteria were defined in the methodology). Table 3.7 presents an example of a comparison matrix for one of the criteria defined in the methodology (Interaction level with humans, effector complexity, space restriction for Cobot operation, and required precision for the effector). Each element of the matrix is obtained by pairwise comparison between the potential applications, considering the specific criteria in question.

TABLE 3.7 – Comparative matrix of potential applications - for one of the criteria.

	Potential application 1	Potential application 2	Potential application 3	...	...	...	...	Potential application n
Potential application 1	A (1,1)	A (1,2)	A (1,3)	...	...	...	...	A (1,n)
Potential application 2	A (2,1)	A (2,2)	A (2,3)	...	...	...	...	A (2,n)
Potential application 3	A (3,1)	A (3,2)	A (3,3)	...	...	...	...	A (3,n)
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
Potential application n	A (n,1)	A (n,2)	A (n,3)	...	...	...	...	A (n,n)

Similarly to the process for the criteria matrix, the matrices for the application comparison must be normalized. The normalization follows the same methodology presented before, resulting in a matrix normalized and the local priority.

TABLE 3.8 – Normalized comparative matrix of potential applications - for one of the criteria.

	Potential application 1	Potential application 2	Potential application 3	...	...	...	...	Potential application n	LOCAL PRIORITY (Criteria X)
Potential application 1	AN (1,1)	AN (1,2)	AN (1,3)	...	...	...	...	AN (1,n)	LPPA (1) - X
Potential application 2	AN (2,1)	AN (2,2)	AN (2,3)	...	...	...	...	AN (2,n)	LPPA (2) - X
Potential application 3	AN (3,1)	AN (3,2)	AN (3,3)	...	...	...	...	AN (3,n)	LPPA (3) - X
...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...
Potential application n	AN (n,1)	AN (n,2)	AN (n,3)	...	...	...	...	AN (n,n)	LPPA (n) - X

It is important to highlight that the consistency must also be verified for these matrices, according to the methodology presented in section 2.6.5.

In this case, the Local Priority elements (LPPA) obtained are related to one of the criteria defined, represented in Table 3.8 by the ”X” element. For each criterion defined,

(Interaction level with humans, effector complexity, space restriction for Cobot operation, and required precision for the effector), a normalized matrix will be defined and as a consequence, there will be four local priority sets of values.

The results obtained by the comparison between the four pillars, and the results obtained by the pairwise comparison between the potential applications must be combined to obtain the final evaluation of the applications. Table 3.9 illustrates how the final score can be obtained by the combination of the matrices defined.

TABLE 3.9 – Global Evaluation Matrix of applications regarding the development complexity

	Interaction level with humans	Effector complexity	Space restriction for Cobot operation	Required precision for the effector	GLOBAL RESULT
	LP 1	LP 2	LP 3	LP 4	
Potential application 1	LPPA (1) - X	LPPA (1) - Y	LPPA (1) - Z	LPPA (1) - W	**GR 1**
Potential application 2	LPPA (2) - X	LPPA (2) - Y	LPPA (2) - Z	LPPA (2) - W	**GR2**
Potential application 3	LPPA (3) - X	LPPA (3) - Y	LPPA (3) - Z	LPPA (3) - W	**GR3**
...	...	...	...	...	**...**
...	...	...	...	...	**...**
...	...	...	...	...	**...**
...	...	...	...	...	**...**
Potential application n	LPPA (n) - X	LPPA (n) - Y	LPPA (n) - Z	LPPA (n) - W	**GR n**

As can be verified, the local priority obtained for the criteria (LP1 to LP4) is presented in the line above the matrix. Each column of the matrices brings the local priorities for the potential applications, obtained by the comparison matrices for each parameter, in this case, each parameter is represented by the letters ”X”, ”Y”, ”Z”, and ”W”.

The global result is the final score for each application regarding the development complexity and is obtained by the direct multiplication of each element of the matrix to the elements of the local priority of the criteria. Equation 3.6 represents this procedure for the global result GR 1.

$$\begin{aligned} GR1 &= (LPPA(1) - X) * LP1 + (LPPA(1) - Y) * LP2 \\ &+ (LPPA(1) - Z) * LP3 + (LPPA(1) - W) * LP4) \end{aligned} \tag{3.6}$$

Finally, the result obtained can be composed with the evaluation of the potential benefits, to complete the potential applications evaluation methodology, presented in the next section.

One important aspect of the methodology defined is that in the case that are mul-

tiple comparison matrices to be defined, the developed methodology can be very time-consuming. For this reason, an alternative process indicated is to use questionnaires to collect the specialist's opinion, and by the analysis of the collected data, perform the application of the methodology and obtain the comparison matrices. However, it is important to guarantee that the specialists are at the same level of understanding of the criteria and the applications that are being evaluated.

3.3.3 Application Selection Methodology

The last stage of the methodology for the selection of the collaborative application is the application selection. At this stage, all the processes were mapped, and the potential applications were identified and evaluated considering both potential benefits and development complexity. In this context, this stage of the process accounts for creating the decision-making matrix, where each potential application is positioned against the two axes regarding the potential benefits and the development complexity. Figure 3.7 shows an example of the matrix obtained.

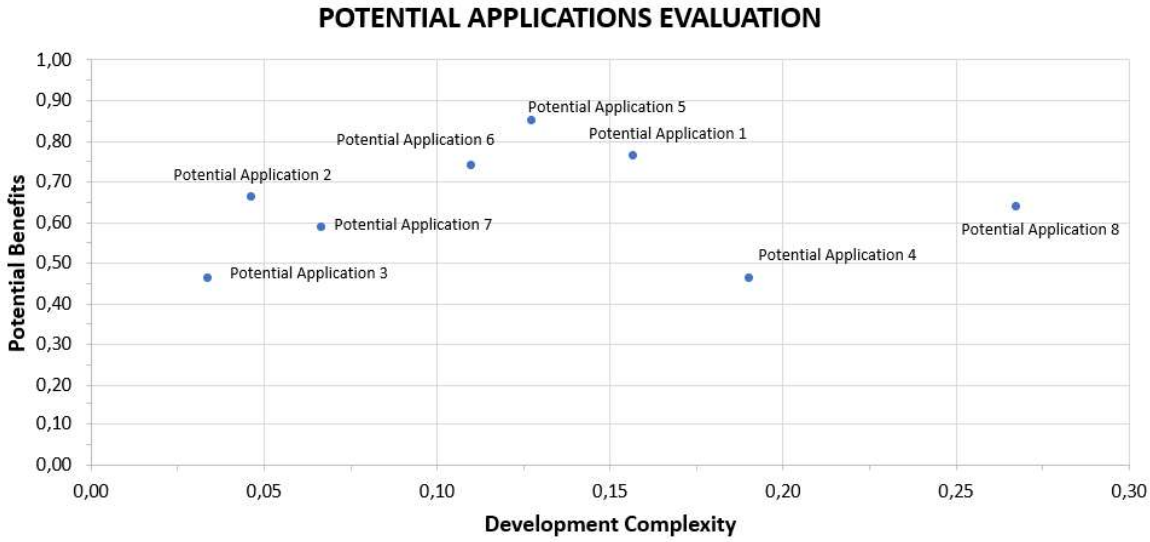


FIGURE 3.7 – Potential Applications Evaluation Matrix.

Several analyzes can be performed over the matrix obtained, to define the application to be prioritized for the development. Considering the median value for each axis, four quadrants can be visualized, represented by the numbers 1 to 4 in Figure 3.7, in which applications mapped are distributed.

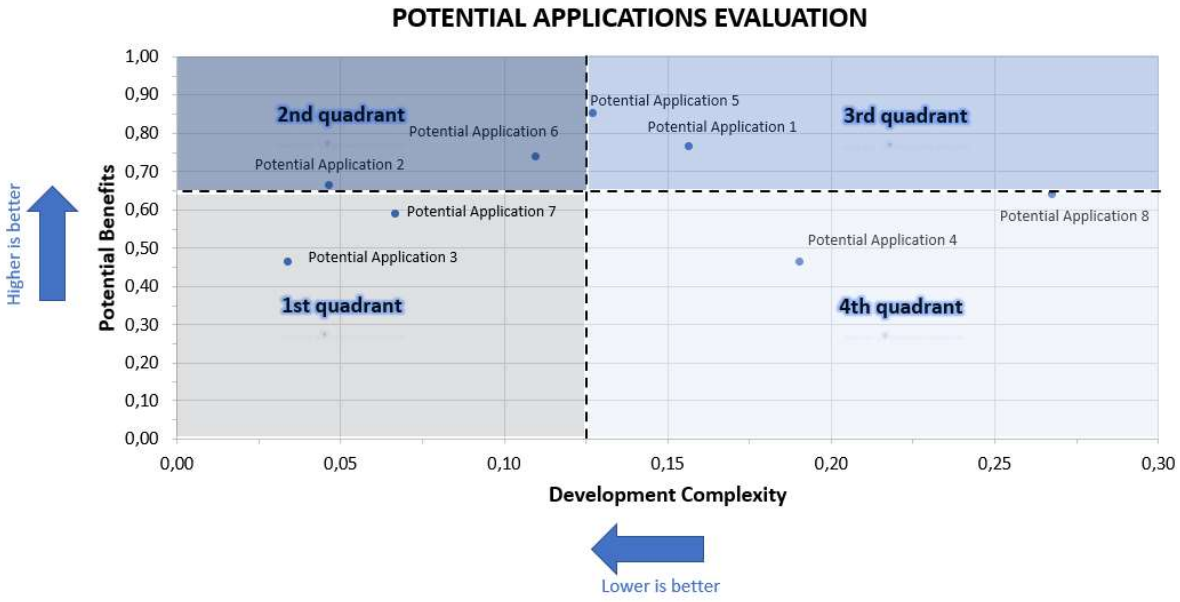


FIGURE 3.8 – Potential Applications Evaluation Matrix - quadrants.

Applications in the fourth quadrant are those with a very low possibility of implementation once present the highest development complexity with a low potential benefit expected. These applications represent the ones that are possible to be benefited by the application of Cobots, however, are not the best indicated.

Similarly, the 1st quadrant contemplates applications that present a low potential benefit, however, presents a low development complexity. In this case, these applications must be considered once may require shorter development times and bring short-term results in the assembly line.

However, if the objective of the company is to seek significant improvements, the applications contained in the second and third quadrants are the ones to be prioritized. The applications presented in the third quadrant present a high potential benefit score, however, are complex to be developed. In this case, these applications probably will require a longer development time and studies to be able to be deployed to the production line.

Applications in the second quadrant are those that present the best balance between potential benefits and development complexity. These applications present a lower development complexity compared to the ones in the 3 and fourth quadrants while trends bring high benefits to the assembly line. For these reasons, applications in this quadrant tend to be the ones selected for development.

Finally, it is important to mention that the developed methodology allows the mapping, identification, and selection of the collaborative application based on a systematic procedure with the most relevant parameters being considered. As a result, the method-

ology allows a robust identification and selection of applications, with a relatively fast execution time. Additional evaluations, tests, and measurements may also be used as a complement to the methodology in further steps if required.

4 Final Assembly Application Selection

In this chapter, the methodology developed in Chapter 3 is applied to the context of the aircraft final assembly, to allow the selection of the collaborative application to be further studied in this work. In the next sections, the results of the four steps of the methodology applied are presented, finalizing the chapter with the selected collaborative application.

4.1 Aircraft's Final Assembly Process Mapping

The first step for selecting the application studied in this work was mapping a real aircraft's final assembly process. The mapping process started with research on academic publications to understand the general assembly process, which was presented in section 2.4. With the general assembly process identified, interviews with specialists were performed to collect deeper information regarding the real aircraft manufacturing process of a single-aisle commercial jet, focusing on the activities performed in the fuselage.

The data collected allowed the identification of the tasks performed in 36 macro activities executed in the 2 macro stages, of the final assembly process: Systems Equipping and Final Equipping. Figure 2.10 presents a workflow of the mapping step performed, however for confidentiality purposes, the information regarding the activities performed in each position and stage is not presented.

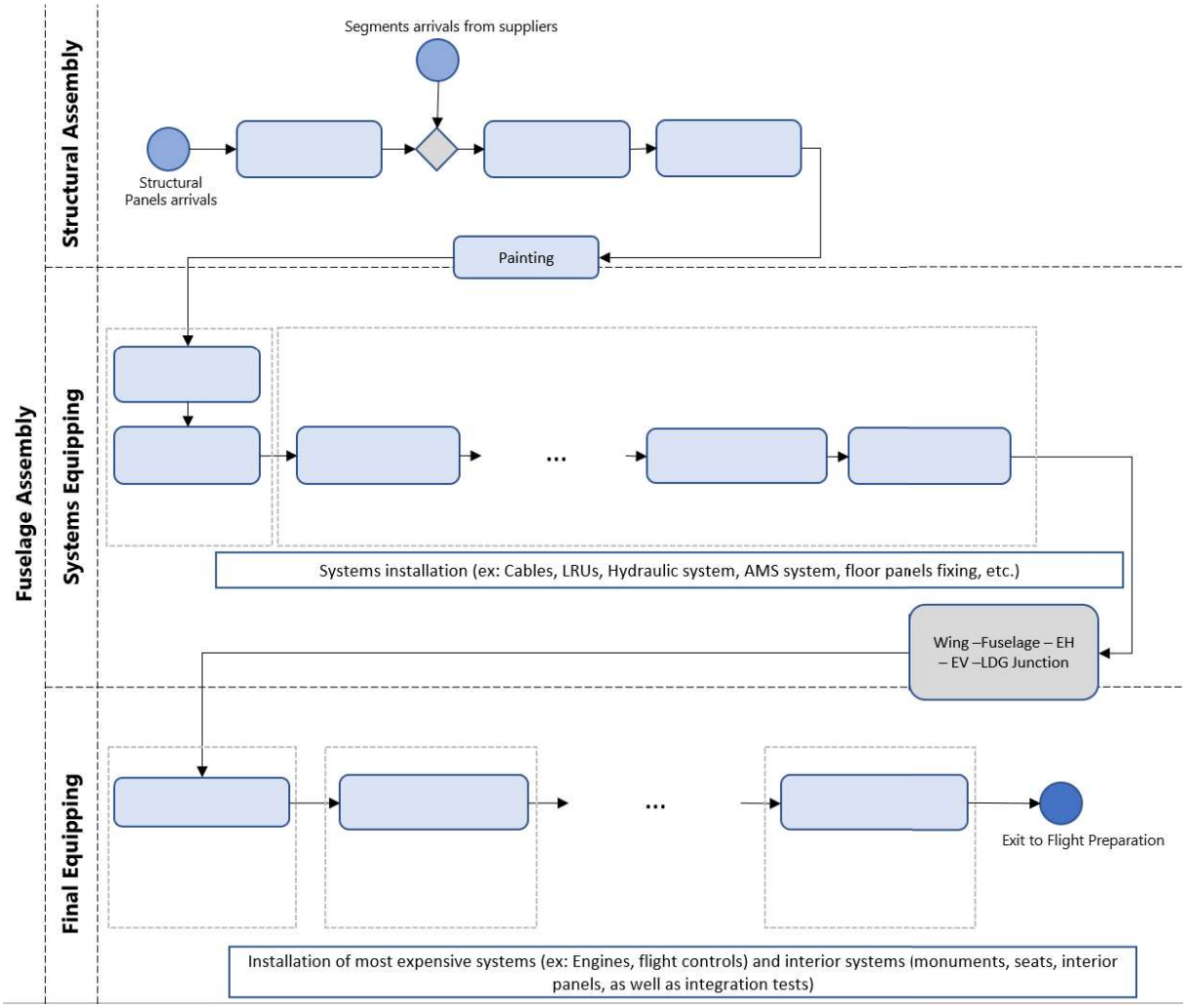


FIGURE 4.1 – Aircraft's final assembly process mapped workflow. The figure is blurred considering confidentiality aspects.

As indicated, in the Systems equipping, the main systems are installed in the aircraft, such as AMS, hydraulic systems, cables, LRUs, and fuel systems, for example. Also, in this stage occurs the installation of some components inside the aircraft cabin such as the insulation blankets and the floor panels.

The final equipping is the phase where the most expensive components are incorporated into the aircraft, such as the engines. Moreover, in this stage, the main components of the interior of the aircraft are installed, including seats, ceiling panels, side panels, overhead bin, and other elements that compose the finishing of the cabin interior.

Finally, as a result of the mapping process, it was possible to verify that the main potential applications would be related to systems equipping and the final equipping stages of the final assembly process. The first is the stage where most of the systems are installed, while the final equipping is the one where the most expensive components are installed such as engines, as well as, the interior components. The next section presents the potential applications identified, detailing the main motivation for the application of

Cobots for each identified opportunity.

4.2 Collaborative Potential Applications

Considering the process mapping, it was possible to identify the applications with the greatest potential for the application of Collaborative Robots. These applications were identified based on the information collected by the interviews with specialists in the main technologies involved in the final assembly process, including sealing, interiors, hydraulics, AMS, and mechanical systems.

The data collection with specialists was performed through the methodology defined in Chapter 3 which allowed the identification of the most promising applications for the deployment of Collaborative Robots. These applications identified are directly related to activities that are currently performed by operators and could be automated through the application of Cobots, aiming for the improvement of ergonomics conditions as well as with the potential for increasing productivity and operator's well-being with the reduction of repetitive tasks. Figure 4.2 presents the eight identified potential applications for Collaborative Robots in the aircraft's final assembly process.

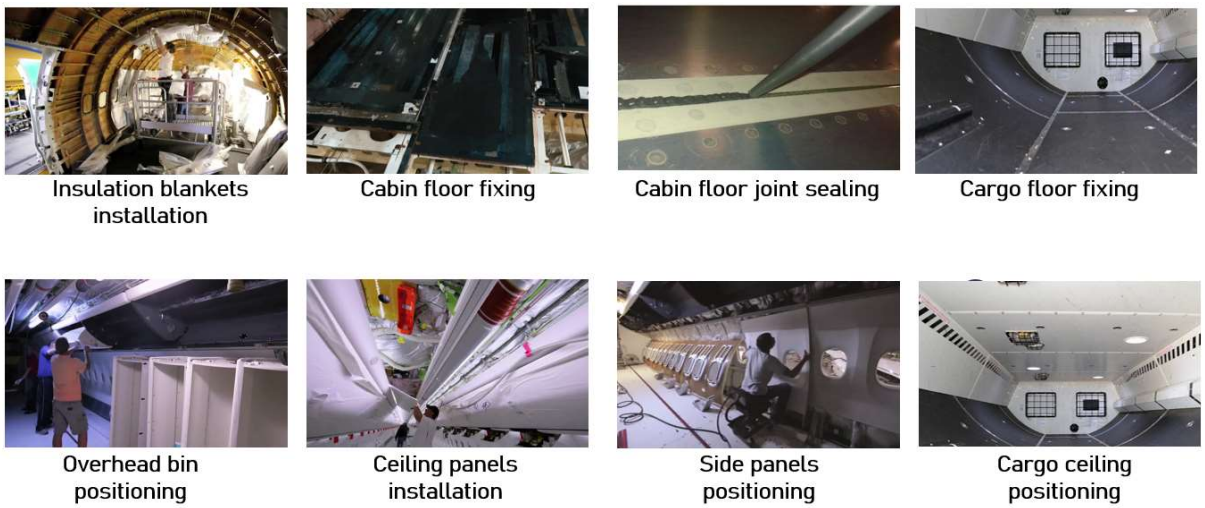


FIGURE 4.2 – Cobots potential applications identified.

Each identified application is further detailed in the next sections.

4.2.1 Insulation Blankets Installation

The first potential application is related to the installation of insulation blankets inside the fuselage passenger cabin. This activity is one of the first operations performed inside the fuselage during the systems equipping stage of the final assembly, currently, installed

manually by human workers. As presented in Figure 4.3(a) the blankets are positioned in specific positions according to an identification code and fixed against the fuselage structure through the insertion of a metal clip. As the blankets are very light (compared to current Cobots payload), there are no ergonomic problems related to carrying heavy objects, however, is a highly repetitive task, which is performed most of the time in awkward postures, as can be seen in the Figure 4.3(b), where the operator is performing the activity with the upper members extended and neck twisted up.

(a) Insulation blanket detail and clips, (AVIAçãO, 2020).

(b) Insulation blankets current installation process, (RYANAIR, 2014).

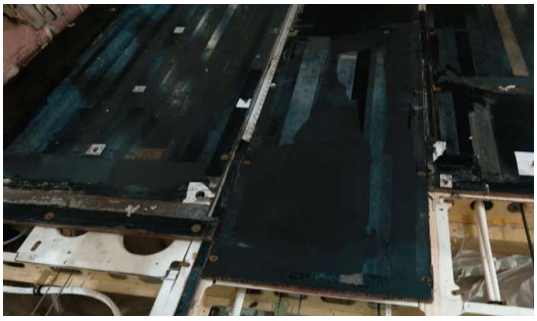
FIGURE 4.3 – Insulation blankets application.

These characteristics turn the application of Collaborative Robots into an interesting solution to reduce the operator's workload. The considered concept is based on the application of a Cobot to pick up and position the blanket as well as insert the clips to fix them in the position, releasing the operator to perform another activity in the same workplace. This scenario was defined considering that the installation of the insulation blankets is performed in seconds, which makes it difficult for the operation to be performed in a sequential or supportive way between human and robot. Furthermore, in this case, is important for the Cobot to perform all the assembly steps since allows the elimination of non-ergonomic conditions for the human worker which in this case is related to awkward postures. The main challenges of the application are based on the flexible characteristics of the blankets, as well as the multiple functions required to be performed by the Cobot.

4.2.2 Cabin Floor Fixing

Another identified application is the floor fixing activity in the passenger cabin. The aircraft floor is composed of composite panels (generally) that are positioned over the floor structures and fixed through the application of fasteners. This activity is currently performed by human operators that must carry on screwdrivers and work crunching in non-ergonomic postures.

In this case, the scenario identified is the application of a Collaborative Robot to perform the identification of the floor holes position, insertion, and torque application on fasteners. As a result, the non-ergonomic activity is performed by the Cobot, freeing the human operator to perform other activities. Figure 4.4(a) shows the floorboards positioned, and the fasteners fixing it over the floor structures.



(a) Cabin floor panels and floor structures, also showing the fasteners, (AVIAçãO, 2020).

(b) View of the cargo compartment, floor and ceiling panels installed, (RYANAIR, 2014).

FIGURE 4.4 – Cargo floor fixing and cargo compartment detail.

4.2.3 Cargo Floor Fixing

The floor fixing activity is similarly performed in the cargo compartment, shown in Figure 4.4(b). For both installations, in the cabin and the cargo, the main challenges related to the deployment of a Cobot are the number of functions required to be performed by the Cobot to perform the activity, including the need of being mobile, have a vision system to identify the position of the holes and control the torque applied. Moreover, the application in the cargo compartment brings an additional challenge related to the space restriction, since the maximum height is about 1 meter.

The floor fixing activity is similarly performed in the cargo compartment, shown in Figure 4.4(b). For both installations, in the cabin and the cargo, the main challenges related to the deployment of a Cobot are the number of functions required to be performed by the Cobot to perform the activity, including the need of being mobile, have a vision system to identify the position of the holes and control the torque applied. Moreover, the application in the cargo compartment brings an additional challenge related to the space restriction, due to the limited height.

4.2.4 Cabin Floor Joint Sealing

After the floor fixing, the possible gap between floor panels is closed by the application of joint sealing, which is another potential application identified. Currently, this activity is

also performed manually and the human operator is exposed to similar working conditions as the floor fixing activity, working almost all the time crunching in a non-ergonomic posture. In this case, the functions required for the Cobot are to identify the position of sealant application and apply the sealant controlling the volume. Figure 4.5 shows an example of a joint sealing application.

FIGURE 4.5 – Joint sealing application, (BOMBARDIER, 2019).

4.2.5 Cargo Ceiling Positioning

The cargo ceiling positioning is another potential application identified, to be performed in the cargo compartment. This activity consists of positioning the ceiling panels of the cargo compartment and inserting fasteners to fix them in the structure. Currently, this activity requires two human workers to be performed: while one operator positions the ceiling panel, another inserts the fasteners. In this case, both operators have to work in a restricted area and in awkward positions, configuring a very non-ergonomic activity, justifying the application of a Collaborative robot.

For this application, the scenario identified considers a supportive operation between the human and the robot, where the Cobot picks up, positions, and holds the panels, while the human worker fixes the panels by inserting the fasteners. This application brings a very high level of complexity, once is performed in a restricted area (as the cargo floor fixing shown in Figure 4.4(b)) with an extra challenge brought by the supportive mode of operation, since the Cobot couldn't perform both activities by itself considering the separation distance between the fasteners' position.

4.2.6 Overhead Bin Positioning

All previously presented potential applications are related to activities performed during the systems equipping stage of the final assembly. Moving forward to the final equipping, three applications were identified as potential for the deployment of Cobots, related to overhead bin and side panels positioning, well as, cabin ceiling panels installation.

Considering the overhead bin installation activity, as presented in Figure 4.6, the overhead bin is a rigid object, that is positioned above the seats in the passenger cabin, where the passenger's carry-on luggage is placed. Currently, this activity is performed by two human operators, that have to lift the bin over their heads and holds it until fixing them. Besides the non-ergonomic postures of the human workers, the overhead bin elements that have to be lifted are very heavy considering a conventional Cobot payload (Kuka LBR iiwa for example), which composes a very non-ergonomic demanding activity to be performed.

In this case, the identified scenario consists of the application of a Collaborative robot to pick up the bin, position and hold it until the human operator fixes it. As a consequence, all the workload related to bin lifting and holding will be performed by the Cobot, reducing human ergonomic problems. It is important to highlight that the fixing activity was considered to be performed by a human operator since the final position of the bin is obtained through an adjustable fixing element, to guarantee its alignment.

(a) Overhead bin installation process exemplification, (RYANAIR, 2014).

(b) Overhead bin detailing installation detail,(ROVEDA *et al.*, 2017)

FIGURE 4.6 – Overhead bin installation.

4.2.7 Side Panels Installation

Another identified potential application is related to the installation of the side panels, shown in Figure 4.7. These elements form the fuselage internal wall that is in contact with the passenger and are significantly lighter than the overhead bins. However, the installation currently is performed manually, requiring the operator to work all the time

crunching in a non-ergonomic posture. In this case, the identified scenario consists of the Cobot being responsible for picking up the panel, positioning it, and holding it until the human operator fixes it through the installation of fasteners situated on the external edge of the panel.

In this application, it was considered a supportive working environment for similar restrictions as the cargo ceiling positioning, since it was considered impracticable for the Cobot to apply the fasteners while holding the panel once the distance between the center of the panel and the position of the fastener installation is high.

(a) Side panels installation, (ROVEDA *et al.*, 2017).

(b) Side panel positioning and installation, (RYANAIR, 2014).

FIGURE 4.7 – Side panels installation.

4.2.8 Ceiling Panels Installation

Finally, the last identified potential application is related to ceiling panels installation. As shown in Figure 4.8, the ceiling panels are very similar in dimensions and weight to the side panels, however, for this activity, the human workers have to pick, position, and hold the panel above their heads, configuring a non-ergonomic activity. For this reason, the application of a Collaborative robot to perform this activity is an important tool to reduce human workload.

In this application, considering the characteristics of the fixing of the ceiling panel, the Cobot can perform all the process by itself, since the panel is docked without the insertion of fasteners. As a consequence, in this scenario, the Cobot is responsible for picking up, positioning, and applying an extra load for docking the ceiling panel against the structure. This is an extra challenge brought by this application, however, allows to free up the human worker from this non-ergonomic activity.

FIGURE 4.8 – Ceiling panel installation example, (RYANAIR, 2014).

As can be seen, all applications identified have a great potential to increase the level of automation of final assembly, reducing the number of non-ergonomic activities necessary to be performed by humans. It should be noted, however, that the expected benefit varies for each application, given its characteristics and particularities, which also lead to different levels of difficulty in developing the solution. focusing on finding the best balance between the expected benefit and the difficulty in the development of the application's selection, the next section will present the results obtained through the methodology developed in Chapter 3.

4.3 Applications Evaluation and Selection

As presented in section 4.2, several opportunities are present in the current aircraft final assembly process, since is highly dependent on human workforce. For the scope of this work, and to guarantee the achievement of the defined objectives, one of the potential applications was chosen to be further studied.

The definition of the application to be developed in this work was based on evaluating the potential application presented in the last section regarding its expected improvements in the current process and its development complexity. This methodology was presented in Chapter 3, section 3.3, and is illustrated by Figure 4.9, also presented in Chapter 3.

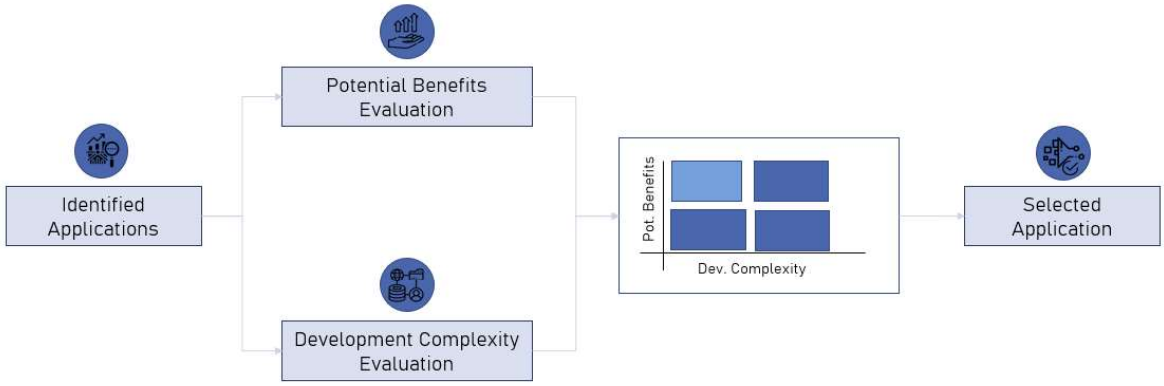


FIGURE 4.9 – Application selection workflow.

The next section presents the results obtained for each pillar of the evaluation performed over the potential applications identified and presented in section 4.2.

4.3.1 Potential Benefits Evaluation

Based on the methodology presented in section 3.3.1 and the information collected during the process mapping presented in sections 4.1 and 4.2, the identified potential applications were evaluated. The information collected during the process mapping was used as a base for the classification of the potential applications. Table 4.1 presents a partition of the information collected and the correspondent score for the Overhead bin positioning application regarding the heavy objects handling element of the ergonomics pillar, as an example of the procedure executed. The complete set of the collected information is not presented in this work considering the confidentiality of the information.

TABLE 4.1 – Example of Potential Benefit evaluation, from the collected data from specialists.

			PROCESS/ APPLICATION	ASPEC EVALUATED
CLASS	QUESTIONS	QUESTION'S MOTIVATION	OVERHEADBIN POSITIONING	Heavy object Manipulation
ERGONOMICS AND SAFETY	Is there heavy objects moving during any activity in this stage? If yes, what weight?	Understand if there are ergonomic problems in activities.	Yes, around 9 Kg.	Weight Score: 3
	What is the frequency of this movement?	Understand if weight movement is constant throughout the work period.	Practically all the time involves this movement.	Frequency Score: 5
				Total Score: 15

The presented process was performed for each parameter of each potential application, for the potential benefits' evaluation. Table 4.2 presents the evaluation performed for

each potential application, regarding the ergonomics aspects and the combined score in the column highlighted in blue, which is a weighted sum of the scores.

TABLE 4.2 – Evaluation of potential applications regarding ergonomics aspects

	Heavy Object handling (Wheight x Frequency)	Repetitive Task (Frequency)	Awkward Working Position (Posture x Exposit. Time)	TOTAL Ergonomics Score
Insulation blankets installation	5	5	16	0,68
Cabin floor fixing	1	5	20	0,68
Cabin floor joint sealing	1	3	20	0,55
Cargo floor fixing	1	3	20	0,55
Overhead bin positioning	20	3	20	0,87
Ceiling panels installation	15	3	20	0,78
Side panels positioning	15	3	8	0,58
Cargo ceiling positioning	15	3	20	0,78

The same procedure was performed for the productivity element of the potential benefit score, which is presented in Table 4.3.

TABLE 4.3 – Evaluation of potential applications regarding productivity aspect

	Productivity (Man-hour scale)	TOTAL Productivity Score
Insulation blankets installation	5	1,00
Cabin floor fixing	3	0,60
Cabin floor joint sealing	1	0,20
Cargo floor fixing	1	0,20
Overhead bin positioning	4	0,80
Ceiling panels installation	3	0,60
Side panels positioning	3	0,60
Cargo ceiling positioning	1	0,20

Finally, the aggregated score is presented in Table 4.4, obtained by combining the results discussed above. As can be seen, the overhead bin positioning was scored with the highest potential for improvement in the current process, mainly because of the bin weight and working positions, followed by the insulation blankets installation (considering the high level of repetitiveness) and the ceiling installation, most impacted by the weight and the working postures.

TABLE 4.4 – Evaluation of potential applications regarding productivity aspect

	TOTAL Ergonomics Score	TOTAL Productivity Score	TOTAL Potential Benefits Evaluation
Insulation blankets installation	0,68	1,00	**0,7625**
Cabin floor fixing	0,68	0,60	**0,6625**
Cabin floor joint sealing	0,55	0,20	**0,4625**
Cargo floor fixing	0,55	0,20	**0,4625**
Overhead bin positioning	0,87	0,80	**0,8500**
Ceiling panels installation	0,78	0,60	**0,7375**
Side panels positioning	0,58	0,60	**0,5875**
Cargo ceiling positioning	0,78	0,20	**0,6375**

It is important to mention that, according to the methodology defined in Chapter 3, the ergonomic aspect was considered with a weight three times higher than the productivity.

4.3.2 Development Complexity Evaluation

Considering the methodology defined and presented in section 3.3.3 an evaluation was performed aiming to classify the identified potential applications with respect to the complexity of their development. As indicated, the methodology applied the AHP (Analytic Hierarchy Process) method, in which the first step, the hierarchical structure of the problem, is presented in Figure 4.10.

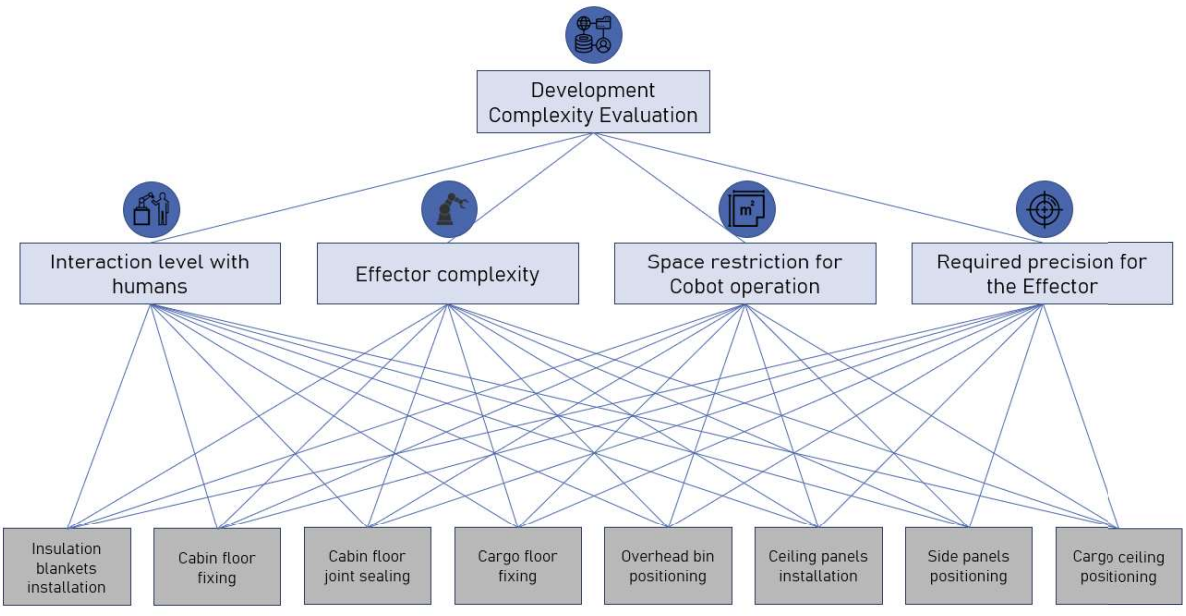


FIGURE 4.10 – Hierarchical structure of the evaluation.

Based on the hierarchical structure defined, the next step was to define the level of importance of each of the criteria, which represents the four pillars of the development complexity evaluation. Therefore, the AHP comparative matrix was obtained as a result of the discussion with specialists, presented in Table 4.5, and normalized according to the method, resulting in the information presented in Table 4.6.

TABLE 4.5 – Comparative matrix of criteria.

	Interaction level with humans	Effector complexity	Space restriction for Cobot operation	Required precision for the effector
Interaction level with humans	1	1/3	1/3	2
Effector complexity	3	1	1/3	3
Space restriction for Cobot operation	3	3	1	3
Required precision for the effector	1/2	1/3	1/3	1

TABLE 4.6 – Normalized comparative criteria matrix.

	Interaction level with humans	Effector complexity	Space restriction for Cobot operation	Required precision for the effector	LOCAL PRIORITY
Interaction level with humans	0,133	0,071	0,167	0,222	**0,148**
Effector complexity	0,400	0,214	0,167	0,333	**0,279**
Space restriction for Cobot operation	0,400	0,643	0,500	0,333	**0,469**
Required precision for the effector	0,067	0,071	0,167	0,111	**0,104**

As can be verified, the obtained result indicated that the most significant pillars for the complexity of the development are related to the space restriction for the Cobot operation and effector complexity, followed by the interaction level with humans, and with the lower weight, the required precision for the effector. Furthermore, the consistency index was calculated according to the method presented in section 2.6.5, which was achieved once the result obtained is inferior to 10%, as presented in Table 4.6.

TABLE 4.7 – Criteria matrix consistency verification.

N (number of criteria)	4
Main Eigen (Max lambda)	4,287
CI	0,087
R	0,9
CR	**9,70%**

Following the method, the identified potential applications are then pairwise compared for each of the pillars that compose the complexity of development evaluation. Considering that there are eight applications identified and four pillars, it would be required to develop four comparative matrices of 8x8 dimensions, leading to 112 pairwise comparisons necessary to be performed. To speed up the comparison process, the specialist's opinions were collected through a questionnaire, presented in Appendix A.1, which was used as a basis for the construction of the comparison matrices.

After a contextualization including the presentation of the potential application characteristics, the questionnaire required that specialists ordinate the potential applications on a scale of complexity of development, for each of the four pillars, where 1 represents the application with the highest complexity and 8 the lowest. For the evaluation, it was possible to set equal values for the potential applications (eg: two applications to be considered equally complex regarding a specific pillar, and both being classified in position 2). Table 4.8 shows the results collected for the first pillar evaluation, as an example, with the remaining data presented in Appendix A.1.

TABLE 4.8 – Results obtained from the questionnaire, for the first criteria - Interaction Level With Humans.

	Answer 1	Answer 2	Answer 3	Answer 4	Answer 5	Answer 6	Answer 7	Answer 8	MEAN
Insulation blankets installation	4	5	2	1	5	7	5	2	3,9
Cabin floor fixing	7	7	5	7	8	5	5	3	5,9
Cabin floor joint sealing	6	7	7	8	8	8	5	6	6,9
Cargo floor fixing	7	7	1	7	8	6	5	5	5,8
Overhead bin positioning	2	1	5	1	1	1	5	5	2,6
Ceiling pannels installation	5	3	6	5	8	4	5	8	5,5
Side pannels positioning	3	5	7	3	8	2	5	6	4,9
Cargo ceiling positioning	1	2	1	4	8	3	4	8	3,9

Taking into account the collected data, the mean value of each potential application was used to perform the pairwise comparison, resulting in the matrix presented in Table 4.9, for the first pillar. Moreover, to perform the comparison, the difference between the mean values obtained for each application was used to define the level of importance between the applications, translated by the Saaty scale, presented in Table 2.4, of section 2.6.5.

TABLE 4.9 – Comparative matrix of potential applications - Interaction Level With Humans criteria.

	Insulation blankets installation	Cabin floor fixing	Cabin floor joint sealing	Cargo floor fixing	Overhead bin positioning	Ceiling panels installation	Side panels positioning	Cargo ceiling positioning
Insulation blankets installation	1	3	4	5	1/6	2	1/2	1/4
Cabin floor fixing	1/3	1	2	1/2	1/7	1/3	1/4	1/6
Cabin floor joint sealing	1/4	1/2	1	1/3	1/8	1/4	1/5	1/7
Cargo floor fixing	1/5	2	3	1	1/6	1/2	1/4	1/5
Overhead bin positioning	6	7	8	6	1	4	3	2
Ceiling panels installation	1/2	3	4	2	1/4	1	1/3	1/4
Side panels positioning	2	5	5	4	1/3	3	1	1/2
Cargo ceiling positioning	4	6	7	5	1/2	4	2	1

Following the procedure indicated by the method, the results were normalized, as presented in Table 4.10. As expected, the applications with higher interaction required between human and Cobot to complete the task, received higher complexity scores for this evaluation element, such as overhead bin positioning, cargo ceiling positioning, and side panels positioning. Furthermore, the results were evaluated to verify its consistency, which was achieved through a consistency ratio of 5.31%, inferior to the limit indicated of 10%.

TABLE 4.10 – Normalized Comparative matrix of potential applications - Interaction Level With Humans criteria.

	Insulation blankets installation	Cabin floor fixing	Cabin floor joint sealing	Cargo floor fixing	Overhead bin positioning	Ceiling panels installation	Side panels positioning	Cargo ceiling positioning	LOCAL PRIORITY
Insulation blankets installation	0,070	0,109	0,118	0,210	0,062	0,133	0,066	0,055	**0,103**
Cabin floor fixing	0,023	0,036	0,059	0,021	0,053	0,022	0,033	0,037	**0,036**
Cabin floor joint sealing	0,018	0,018	0,029	0,014	0,047	0,017	0,027	0,032	**0,025**
Cargo floor fixing	0,014	0,073	0,088	0,042	0,062	0,033	0,033	0,044	**0,049**
Overhead bin positioning	0,420	0,255	0,235	0,252	0,373	0,265	0,398	0,444	**0,330**
Ceiling panels installation	0,035	0,109	0,118	0,084	0,093	0,066	0,044	0,055	**0,076**
Side panels positioning	0,140	0,182	0,147	0,168	0,124	0,199	0,133	0,111	**0,150**
Cargo ceiling positioning	0,280	0,218	0,206	0,210	0,186	0,265	0,265	0,222	**0,232**

The same procedure was performed for the pillar “Effector complexity”, presented in Tables 4.11 and 4.12, also consistent, with a consistency ratio of 3.08%. In this case, the insulation blankets installation received the highest complexity score, mainly influenced by the number of functions required to be performed and the characteristics of the blankets (flexible). Ceiling panels installation and cargo ceiling positioning also received high complexity scores, once the first requires controlled force application to install the panels, and the second is in a restricted space, limiting the dimensions of the effector to be developed. Finally, the overhead bin installation application also received a considerable complexity score, mainly motivated by the dimensions and weight of the bins.

TABLE 4.11 – Comparative matrix of potential applications - Effector Complexity.

	Insulation blankets installation	Cabin floor fixing	Cabin floor joint sealing	Cargo floor fixing	Overhead bin positioning	Ceiling panels installation	Side panels positioning	Cargo ceiling positioning
Insulation blankets installation	1	7	8	5	4	2	6	3
Cabin floor fixing	1/7	1	1	1/2	1/3	1/5	1/2	1/4
Cabin floor joint sealing	1/8	1	1	1/3	1/4	1/5	1/2	1/4
Cargo floor fixing	1/5	2	3	1	1/2	1/4	2	1/3
Overhead bin positioning	1/4	3	4	2	1	1/3	3	1/2
Ceiling panels installation	1/2	5	5	4	3	1	4	2
Side panels positioning	1/6	2	2	1/2	1/3	1/4	1	1/4
Cargo ceiling positioning	1/3	4	4	3	2	1/2	4	1

TABLE 4.12 – Normalized Comparative matrix of potential applications - Effector Complexity.

	Insulation blankets installation	Cabin floor fixing	Cabin floor joint sealing	Cargo floor fixing	Overhead bin positioning	Ceiling panels installation	Side panels positioning	Cargo ceiling positioning	LOCAL PRIORITY
Insulation blankets installation	0,368	0,280	0,286	0,306	0,350	0,423	0,286	0,396	**0,337**
Cabin floor fixing	0,053	0,040	0,036	0,031	0,029	0,042	0,024	0,033	**0,036**
Cabin floor joint sealing	0,046	0,040	0,036	0,020	0,022	0,042	0,024	0,033	**0,033**
Cargo floor fixing	0,074	0,080	0,107	0,061	0,044	0,053	0,095	0,044	**0,070**
Overhead bin positioning	0,092	0,120	0,143	0,122	0,088	0,070	0,143	0,066	**0,106**
Ceiling panels installation	0,184	0,200	0,179	0,245	0,263	0,211	0,190	0,264	**0,217**
Side panels positioning	0,061	0,080	0,071	0,031	0,029	0,053	0,048	0,033	**0,051**
Cargo ceiling positioning	0,123	0,160	0,143	0,184	0,175	0,106	0,190	0,132	**0,152**

Tables 4.13 and 4.14, present the results obtained regarding the pillar “space restriction for the Cobot operation” also consistent, with a consistency ratio of 5.46%. Analyzing the results obtained, as expected, both applications performed in the cargo compartment received the highest complexity score, once the cargo compartment presents a considerable space restriction for the Cobot operation.

TABLE 4.13 – Comparative matrix of potential applications - Space Restriction for the Cobot Operation.

	Insulation blankets installation	Cabin floor fixing	Cabin floor joint sealing	Cargo floor fixing	Overhead bin positioning	Ceiling panels installation	Side panels positioning	Cargo ceiling positioning
Insulation blankets installation	1	2	2	1/6	1/3	1/2	1/2	1/8
Cabin floor fixing	1/2	1	1	1/8	1/3	1/2	1/2	1/9
Cabin floor joint sealing	1/2	1	1	1/8	1/3	1/2	1/2	1/9
Cargo floor fixing	6	8	8	1	7	8	8	1/2
Overhead bin positioning	3	3	3	1/7	1	2	2	1/7
Ceiling panels installation	2	2	2	1/8	1/2	1	1	1/7
Side panels positioning	2	2	2	1/8	1/2	1	1	1/7
Cargo ceiling positioning	8	9	9	2	7	7	7	1

TABLE 4.14 – Normalized Comparative matrix of potential applications - Space Restriction for the Cobot Operation.

	Insulation blankets installation	Cabin floor fixing	Cabin floor joint sealing	Cargo floor fixing	Overhead bin positioning	Ceiling panels installation	Side panels positioning	Cargo ceiling positioning	LOCAL PRIORITY
Insulation blankets installation	0,043	0,071	0,071	0,044	0,020	0,024	0,024	0,055	**0,044**
Cabin floor fixing	0,022	0,036	0,036	0,033	0,020	0,024	0,024	0,049	**0,030**
Cabin floor joint sealing	0,022	0,036	0,036	0,033	0,020	0,024	0,024	0,049	**0,030**
Cargo floor fixing	0,261	0,286	0,286	0,263	0,412	0,390	0,390	0,220	**0,313**
Overhead bin positioning	0,130	0,107	0,107	0,038	0,059	0,098	0,098	0,063	**0,087**
Ceiling panels installation	0,087	0,071	0,071	0,033	0,029	0,049	0,049	0,063	**0,057**
Side panels positioning	0,087	0,071	0,071	0,033	0,029	0,049	0,049	0,063	**0,057**
Cargo ceiling positioning	0,348	0,321	0,321	0,525	0,412	0,341	0,341	0,439	**0,381**

The last comparison performed is presented in Tables 4.15 and 4.16, regarding the pillar "Required precision for the effector". The obtained result also was consistent, with a consistency ratio of 5.72%. Furthermore, the results obtained showed that the insulation blankets installation received the highest score for this element, once the blankets must be installed in specific positions, and the clips need to be inserted in the correct position to fix the blankets against the structure. Also, considering the need for torque control, the floor fixing applications, in both, cabin and cargo, received a considerable complexity score, followed by ceiling panels installation, since requires the application of controlled force to fit the panels on the structure.

TABLE 4.15 – Comparative matrix of potential applications - Required Precision for the Effector.

	Insulation blankets installation	Cabin floor fixing	Cabin floor joint sealing	Cargo floor fixing	Overhead bin positioning	Ceiling panels installation	Side panels positioning	Cargo ceiling positioning
Insulation blankets installation	1	2	3	2	3	3	5	3
Cabin floor fixing	1/2	1	3	1	2	2	4	2
Cabin floor joint sealing	1/3	1/3	1	1	1/2	1/3	2	1/3
Cargo floor fixing	1/2	1	1	1	3	2	5	2
Overhead bin positioning	1/3	1/2	2	1/3	1	1/2	3	1/2
Ceiling panels installation	1/3	1/2	3	1/2	2	1	4	1
Side panels positioning	1/5	1/2	1/2	1/5	1/3	1/4	1	1/5
Cargo ceiling positioning	1/3	1/2	3	1/2	2	1	5	1

TABLE 4.16 – Normalized Comparative matrix of potential applications - Required Precision for the Effector.

	Insulation blankets installation	Cabin floor fixing	Cabin floor joint sealing	Cargo floor fixing	Overhead bin positioning	Ceiling panels installation	Side panels positioning	Cargo ceiling positioning	LOCAL PRIORITY
Insulation blankets installation	0,283	0,316	0,182	0,306	0,217	0,298	0,172	0,299	**0,259**
Cabin floor fixing	0,142	0,158	0,182	0,153	0,145	0,198	0,138	0,199	**0,164**
Cabin floor joint sealing	0,094	0,053	0,061	0,153	0,036	0,033	0,069	0,033	**0,067**
Cargo floor fixing	0,142	0,158	0,061	0,153	0,217	0,198	0,172	0,199	**0,163**
Overhead bin positioning	0,094	0,079	0,121	0,051	0,072	0,050	0,103	0,050	**0,078**
Ceiling panels installation	0,094	0,079	0,182	0,077	0,145	0,099	0,138	0,100	**0,114**
Side panels positioning	0,057	0,079	0,030	0,031	0,024	0,025	0,034	0,020	**0,037**
Cargo ceiling positioning	0,094	0,079	0,182	0,077	0,145	0,099	0,172	0,100	**0,118**

Finally, the results obtained by the comparison between the four pillars, and the results obtained by the pairwise comparison between the potential applications can be combined. Table 4.17 presents the final results obtained for the complexity of the development for each application identified, where the higher the global value the higher the complexity.

TABLE 4.17 – Global Evaluation Matrix of potential applications regarding the development complexity.

	Interaction level with humans	Effector complexity	Space restriction for Cobot operation	Required precision for the effector	GLOBAL RESULT
	0,1484	**0,279**	**0,4690**	**0,1040**	
Insulation blankets installation	0,103	0,337	0,044	0,259	**0,157**
Cabin floor fixing	0,036	0,036	0,030	0,164	**0,047**
Cabin floor joint sealing	0,025	0,033	0,030	0,067	**0,034**
Cargo floor fixing	0,049	0,070	0,313	0,163	**0,191**
Overhead bin positioning	0,330	0,106	0,087	0,078	**0,127**
Ceiling pannels installation	0,076	0,217	0,057	0,114	**0,110**
Side pannels positioning	0,150	0,051	0,057	0,037	**0,067**
Cargo ceiling positioning	0,232	0,152	0,381	0,118	**0,268**

As can be verified, the cargo ceiling positioning, cargo floor fixing and insulation blankets installation were the potential applications evaluated with the highest complexity development complexity score. Considering the Cargo ceiling positioning, the main elements of the application that lead to the highest complexity score are related to the restricted area, once the cargo compartment has a low height and limited width, which difficult the insertion of a Cobot. Also, the complexity is high for interaction with humans, and effector complexity, considering that this application would still require human operators working alongside the Cobot to perform the installation.

For the cargo floor fixing, the complexity is similarly influenced by the restricted space of the compartment, leading to higher scores for space restriction. Also, this application presented a high complexity related to the precision required, a characteristic of the floor-fixing process.

Finally, the insulation blankets installation received a high development complexity score mainly by the characteristics of the application. First, once the insulation blankets are malleable the complexity of the development of the end effector increases significantly. Furthermore, once the blankets are fixed by the insertion of clips, it would require additional functionality for the end effector. Moreover, once each insulation blanket has a specific installation position, an complexity related to the required precision also accounted for the development complexity score.

4.3.3 Application Selection

As presented at the beginning of section 4.3, the evaluation of each potential application regarding the expected benefits for the current processes (Section 4.3.1.) could be combined with the analysis of the complexity of the development, presented in the last section. As a result, it was obtained the evaluation matrix presented in Figure 4.11, delimited by the average values for both variables.

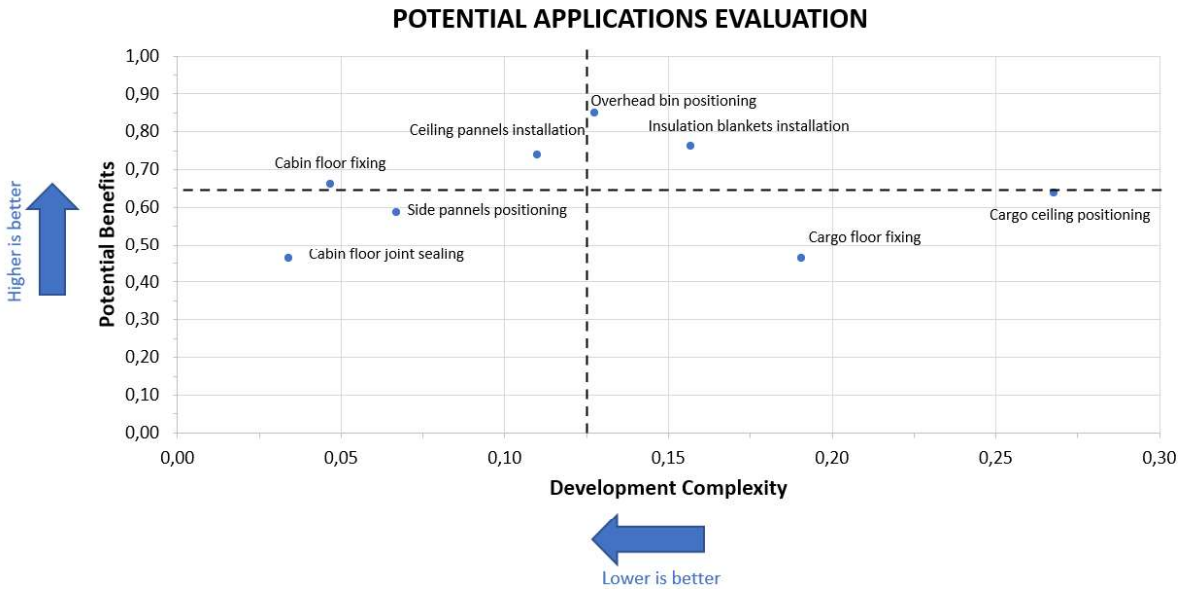


FIGURE 4.11 – Evaluation Matrix of Potential Applications.

As expected, the applications related to the cargo compartment showed the highest complexity, mainly influenced by the space restrictions. Also, presented an inferior potential benefit compared to others, mainly influenced by the short duration of the activities

when compared to that performed in the passenger cabin, and as a result, were not selected for the scope of this work.

With similar potential improvement as the activities performed in the cargo compartment, is the cabin floor joint sealing. This is also motivated since the activity does not involve the handling of heavy objects and the duration of the activity is shorter and less repetitive than others. Therefore, even with the lowest complexity of the development, this activity was not considered for the present work.

Another result aligned with the collected information of the current process is the classification of the overhead bin positioning and insulation blankets installation. As presented in section 4.2, both activities present a high level of human workload, since the first encompasses the handling of heavy components in awkward positions, and the second, also involves awkward postures and is highly repetitive. However, as expected, both applications present a high complexity in their development, considering the characteristics of the objects (heavy bin and flexible insulation blanket) and other factors such as interaction with humans in the case of the bin positioning and the multi-functionalities required for the blankets installation.

Finally, considering the whole evaluation, it was concluded that the best application to be further developed in the scope of this work is related to the ceiling panels installation, considering that this application presented one of the highest potential improvements indexes while showing an intermediate complexity score. Furthermore, there were not found in literature studies related to this application, for the cabin floor fixing (presented in section 2.4), which complies with the target for the work to contribute to the increase in the Cobot deployment in aircraft's final assembly. Additionally, considering the characteristics of the ceiling panels, the solution developed may also be expanded to the side panel positioning application, which can increase the financial viability of this application once the same equipment may be used for different applications.

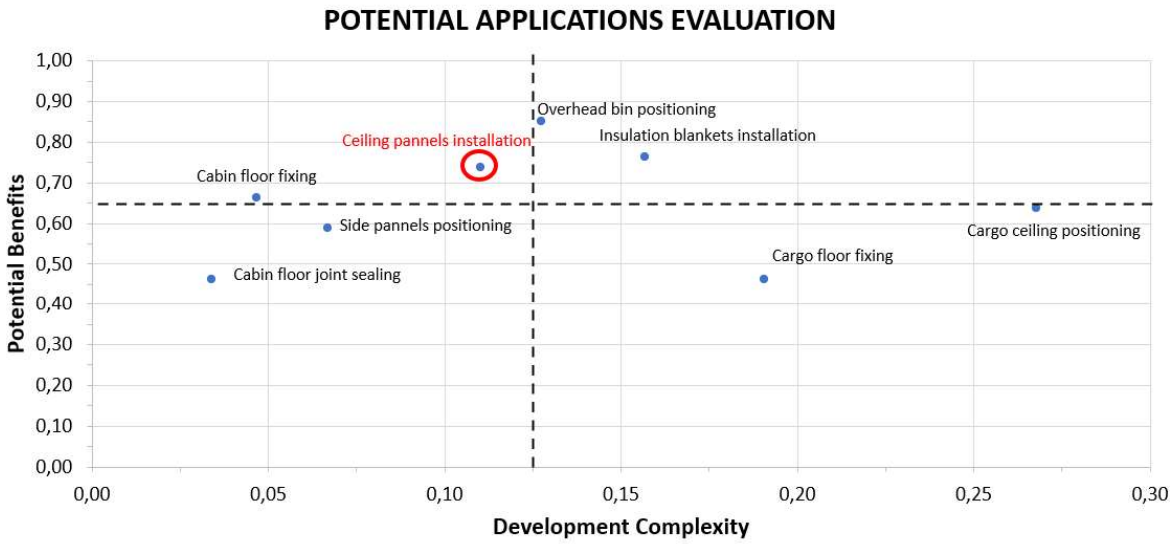


FIGURE 4.12 – Selected application in the Evaluation Matrix.

5 Conceptual Design of the Solution

This chapter presents the process and results obtained during the development of the conceptual solution for the ceiling panels installation, as presented in Chapter 4. The solution definition process followed the system design process consolidated in the systems engineering approach presented in Section 2.7.2.

After the selection of the application, the first step followed was to map the current situation as the basis for the definition of the envisioned operation. In this step, the collection of the stakeholder's expectations allowed the identification of needs, goals, and constraints that guided the definition of the envisioned scenario, using the CONOPS framework.

The Concept of the envisioned operation allowed the definition of the solution's requirements, which allowed the definition of architecture for the solution, as part of the Logical decomposition step of the system design process. Finally, different solutions were identified for each function identified in the solution. Based on the requirements, the solutions were evaluated, and the recommended conceptual solution was selected. Figure 5.1 resumes the process applied to the definition of the conceptual solution, and each step is detailed in the next sections.

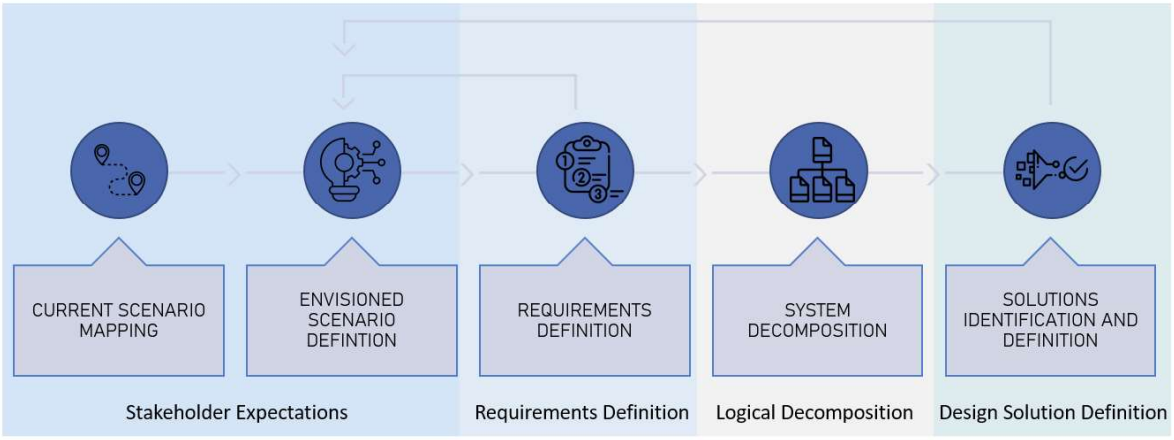


FIGURE 5.1 – Conceptual solution process, derived from the system design process.

5.1 Current Scenario Mapping

According to the system design process, the first step performed was the capture of the stakeholder's expectations. This process was interactively performed with the current scenario mapping, allowing the identification of the main stakeholders involved with the solution. Table 5.1 presents a list of identified stakeholders.

TABLE 5.1 – Identified Stakeholders for the project.

Stakeholder	Description
Operator	Responsible for performing assembly operations
Production Planner	Responsible for define the sequence between the assembly operations, and schedule the excution of activities inside the aircraft
Automation Engineer	Responsible for design of automation solutions, aims for the increase of Level of Automation.
Manufacturing Engineer	Responsible for defining the production process, assembly sequence, resources and instructions
Quality Inspector	Responsible for assuring that the assembly is according to the stablisehd quality levels.
Occupational Safety Technician	Responsible for assuring that the activities performed by the operators are safe and cannot compromise their health.
Production Leader / Manager	Responsible for the aircraft production results

The identified stakeholders were then evaluated through the power and interest matrix and the onion diagram tools, with the objective of classifying the stakeholders to correctly involve in the development of the solution.

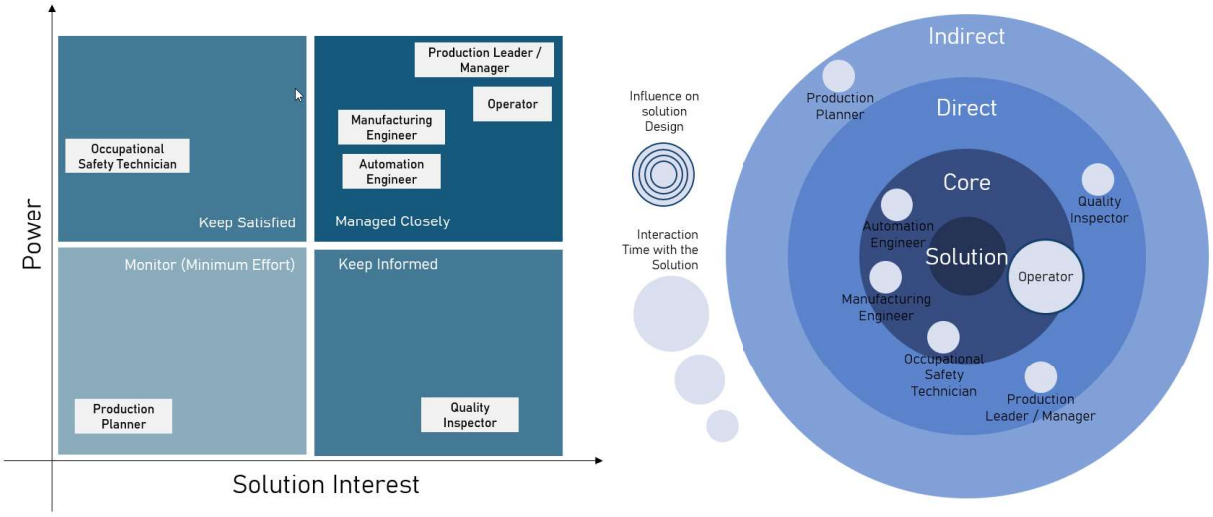


FIGURE 5.2 – Stakeholders evaluation. Power x Interest diagram (left), Onion diagram (right).

As can be verified in Figure 5.2 the operator is the stakeholder with the most interac-

tion time with the solution, with a high influence on solution design. This result shows that the operator is the main responsible for bringing desires and solution requirements to be considered in the solution. Furthermore, the operator has a high power once is the one who will be using the solution in the future. If the solution does not meet operators' expectations, it will hardly have continuous use in the production line.

The Manufacturing is responsible for defining the processes and resources required to manufacture the aircraft, and is also, responsible for assuring the efficiency of the designed processes. For this reason, this stakeholder has a high interest and power in defining the solution, in this case, a high influence on solution design.

The Automation engineer is responsible for the implementation of automation solutions to the process, and together with the manufacturing engineer, assists in defining the production processes. For this reason, this stakeholder has a high interest and power in the definition and deployment of the solution, as well as a high influence on the solution's design.

The quality inspector is responsible for assure that the assembly process performed results in a product with the expected quality. Despite the fact that this stakeholder has a lower influence on the design and power on the implementation, has an interest in the solution once this can increase the quality of the production process. Also, this stakeholder may bring quality requirements that impact the solution.

The occupational safety technician is responsible for assuring operators' safety during the execution of the manufacturing activities, through the implementation and inspection of safety rules and procedures. Since the solution will introduce the operation of human operators alongside robots, this stakeholder has an influence on the solution and the power to allow the implementation in the production line.

Since it will be inserted a Cobot inside the fuselage, it may impact the activities schedule, and it will be required to consider a schedule/time for inserting the Cobot inside the aircraft. This impacts the production planner activity which is responsible for the production plans and resource allocation. However, to define the solution Production planner has low power and interest, and in this case, a minimum effort is required to manage the stakeholder.

Finally, the production leader/manager is the stakeholder responsible for the results related to production and generally is the sponsor for these types of projects. For this reason, this stakeholder has high power and interest in the solution, bringing requirements related to the solution's cost and benefits.

After the identification and classification of the stakeholders, the next step performed was the collection of expectations through needs and desires mapping. For this step, GUENBA in the production line was performed, which allowed the collection of the in-

formation presented in Table 5.2. The needs or desires were collected and associated with the stakeholders with direct interest, and classified as Must have or a desirable aspect for the solution.

TABLE 5.2 – Identified needs from stakeholders.

ID	Need / Desire	Main Stakeholder	Stakeholder 2	Stakeholder 3	Stakeholder 4	Classification
1	Operator need to feel safe to work with the COBOT	Operator				MUST HAVE
2	Improve operator's ergonomics	Operator	Ocupacional Safety Technician	Manufacturing Engineer		MUST HAVE
3	Solution must allow a collaborative workspace	Operator	Production Planner			MUST HAVE
4	Solution must allow other activities to be performed in the same workspace	Production Planner	Operator			MUST HAVE
5	Attractive implementation cost	Production Manager	Manufacturing Engineer			DESIRABLE
6	Return on Investiment compatible with the company's vision	Production Manager	Manufacturing Engineer			DESIRABLE
7	The solution can be used for more than one application inside the aircarft cabin	Production Manager	Operator	Manufacturing Engineer	Production Planner	DESIRABLE
8	The operational time with the application of COBOT will be reduced.	Production Planner	Operator			DESIRABLE
9	The COBOT must be able to handle the components with no damage	Quality Inspector	Operator			MUST HAVE
10	The COBOT must be abble to reach and install all the ceiling pannels along the fuselage cabin	Operator	Production Planner			MUST HAVE
11	Solution able to identify the ceiling panels and interior components	Operator	Automation Engineer			MUST HAVE
12	Solution able to pick up and handle the ceilling pannels	Operator	Automation Engineer			MUST HAVE
13	Solution abble to positioning the ceiling panels/ components in the correct position	Operator	Automation Engineer			MUST HAVE
14	Solution able to manage the correct force to fit the ceiling pannel in the position.	Operator	Automation Engineer			MUST HAVE
15	Easy to program	Operator	Automation Engineer			DESIRABLE
16	Allows the stop of operation by operator	Operator	Ocupacional Safety Technician			DESIRABLE
17	Abble to communicate with human workrs	Operator	Automation Engineer			MUST HAVE
18	Installation must result in an acceptable Level of Gap between panels	Quality Inspector	Operator			MUST HAVE
19	Installation must result in an acceptable Level of Step between panels	Quality Inspector	Operator			MUST HAVE
20	Installation must result in a acceptable Level of alignment between panels	Quality Inspector	Operator			MUST HAVE

Additionally, to the needs and desires mapping, the current process was mapped through GUENBA in the production line. Figure 5.3 resumes the current process.

The general sequence of the installation of the ceiling panels starts with the arrival of the components from logistics to the manufacturing hangar, which are loaded into the fuselage cabin by the logistics operators through a conveyor. Operators, inside the aircraft cabin, take the panels from the conveyor and position them above the cabin floor, and perform the installation. After installation is completed, the operator registers the results and main information of the assembly process performed. Finally, after installation, a quality assessment is performed by the quality team to assure that the process was performed according to the standards.

Once the overall sequencing was mapped, a deep investigation was performed in the process steps of interest for this work. The operational sequence of the activities performed

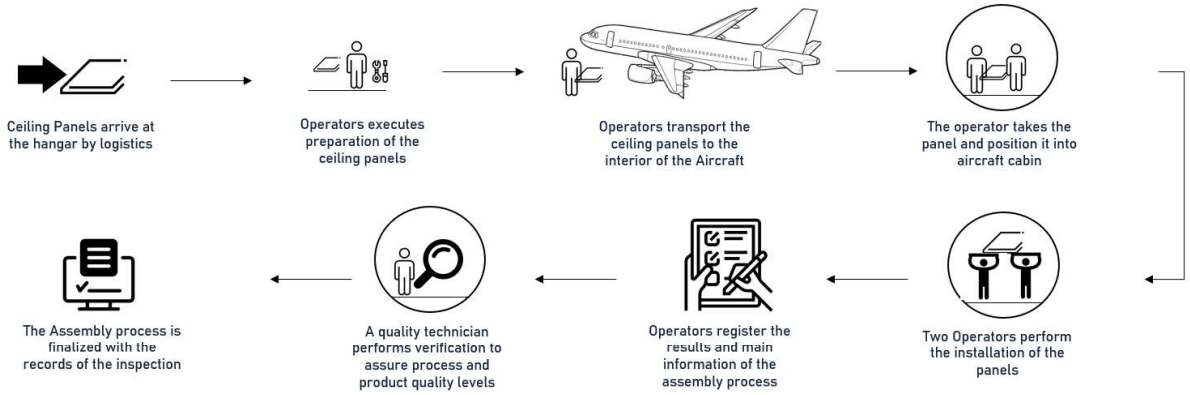


FIGURE 5.3 – Current process operational sequence.

by the production operators was mapped and is presented in Figure 5.4. In addition, the required resources were also mapped as well as the duration of the activities.

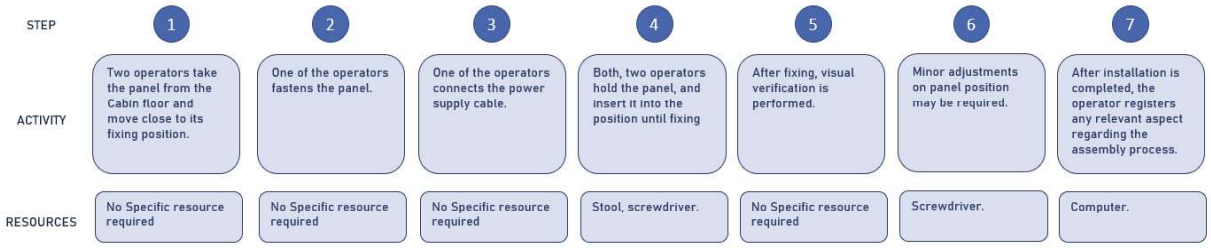


FIGURE 5.4 – Operational sequence - Assembly of ceiling panels.

During the process mapping and GUENBA, it was possible to identify the main issues and challenges faced by the operator during the execution of the activity. As expected, the higher difficulties faced by operators in the current process are related to the installation, verification, and adjustments.

The main challenge verified during this step is related to the ergonomics conditions faced by the operator to execute the activities, confirming what was previously identified during the selection of the application in Chapter 4. During the installation, the operator needs to work with arms raised above the head, with the neck twisted and sometimes need to use a stool to perform the activity, raising the risks of falls and injuries.

Additionally, operators are unable to see the elements used to fit the panels, increasing the installation complexity and execution time. Therefore, it is required that the operator have experience in this installation to be able to perform it, which leads to another problem for the process which is the availability of a capable workforce.

In this sense, it was verified that the complexity of the installation is increased once some fixtures are currently adjustable. This implies that the operator needs to work in non-ergonomic conditions for longer periods than the installation of the panel.

After the installation, the panel installation verification is performed visually by the operator, also depending on the experience of the operator. Furthermore, in case of the operator verifies that the panel assembled presents a gap, step, or alignment considered not visually acceptable, the panel must be removed, and washers must be manually inserted, significantly increasing the operation times.

After the installation, the panel installation verification is performed by the operator, also depending on the experience of the operator. Furthermore, in case of the operator verifies that the panel assembled presents any not acceptable quality characteristics, the panel must be removed, and washers must be manually inserted, significantly increasing the operation times.

Finally, with the mapping of the current process, it was possible to identify constraints for the implementation of the automation solution. Table 5.3 presents the main constraints verified, which were classified by this origin as environment or process, all being considered non-negotiable constraints that must be met by the solution design. The real values were suppressed and substituted by symbols, considering confidentiality.

TABLE 5.3 – Identified Constraints.

ID	Constraint	Constraint Class	Level of Acceptance Class
1	The entrance of the aircraft for the cobot deployment is ***H_D*** m from the ground.	ENVIRONMENT	**HARD (NON-NEGOTIABLE)**
2	The COBOT must be able to enter the aircraft by the pax door (***h_D*** m height x ***w_D*** m width)	ENVIRONMENT	**HARD (NON-NEGOTIABLE)**
3	COBOT Must be able to operate inside the cabin (Circular with ***W_F*** m floor width and ***H_F*** m height in the centerline)	ENVIRONMENT	**HARD (NON-NEGOTIABLE)**
4	To install the Ceiling panels, the COBOT must be able to avoid the Overhead bins that create a ***W_OB*** m width space in the middle section of the cabin to position the ceiling panel	ENVIRONMENT	**HARD (NON-NEGOTIABLE)**
5	Cobot must be able to handle panels of ***l_p*** m length x ***w_p*** m width and ***t_p*** mm thin	PROCESS	**HARD (NON-NEGOTIABLE)**
6	Cobot must be able to handle panels of **l_p2** m length x **w_p2** m width	PROCESS	**HARD (NON-NEGOTIABLE)**
7	COBOT must be able to handle ceiling panels with **W_p** kg weight.	PROCESS	**HARD (NON-NEGOTIABLE)**
8	The current process requires connection wiring related to the panel's lights during the installation.	PROCESS	**HARD (NON-NEGOTIABLE)**

The identified constraints in conjunction with the verified problems and challenges, as well as the stakeholder's needs and desires identified, formed the basis for the mapping of the current scenario, since are the main elements of the CONOPS. This information allowed the definition of the envisioned operating system as well the solution's requirements. The next section will discuss the defined future operating scenario for the application.

5.2 Envisioned Scenario

Following the system design process and composing the CONOPS, the definition of the envisioned scenario was the next step performed. The general concept to be adopted is the insertion of a Collaborative robot solution inside the aircraft cabin to perform the ceiling panels installation. In this sense, Figure 5.5 presents the envisioned operational scenario, highlighting the added or changed steps of the original process steps.

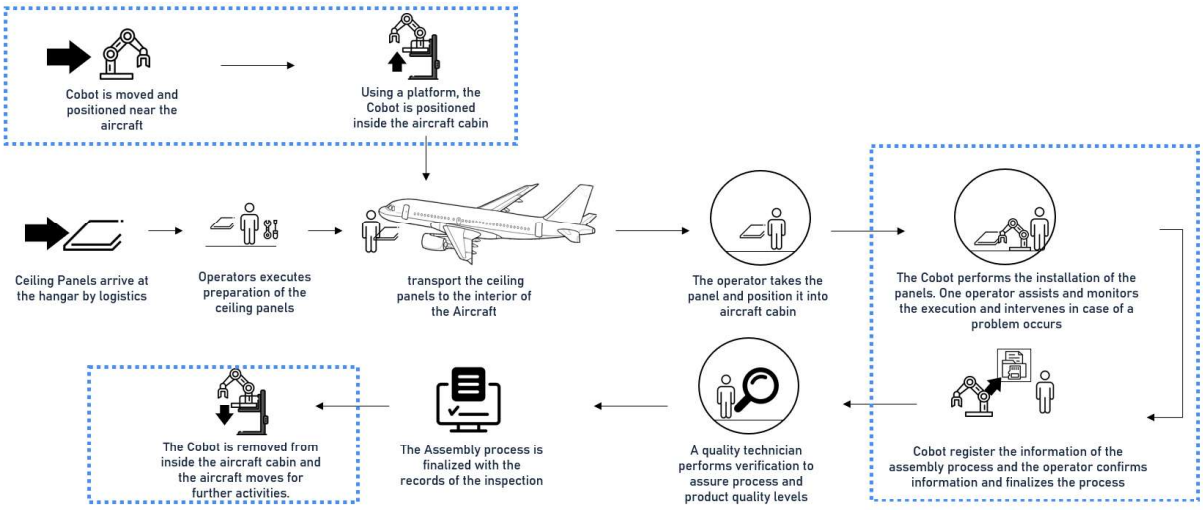


FIGURE 5.5 – Envisioned process operational sequence.

In the production stage when the ceiling panels are installed, the aircraft is in the final assembly, where all structural segments are already assembled and the operators and material access the aircraft cabin through the passengers or service doors. This leads to the necessity of suspending the collaborative solution by a lifting platform to insert it inside the aircraft cabin, as identified in the last section.

In the envisioned scenario, the assembly steps with the highest problems identified were the focus for the implementation of the collaborative solution. After being inserted inside the aircraft cabin, the robot will be responsible for the execution of most parts of the ceiling installation process, reducing the human workload and ergonomics issues.

For the execution of the installation process, different arrangements, operation modes, and tasks sharing configurations are possible to be implemented, with different levels of complexity and collaboration. For this reason, these different arrangements were synthesized through five use cases presented below.

The first use case defined was the so-called Towards Independent Automation which is presented in Figure 5.6. The name refers to the fact that the Cobot tends to perform most of the assembly steps independently, while the human operator is responsible for the final verification of the procedure. This scenario basically eliminates the ergonomics issues faced by the human operator, however, presents the highest complexity of implementation.

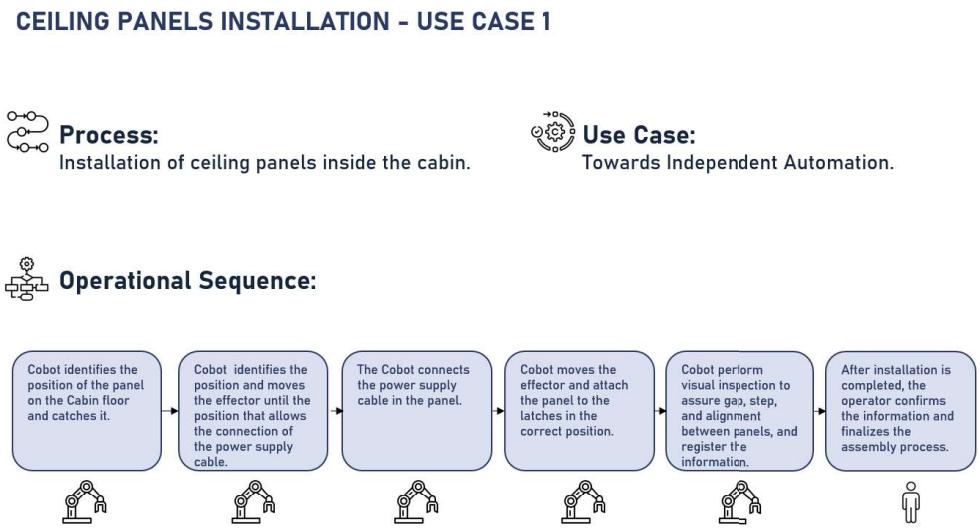


FIGURE 5.6 – Use case 1 - Towards Independent Automation

The second use case, Towards Supportive-Sequential Automation, differs from the first only in the first three steps of the assembly process, where the human operator starts to perform the activity collaboratively with the Cobot, through hand guiding and the human operator becomes responsible for the power supply connection step. This scenario simplifies the programming aspects of the application and avoids problems caused by positioning errors of the panels, however, increases the workload of the operator.

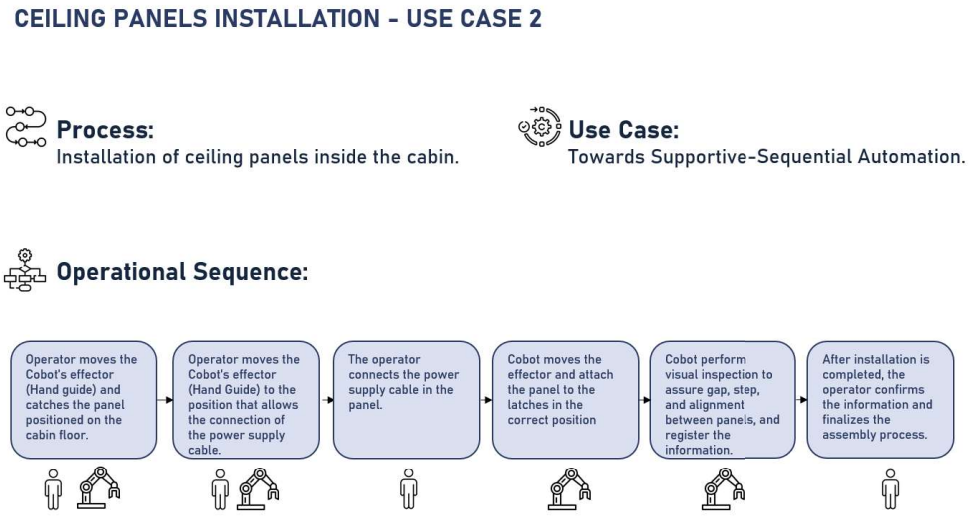


FIGURE 5.7 – Use case 2 - Towards Supportive-Sequential Automation

The third scenario is an evolution of the second one, increasing the level of collaboration between the human operator and the Cobot, presented in Figure 5.8. In this scenario, Towards Supportive Automation, the human operator uses hand guiding also in the panel installation activity. This scenario simplifies the programming aspects of the application, however, since the operator guides the collaborative arm, only the aspects related to the

weight handling would be solved, once the operator would still need to work with arms raised.

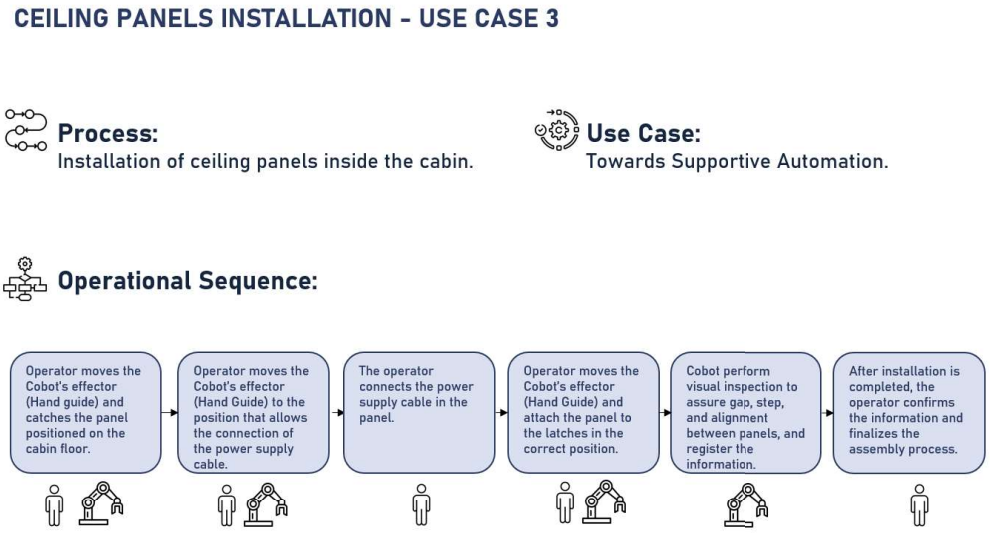


FIGURE 5.8 – Use case 3 - Towards Supportive Automation

The fourth use case tends to a reduction of the automation complexity, increasing the workload on the human operator. In this scenario, Towards Supportive-Reduced Automation, the human operator passes to be responsible for the visual inspection of the assembled panel, which leads to a significant reduction of complexity for the effector. However, in this scenario, the analysis of the correct installation would be still dependent on the operator's expertise.

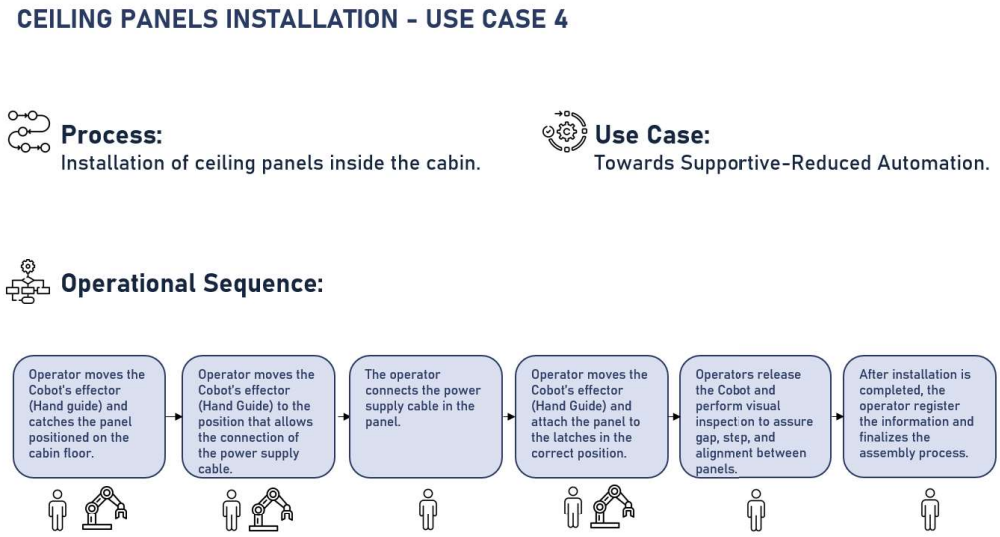


FIGURE 5.9 – Use case 4 - Towards Supportive-Reduced Automation

The last identified scenario was the so-called Towards Sequential Automation which is presented in Figure 5.10. The name refers to the level of collaboration between the human

and the Cobot, where the Cobot was placed to perform the most non-ergonomic activities while the human worker performs the activities that are most complex to automate. In this scenario, the human workload is reduced in comparison to use cases 2, 3, and 4. Furthermore, the human operator is only responsible to perform the steps with a high complexity to automate, which may lead to a very complex end effector that could turn the application unfeasible. For these reasons, this use case was selected as the basis for the application development.

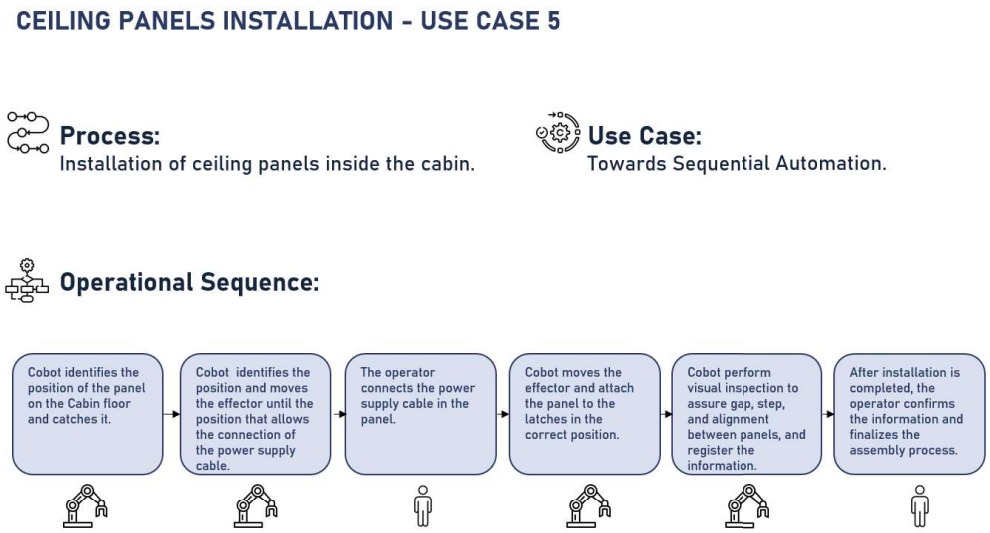


FIGURE 5.10 – Use case 5 - Towards Sequential Automation

With the operation scenario defined, it was possible to define the design reference mission, which is an important aspect of the CONOPS, composing the envisioned operating scenario. Figure 5.11 presents the reference operation in nominal condition, which is the operational configuration when the system is working without any issues or problems. As can be seen, the operator is responsible for bringing the panels to the position for the Cobot, which takes the panel and positions it for the power supply cable installation by the human operator. In the sequence, the Cobot performs the panel installation and the verification of the installation. Finally, the human operator confirms that the installation was correctly performed and makes the robot move for the installation of the next panel. This procedure is performed until the last panel installation, where the Cobot is removed from inside the aircraft cabin.

As explained before, the presented diagram refers to the nominal operating condition. Off-nominal conditions can be assessed but this was out of the scope of this work. However, it can be verified that other previously identified use cases can be applied or can be used to operate the system at an off-nominal condition. As an example, if the system presents any difficulty in correctly identifying and catching the ceiling panel, this activity may be performed temporarily as a collaborative one, with the human operator's hand guiding the effector.

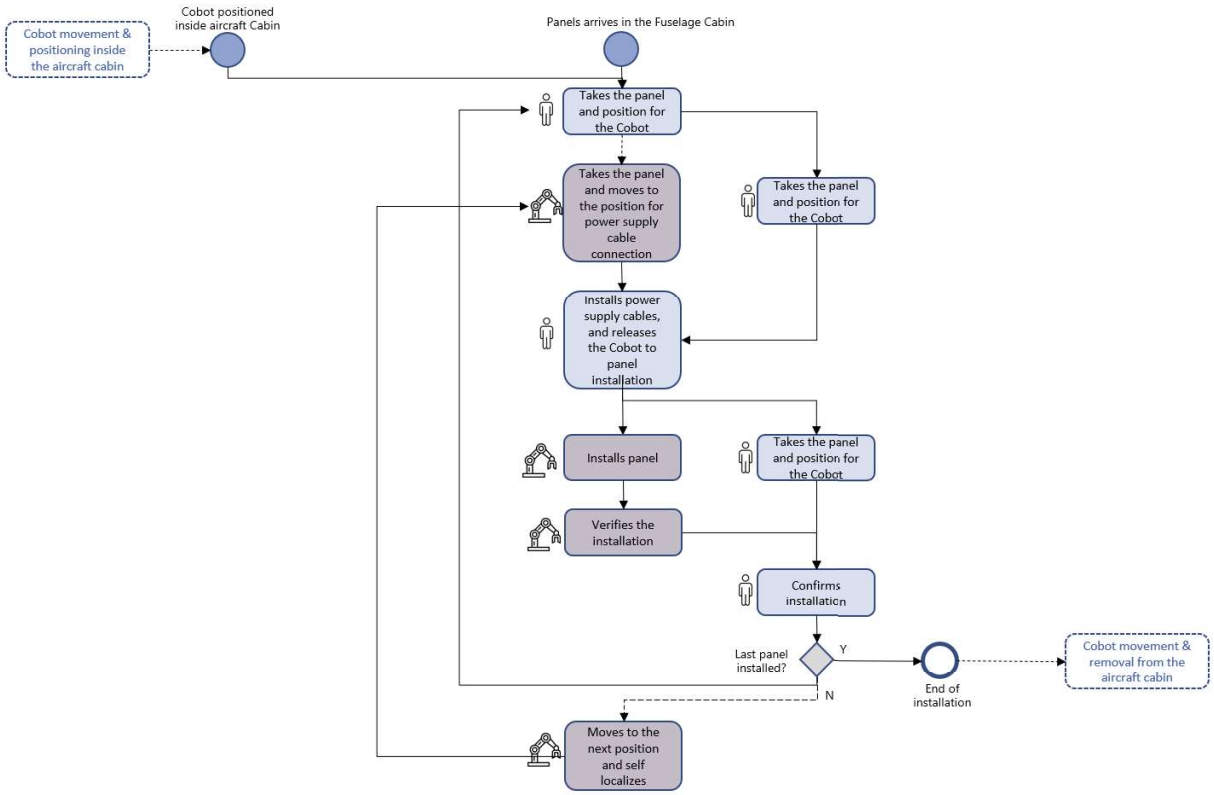


FIGURE 5.11 – Design Reference Mission: Nominal Conditions

Considering the defined reference mission, the safe operation mode most indicated is Speed and Separation Monitoring (SSM), described in Section 2.3. This is a consequence of the fact that the operator will be moving around the solution, entering and leaving near the base. In this case, the solution may operate at lower speeds when the operator enters the collaborative space, while, increases the speed as the operator moves away.

With the reference design mission identified, the concept of operation in the context of the present work was finalized, and the next step of the system design process was able to be performed, which is the requirements definition and is presented in the next section.

An important point to be mentioned is the recommendation to explore the other use cases identified for the application in further phases of the work. In this case, the objective is to compare this scenario with the baseline use case in terms of the ergonomics impact on the operator versus the solution design simplification, mainly for the scenarios with hand-guiding activities such as use cases 2 and 3.

5.3 Requirements Definition

The requirements definition process is the second step in the system design framework and is the stage where the stakeholder's expectations are translated into a clear definition

of the problem through the system requirements. Basically, the requirements are a set of technical conditions expressed in statements that are used for defining a design solution for the system.

The process of summarizing all the information collected during the previous system design phase and generating the requirements is interactive and complex, once must result in a set of statements that correctly address the stakeholder's expectations and lead to a design that is suitable for the operation. For this reason, the requirements definition for this work was based on the process indicated by the system design framework and is resumed in Figure 5.12.

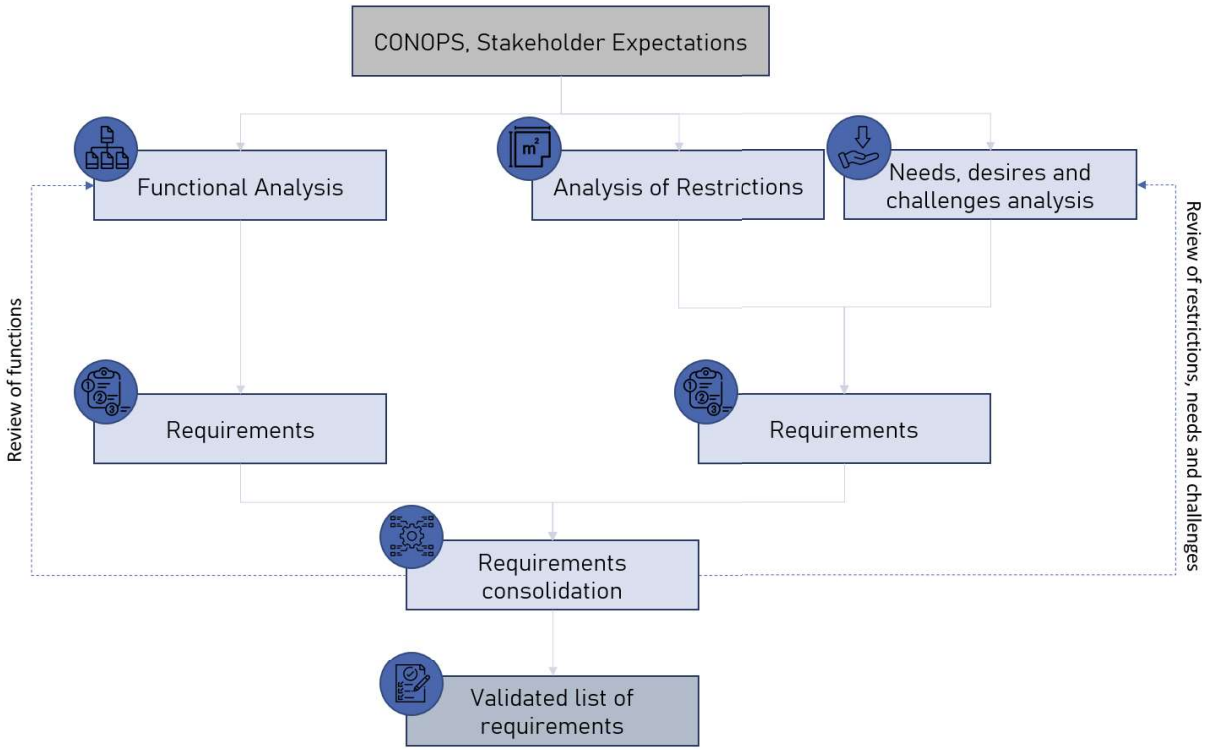


FIGURE 5.12 – Requirements definition process

Based on the CONOPS and the stakeholders' expectations the requirements for the application were derived. Firstly, a functional analysis was performed considering mainly the CONOPS and the reference mission, to allow an understanding of the required functions for the system, which directly derived functional requirements. Figure 5.13 presents the functional decomposition performed, breaking down the ceiling panel assembly into the primary functions required.

Considering the function of installing panels, it was possible to break it into three sub-functions in the first level: Move to the position, install panels, and verify the installation, which were functions aligned with the use case defined in the CONOPS. Each of these three sub-functions was broken into a lower functional level, and for each function identified, different requirements were defined.

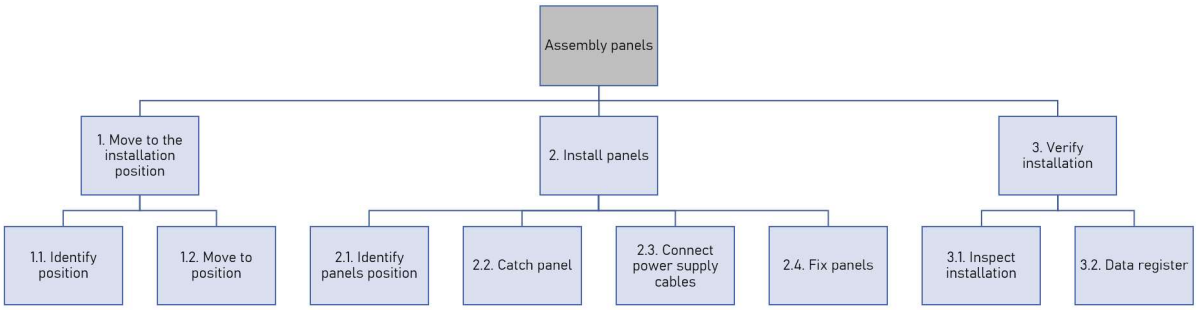


FIGURE 5.13 – Functional decomposition for the ceiling panel installation process.

In addition to the functional analysis, collected needs, desires, and restrictions were also analyzed and translated into statements, composing another set of requirements, as represented by the left side of the workflow shown in Figure 5.12.

After this first definition, the requirements were consolidated and confronted, to verify if all aspects were being correctly addressed. Based on the analysis of consolidated requirements, a review of the functional analysis and the stakeholder's expectations analysis were performed, feeding back the set of requirements. After rounds of analysis, a set of validated requirements were defined. Table 5.4 presents the identified requirements for the system design.

TABLE 5.4 – High-Level requirements defined for the solution design.

ID	Requirement	Associated function
1	The Cobot System shall provide means to identify the target position to move during the panels assembly.	1.1. Identify position
2	The Cobot System shall provide means to move inside the aircraft cabin during ceiling panel assembly.	1.2. Move to position
3	The Cobot System shall be able to avoid collisions with the cabin interior elements during its movimentation inside aircraft cabin.	1.2. Move to position, 2.2. Catch panel, 2.4. Fix panels
4	Solution must be able to identify the correct position of the panel to be taken.	2.1. Identify panels position
5	The Cobot System shall provide means to identify the target position to move the panels for power suply connection.	2.1. Identify panels' position, 2.3. Connect power supply cables
6	The Cobot system shall provide means to pick up the panels without damaging their surface.	2.2. Catch panel
7	The Cobot system shall provide means to pick up panels with at least **W_p** kg of weight.	2.2. Catch panel, 2.4. Fix panels
8	The Cobot system shall be able to move panels at least **H_F** m high from the ground and at least 0.95 m circumferentially from the Cobot center point at the ground level.	2.4. Fix panels
9	The Cobot System shall provide means to identify the target position to install the panels.	2.1. Identify panels position, 2.4. Fix panels
10	The Cobot System shall be able to move the panel to the installation position.	2.2. Catch panel, 2.4. Fix panels
11	The Cobot System shall be able to apply the correct force to fit the panels in their position.	2.4. Fix panels
12	The Cobot System shall collect visual parameters related to the relative position of the installed panel.	3.1. Inspect installation, 3.2. Data register
13	The Cobot System shall be able to process the collected visual information indicating the quality of the assembly.	3.1. Inspect installation, 3.2. Data register
14	The Cobot system shall register the information of the assembly process.	3.2. Data register
15	The Cobot system shall provide means to allow full operational control by the human operator.	All functions
16	The Cobot system shall provide indicative interfaces, status, and warning messages to the operator.	All functions
17	Solution must reduce operators non ergonomics activitie	All functions
18	The Cobot system shall be able to operate alongside humans operators in the same workspace without separation structures.	All functions
19	The Cobot system shall be able to perform at least one different assembly activity inside the aircraft cabin	Additional functionality
20	The mean installation time for a panel should be lower than current mean installation time for panels without any non-conformity	All functions
21	The Cobot system must be able to perform the installation resulting in an acceptable level of Gap between panels.	2.4. Fix panels, 3.1. Inspect installation
22	The Cobot system must be able to perform the installation resulting in an acceptable level of Step between panels.	2.4. Fix panels, 3.1. Inspect installation
23	The Cobot system must be able to perform the installation resulting in an acceptable level of alignment between panels.	2.4. Fix panels, 3.1. Inspect installation
24	The Cobot system must be able to be moved inside the aircraft by the entrance door, **H_D** m high from the ground.	Design requirement
25	The Cobot system must be able to enter inside the aircraft by the pax door (**h_D** m height x **w_D** m width).	Design requirement
26	The Cobot system must be able to operate inside the cabin (Circular with a **W_F** m floor width and **H_F** m height in the centerline).	Design requirement

The derived requirements encompass all technical aspects required for the application, based on the information collected in the stakeholder expectations phase. It is important to highlight that as presented in the research delimitation in Chapter 1, this work focused on technical and operational aspects of conceptual solution development. Economic aspects related to the application, such as ROI (Return on investment), payback time, and implementation costs were not derived through requirements and must be addressed in a detailed development phase, which is out of the scope of this work. However, considering the importance of these aspects, the solution costs were qualitatively considered as an additional aspect during the solution's trade-offs in cases where more than one option complies with all technical requirements.

The set of requirements characterized the second step of the system design process framework and allowed the definition of the solution design, which will be presented in

the next sections.

5.4 System Decomposition

The system decomposition is the third step of the system design process and has as its main objective the definition of the system architecture, which connects the technical requirements with the possible solutions (software and hardware) to comply with them.

The system architecture defines the structure and relationships of hardware, software, humans-in-the-loop, operations, etc., required by the system to perform the mission defined in the CONOPS. Also, the system architecture can be seen as a partitioning of the system into subsystems and elements, with specific functions to address the identified requirements. The connection between subsystems and elements is also important to be mapped in the architecture once influences the design solutions.

For this work, the definition of the system architecture was based mostly on the analysis of the requirements defined in the last section. For each requirement defined, an analysis was performed to identify subsystems or elements that were necessary to comply with the requirement. Figure 5.14 schematically represents the process executed.

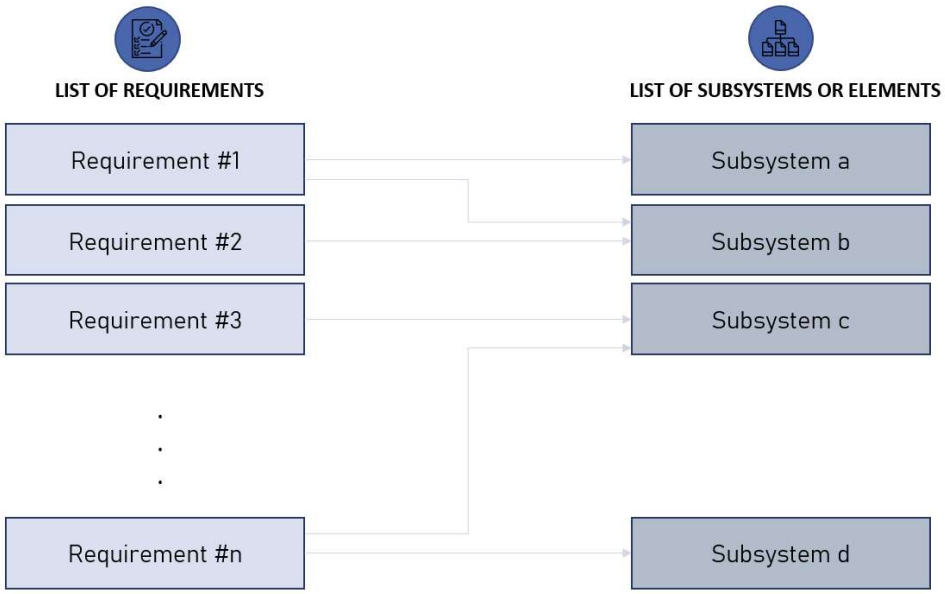


FIGURE 5.14 – System decomposition process.

As an example of the process performed, considering the requirement ID 7, “The Cobot system shall provide means to pick up panels with at least W_p kg of weight.”, indicates the need for a subsystem that would be responsible for picking up the panels for installation, in this case, an end effector. However, this requirement influences an additional subsystem required, the robotic arm, once this subsystem must have a payload that meets this requirement of being able to pick up panels with indicated weight.

After the analysis a preliminary systems architecture for the system was defined, presented in Figure 5.15. As can be verified the preliminary system architecture identified was composed of four main subsystems: Collaborative robot, End effector, Mobile base, and control system. Each of the first-level subsystems was composed of subsystems that can be verified. The number e division of the subsystems was based on the correct addressing of the requirements.

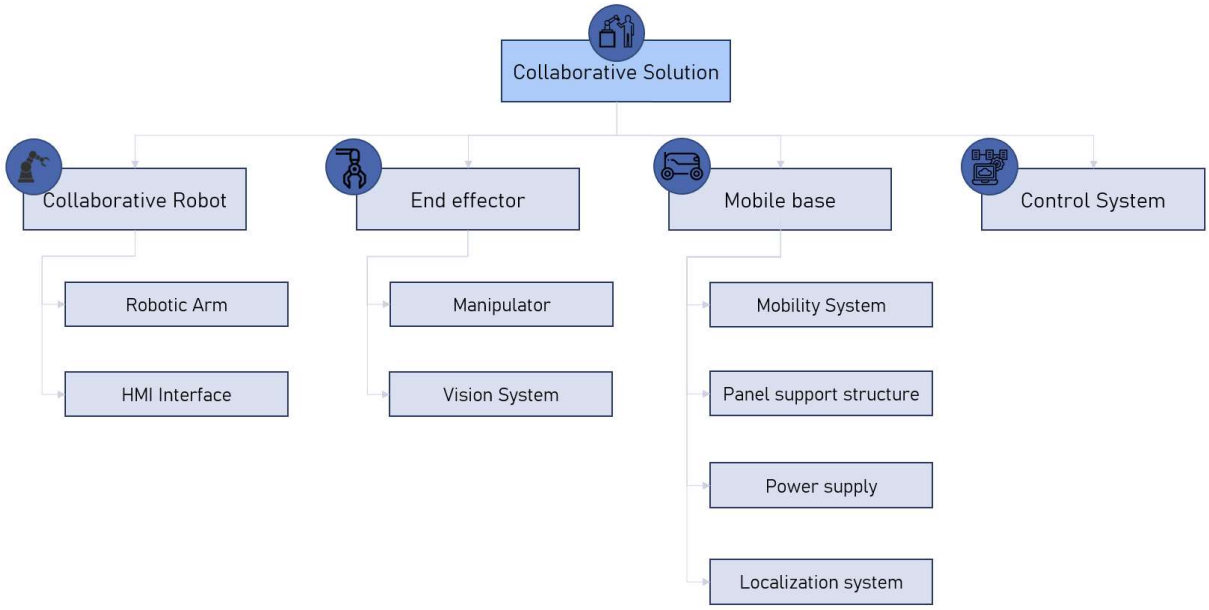


FIGURE 5.15 – Preliminary solution architecture.

The definition of the system's architecture is a recursive process. After a first description of the system components by the methodology described before, the architecture is confronted with previously defined requirements to guarantee that all functions and operational aspects are being covered.

The process of identifying subsystems and elements from the defined requirements was also complemented by the analysis of articles, publications, and research on automation solutions companies' products. A set of architectures for different applications was identified and used to compare and validate the systems architecture in construction. Based on these studies, the final architecture of the system for the ceiling panel installation solution was defined and is presented in Figure 5.16.

As presented, the collaborative solution architecture is based on four main subsystems for the ceiling panels installation. The first is the Collaborative robot, which is composed of the robotic arm and the HMI Interface. Robotic Arm is the component responsible for moving the ceiling panel during the installation process and characterizes the automation solution. The main requirements related to this component are the payload and reach required to move and install the ceiling panels. Composing the collaborative robot is the

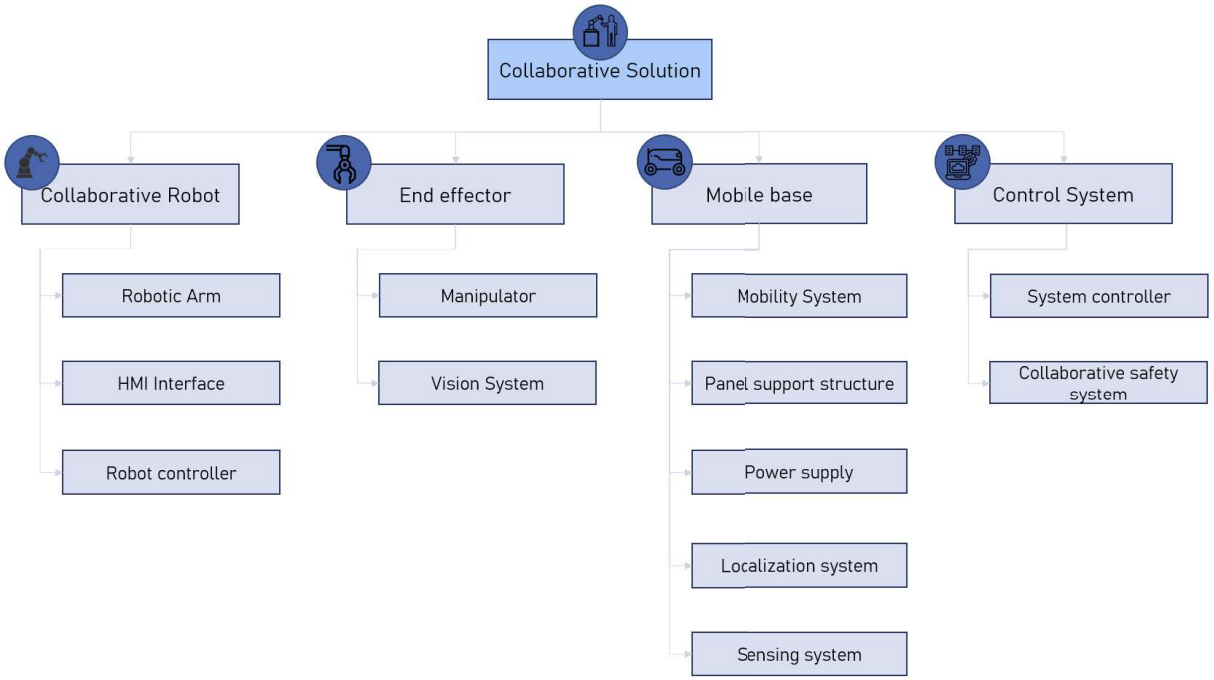


FIGURE 5.16 – Final solution architecture.

HMI (Human machine interface), which is the component that allows the communication between human and robot for all the operation phases, including safety aspects. Finally, the collaborative robot subsystem is composed of the controller, which is responsible for gathering input data, processing, and executing routines to control the robot's movement. Also, this controller is responsible for monitoring the safety aspects of the operation and guaranteeing a safe collaboration with a human operator.

The second subsystem defined in the solution's architecture is the end-effectors, which is the component responsible for the handling of the ceiling panels. There are several different end-effectors available in the market, which are designed to comply with different applications. For this application, the end-effector structure is composed of the manipulator, which is the structure that is attached to the robotic arm and has the function of Picking and placing the panels.

Additionally, according to the defined CONOPS, the collaborative solution is responsible for assembling and verifying the quality result of the process. These characteristics led to the need for a vision system composing the structure of the end effector.

The third subsystem of the collaborative solution is the mobile base, which has the primary function of allowing the system to move along the fuselage length to perform ceiling panel installation. For this reason, the first element of the mobile base is the mobility system, which provides the capacity of moving the solution along the fuselage.

Composing the mobile base subsystem is the support for the ceiling panels. This element is responsible for supporting the placement of the panel for the robot to perform the

installation. The design solution for this component is hardly connected to the mobility system and will be discussed in the next section.

To allow the execution of all functionalities of the solution a power supply system is required and composes the mobile base subsystem. The power supply includes the energy supply for movement and other sources of energy eventually required for the end effector, for example.

To complete the set of required components to allow the mobile base to operate is the localization system. This component is responsible for providing the means for the correct positioning of the mobile base in the aircraft cabin to allow the installation of the ceiling panels.

Safety aspects are a major concern when designing a collaborative solution. For this reason, the last component of the mobile base is the sensing system. Basically, this component is responsible for guaranteeing a safe collaboration with humans, and in this case no collision with the mobile base.

Finally, to orchestrate the operation of the solution integrating all the subsystems it was defined the subsystem Control System. The first component of this subsystem is the system controller, which is responsible for monitoring all parameters, connecting all the elements of the solution, and executing the defined logic to allow the operation. In addition to the system controller, it was defined the Collaborative Safety system component, which is responsible for guaranteeing safe collaboration with humans during robot operation. This strategy of segregating the control system into two components was defined to guarantee that safety aspects can be correctly addressed independently of the system controller operation.

With the definition of the system architecture, the third phase of the system design process presented in Figure 5.1 was complete, allowing the start of the definition of the design solutions. The process of defining the conceptual solutions for the system will be presented in the next section.

It is important to highlight that this system architecture was defined in the context of a conceptual design phase of the solution and may be adjusted according to the detailing of the project.

5.5 Solutions Identification and Definition

Based on the definition of the solution architecture, the final stage of the system design process was performed. This section presents the results of the solutions identification and definition phase which culminates in the definition of the conceptual solution for the

application studied in this work.

The process of identifying solutions and defining the conceptual model for the application is the transformation of the system architecture into alternative solutions to comply with the identified solution's requirements. The identification of the possible solutions for each component of the solution was based on brainstorming with experts as well as a literature review of published works and research on systems supplier's websites.

Based on the architecture defined in the last section and the design alternatives identified it was possible to define the morphological chart for the solution, which consolidates all solutions alternatives for the solution and is presented in Figure 5.17.

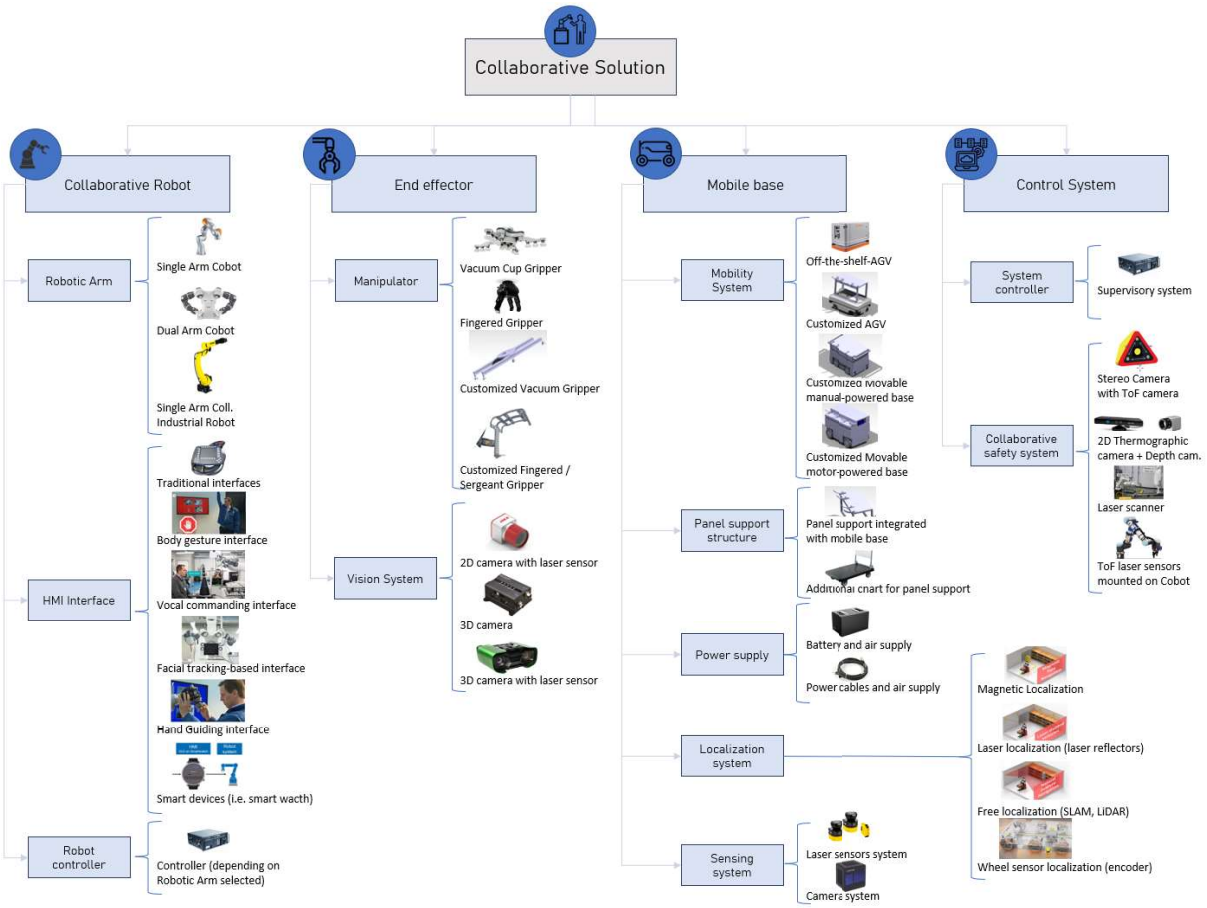


FIGURE 5.17 – Identified solutions for the solution architecture.

As can be verified different alternatives were defined for each component of the solution structure and function. The alternative solutions identified were analyzed considering compliance with the requirements identified, restrictions for the application, and trade comparisons. For some elements of the system architecture, the selection of the most indicated solution was performed considering another element together, depending on the connection level between elements.

Identified solutions for each component are discussed in more detail in the further

sections. All solutions aspects and trade studies performed were also detailed, allowing the consolidation of the conceptual solution for the application.

5.5.1 Collaborative Robot

5.5.1.1 Robotic Arm

The main component of the solution is the robotic arm, which is responsible for providing the means of moving the ceiling panels for its installation. As presented in Figure 5.17, three different options for this component were identified for the application.

The first option is the use of an industrial robotic arm as a collaborative one. As presented in Chapter 2, industrial robots have been applied to industrial applications since the 1960s, becoming nowadays robust solutions that can work at high speeds can carry high payloads with high accuracy and repeatability, which motivated the evaluation of this solution for the scope of this work.

Additionally, several studies have been developed to allow the application of industrial robots in collaborative applications, such as (COSTANZO *et al.*, 2021) where a FANUC M-20 series was applied to an aeronautics application in a cooperative assembly operation in which the robot performs a quality inspection task, while the human worker is dismantling a metal part for deburring after hole drilling. Another example is presented by (BRUNO; ANTONELLI, 2018) where a robot KUKA KR 300 R2500 is applied for a mechanical assembly.

These implementations of industrial robots in collaborative workspaces, require the application of several technologies to allow a safe operation. Most of the solutions are based on vision systems such as thermal and depth cameras combined with IA to allow a real-time identification of the operator and feedback to the robot controller to execute the programmed logic and avoid collisions.

Figure 5.18 presents an example of an industrial robotic arm applied to a collaborative application, where a multi-modal perception system that fuses depth and thermal images is applied for reliable human detection and tracking in the robotic work cell. Image processing data are used as input for the robot operating in the ”SSM” operating mode, where the robotic arm reduces or increases the operating speed according to the distance from the human operator.

The application of industrial robots to collaborative work cells is hardly motivated for applications that require a solution capable of working with higher payloads compared to collaborative robotic arms, in a collaborative workspace. Another advantage is the lower implementation cost of the robotic industrial arm when compared to collaborative ones.

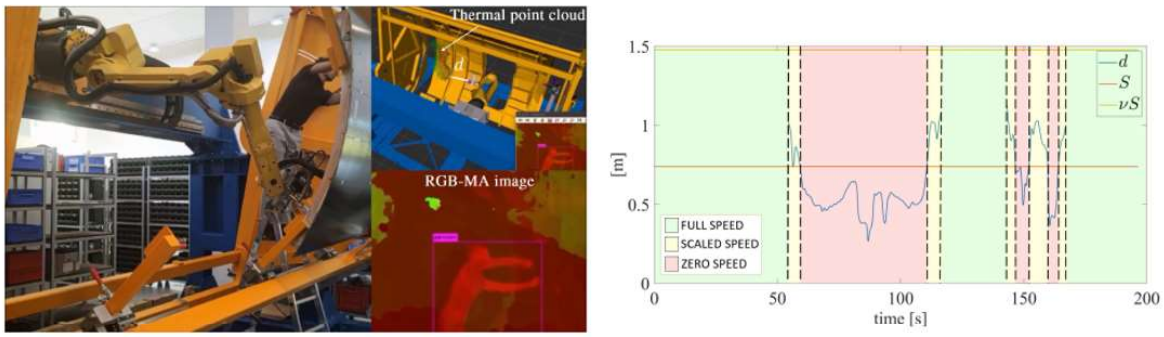


FIGURE 5.18 – Industrial Collaborative Robot.

Another identified option for the robotic arm to be implemented for the ceiling panels installation is the Collaborative robots. As presented in Chapter 2, Collaborative robots are an evolution of industrial robots and are conceived to work in the same workspace as human workers, to collaborate and reduce their strain, rather than replacing human labor. This means that collaborative robots present native technology that allows a safe collaboration with humans.

Considering Cobots, two different concepts were identified as potential for this application: the single-arm and the dual-arm collaborative robots. Dual arm Cobots are solutions that consist of two robotic arms that can work synchronized for an application, having as the most relevant differential the number of degrees of freedom. Once generally one cobot arm is present between six and seven degrees of freedom, this number is duplicated when considering a dual arm Cobot. Considering the aspects of the application that is the scope of this work, the dual arm Cobot and the number of degrees of freedom could present an advantage to facilitate the movement of the ceiling panel. Also, with the double arms, a simpler end effector could be applied for the robotic arms, once the robot can catch the ceiling panel at two different points, which may facilitate the installation.

Figure 5.19 presents a set of Collaborative robots, including the dual-arm Cobots available nowadays in the market, with the main specifications.

Manufacturers, robot models and specifications	*Manufacturers, robot models and specifications*
ABB (Switzerland) ⟅ YuMi - IRB 14000 *DOFs:* 7+7 *Payload:* 0.5 kg *Reach:* 559 mm *Repeatability:* ±0.02 mm *Weight:* 38 kg *Velocity:* 1500 mm/s	**ABB (Switzerland) ⟅ Roberta 1 / Roberta 2 / Roberta 3** ell *DOFs:* 6 *Payload:* 4 kg ‖ 8 kg ‖ 12 kg *Reach:* 600 mm ‖ 800 mm ‖ 1200 mm *Repeatability:* ±0.1 mm *Weight:* 14.5 kg ‖ 19.5 kg ‖ 30.5 kg *Velocity Joints:* 110°/s
FANUC (Japan) ⟅ CR-35iA *DOFs:* 6 *Payload:* 35 kg *Reach:* 1813 mm *Repeatability:* ±0.04 mm *Weight:* 990 kg *Velocity:* 750 mm/s	**FANUC (Japan) ⟅ CR4iA / CR-7iA / CR-7iA/L** *DOFs:* 6 *Payload:* 4 kg ‖ 7 kg ‖ 7 kg *Reach:* 550 mm ‖ 717 mm ‖ 911 mm *Repeatability:* ±0.02 mm ‖ ±0.02 mm ‖ ±0.03 mm *Weight:* 48 kg ‖ 53 kg ‖ 55 kg *Velocity:* 1000 mm/s
Rethink Robotics (Boston-USA) ⟅ Baxter / Sawyer *DOFs:* Baxter 7+7 ‖ Sawyer 7 *Payload:* 2.2 kg per arm ‖ 4 kg *Reach:* 1210 mm per arm ‖ 1260 mm *Repeatability:* ±0.1 mm *Weight:* 75 kg ‖ 19 kg *Velocity:* 1500 mm/s	**UNIVERSAL ROBOT (Denmark) ⟅ UR 3 / 5 / 10** *DOFs:* 6 *Payload:* 3 kg ‖ 5 kg ‖ 10 kg *Reach:* 500 mm ‖ 850 mm ‖ 1300 mm *Repeatability:* ±0.1 mm *Weight:* 11 kg ‖ 18.4 kg ‖ 28.9 kg *Velocity:* 1000 mm/s
MABI Robotics (Switzerland) ⟅ SPEEDY 6 / 10 / 12 *DOFs:* 6 *Payload:* 6 kg ‖ 10 kg ‖ 12 kg *Reach:* 800 mm ‖ 1384.5 mm ‖ 1250 mm *Repeatability:* ±0.1 mm *Weight:* 28 kg ‖ 28 kg ‖ 35 kg *Velocity Joints:* 145 → 275°/s ‖ 120 → 180°/s ‖ 75 → 275°/s	**KUKA (Germany) ⟅ LBR IIWA** *DOFs:* 7 *Payload:* 7 kg ‖ 14 kg *Reach:* 800 mm ‖ 820 mm *Repeatability:* ±0.1 mm ‖ ±0.15 mm *Weight:* 22 kg ‖ 30 kg *Velocity Joints:* 90 → 180°/s ‖ 70 → 180°/s
Techman Robot (Taiwan) ⟅ TM5-900 / 700 *DOFs:* 6 *Payload:* 4 kg ‖ 6 kg *Reach:* 900 mm ‖ 700 mm *Repeatability:* ±0.05 mm *Weight:* 22.5 kg ‖ 22 kg *Velocity Joints:* 180 → 225°/s	**Productive Robotics (Carpinteria-USA) ⟅ OB7** *DOFs:* 7 *Payload:* 5 kg *Reach:* 1000 mm *Repeatability:* ±0.1 mm *Weight:* 24 kg *Velocity:* 2000 mm/s
Yaskawa (Japan) ⟅ Motoman HC10 *DOFs:* 6 *Payload:* 10 kg *Reach:* 1200 mm *Repeatability:* ±0.1 mm *Weight:* 45 kg *Velocity Joints:* 130 → 250°/s	**AUBO Robotics (China) ⟅ AUBO-i5** *DOFs:* 6 *Payload:* 5 kg *Reach:* 880 mm *Repeatability:* ±0.05 mm *Weight:* 24 kg *Velocity:* 2800 mm/s
FRANKA EMIKA (Germany) ⟅ FRANKA ARM *DOFs:* 7 *Payload:* 3 kg *Reach:* 855 mm *Repeatability:* ±0.1 mm *Weight:* 18 kg *Velocity Joints:* 2000 mm/s	**Precise Automation (Fremont-USA) ⟅ PP100 - Cartesian** *DOFs:* 3 *Payload:* 1 kg *Reach:* X 635 mm - Y 300 mm - Z 225 mm *Repeatability:* ±0.1 mm *Weight:* 20 kg *Velocity:* 1500 mm/s
Kawasaki Robotics (Japan) ⟅ duAro – Dual-Arm SCARA Robot *DOFs:* 4+4 *Payload:* 2 kg *Reach:* 760 mm *Repeatability:* ±0.05 mm *Weight:* 200 kg *Velocity:* N/A	**BOSCH (Germany) ⟅ APAS** *DOFs:* 6 *Payload:* 2 kg *Reach:* 911 mm *Repeatability:* ±0.03 mm *Weight:* 230 kg *Velocity:* 500 mm/s

FIGURE 5.19 – Examples of Cobots available, (VILLANI *et al.*, 2018).

Considering yet the collaborative robots, the last identified solution option is the single-arm Cobot. There are currently more than a hundred different single-arm Cobot models in the market, presenting different payloads, reach, repeatability, and operational velocity. Single-arm Cobots were the first ones to be developed and the most used for industrial applications. When compared to dual-arm Cobots, the single ones have generally a lower acquisition cost, and present higher payload and reach capacity.

Considering the characteristics of the application that is the scope of this work, the ceiling panels installation requires a low payload capacity, being served by several models of collaborative robots in the market. Additionally, the range required for the application is also achievable by several models of Collaborative robots, considering that the robotic arm will be mounted over a mobile base.

In this sense, although there are Industrial Robots that can be used as collaborative robots since there are already collaborative robots with payload, reach, and other characteristics capable of meeting the needs of the project, this alternative did not present any relevant characteristics to be applied in this project. Another important aspect is that Cobots are designed to work alongside the operator and are naturally equipped with dedicated sensing, controlling, and measuring systems that allows a precise and safe operation. In the case of applying an industrial robot, would be necessary to implement sophisticated perception systems for safety aspects, which besides the higher implementation costs would also present a technological no-go challenge considering the space restriction inside the fuselage cabin.

When comparing the dual-arm Cobot option, to the traditional single ones, the main motivation was based on the possibility of simplifying the end effectors for the ceiling panels installation. However, these robots have lower payload and range (The ABB YuMi, for example, has only 0.5 kg of payload capacity and a reach of only 559 mm), and based on the requirements identified for the Cobot payload, only one of the current solutions available in the market could comply, considering the estimated end effector weight and payload-range capacity. This solution, which is the DSCR5, however, presents a reach lower than the required for the installation (H_F m height from the floor) and has a different architecture that not allows it to be mounted over a mobile base. On the other hand, even if the payload of the other models would be sufficient for the application, it would require a lifting system on the mobile base to address the reach requirement, increasing the cost, complexity, and operational time.

Based on the analysis performed above, the eventual gains expected by both the dual-arm Cobot and the industrial robotic arm have not been confirmed, and for this reason, the single-arm Cobot was selected as the robotic arm solution for the application. As presented, there are currently several models available in the market, among which there are models capable of meeting the payload requirements and with a sufficient range to meet the range requirements, with a mobile base without the need for an elevation system.

Once the scope of this work is at a conceptual level, the selection of the most appropriate model of Cobot must be addressed in a further detailed project phase, considering technical aspects as discussed before (such as payload, range, repeatability, velocity) and as well as financial aspects such as implementation and recurring costs. However, once the capabilities of the Cobot interfere with all the other elements of the solution (e.g.

the height of the mobile base, depends on the reach of the Cobot), the Universal Robots UR10e model was selected as a baseline for the scope of this work. The preliminary selection of this Cobot model was based on the good balance between technical capacities for this application beyond the fact that is one of the most successful Collaborative robots models in the market.

5.5.1.2 HMI Interface

One of the most important aspects of a collaborative application is the interaction between humans and robots, once above all, this communication is connected directly to operational safety. There are currently several studies regarding the development of different communication strategies and technologies for collaborative applications as presented in some publications described in Chapter 2, most of them, looking for solutions to increase the perception of safety by the operator.

Based on the studies identified, different options for Human-robot interaction solutions were identified for the application in this work, including traditional interfaces (e.g., Teach pendant, buttons, lights), body gesture recognition interface, vocal commanding interface, facial tracking-based interface, hand guiding interface and the use of smart devices (e.g., smartwatch).

Traditional interfaces of communication, such as teach pendants, buttons, and lights are the simplest communication option, which requires the lowest systems and infrastructure investments, and in almost all cases composes the Collaborative robot package. For these reasons, this solution is still the most used in collaborative applications. However, it is important to highlight the increase in interest in other types of interaction and communication interfaces, such as body gestures, hand guiding, and the use of smart devices. This result is aligned with the indicated by Villani *et al.* (2018) in which the use of more interactive ways to interact and program robots is a key enabler to allow the adoption of this technology in the industrial environment.

Body gesture recognition interface is a technology that has been gaining ground in industrial applications, mainly for the ones that require the identification and anticipation of human movements to guarantee a safe operation. However, considering the scope and type of interaction between human and robot for this work (based on the CONOPS use case) which is expected to use predefined movement trajectories, no additional benefit of using this technology was found. This is also justified because this technology can bring greater benefits in operations with a greater degree of collaboration between the human and the robot compared to the defined interaction strategy for the application of this work.

Considering the vocal commanding interface, this technology can bring higher produc-

tivity for the operating cell, once the human operator is able to communicate by voice with the Cobot. This technology has been applied to specific applications such as electronic assembly or medical solutions, however, as verified through bibliographical review, is not recommended to be used in industrial applications. This can be explained by the fact that, in most cases, the industrial environment generally presents a high level of noise, which difficult the implementation of a vocal recognition system without the risk of errors. Considering also the strategy defined by the CONOPS, and the challenges identified, this solution also was not selected as feasible for this application.

The facial-tracking-based interface is a technology that is indicated when the collaborative robot must be aware of where the human operator is focusing its attention. Similar to the body gesture one, this technology aims to bring a higher level of interaction and safety for a collaborative application. However, as verified in bibliographical research, this communication strategy has a low number of applications, mainly due to the complexity of the processing required to perform communication without errors. Considering the aspects of the application, no advantages of this technology compared to the traditional one are identified. Also, according to the CONOPs, since both the collaborative system and the human operator will be on the move (not at a fixed workstation) such technology may not be adequate, and for these reasons was not selected for the application of this work.

Hand guiding strategy is based on native technologies present in collaborative robots that allow tactile communication between human and robot in a safe condition. This communication could be applicable if the operating mode defined between human and Cobot was different, however, for the nominal operating mode, no advantages of using this strategy are found. It should be noted that once collaborative robots already present these modes of operation natively, this solution can be used in other off-nominal operation conditions. However, for the solution that is the scope of this work, this solution was not selected.

Finally, as verified in research, the use of smart devices for communication between human and robot has been widely applied. The main reason that supports the use of this technology is that the advantages of the traditional interfaces can be used with the additional benefit of mobility. In this sense, for the application of this work this technology can bring benefits considering the CONOPs and reference mission defined, once there are moments during the operation where the human operator will not be by the side of the collaborative solution, and in any off-nominal eventual situation would require a time for the operator reach the interface to interfere into the Cobot operation. For these reasons, the application of this technology in conjunction with traditional interfaces is suggested for this application, in order to allow the human operator to control the Cobot from a distance. As a result, the human operator will always have the power of controlling the

Cobot without needing to be next to the system to release/paralyze the process, which is an important aspect identified in the system requirements.

5.5.1.3 Robot Controller

The robot controller is another important element of the solution design, which is responsible for receiving, processing, and controlling all the information for the Cobot operation. Basically, the controller receives all the input data from sensors, such as the torque and velocity of the joints, and executes all the routines programmed for the Cobot.

Another important function of the controller is to execute the routines and manage the sensor's data to guarantee the safety aspect of the operation, which has generally segregated processors to guarantee. The controller of the robot also receives and processes the information from the HMI.

Considering that the recommendation presented before is to apply a collaborative robotic arm to the solution, this type of robot already contains its own controller which is part of the hardware package acquired. Despite some specific characteristics of each Cobot model, the controller from each manufacturer has the same basic function, and for this reason, the final model and specifications of the controller must be defined in a further step of the project, when selecting the collaborative robot model.

FIGURE 5.20 – Example of Cobot controller - KUKA Sunrise Cabinet, used in the LBR iiwa products line, (CORPORATION, 2023b).

5.5.2 End Effector

The end-effector is the element of the solution that basically connects the robotic arm to the product that is being handled. Consequently, the design of this component must

be adequate to allow the connection to the robot flange and to be able to correctly catch and hold the ceiling panels.

Based on the concept of operation and the requirements defined for the application, it was identified that the end-effector would be responsible for performing three main functions: Pick and place the panels; Identify the position for panel installation; and perform quality inspection of the installation performed. As a result, the architecture of the end-effector defined was based on two main components as presented in Figure 5.17, the manipulator, and the vision system

5.5.2.1 Manipulator

The design of the manipulator solutions was driven by the requirements defined for the solution. Based on the requirements, several solutions were identified through research on supplier catalogs, technical publications, and research works. Considering the basic function required for the manipulator of picking and placing the ceiling panels, the manipulator must be able to catch and hold the product during all phases of installation. Considering the characteristics of the product, two main philosophies for the manipulator were identified: The use of a fingered-based gripper or the use of a Vacuum cup-based gripper.

Fingered grippers are the most common type of manipulator used in collaborative robots, once this type of component is the most indicated for picking and placing activities, the main application of Cobots. For this type of manipulator, solutions may rely on different types of power sources to operate. One of the most common is the servo-electric grippers, which rely on electric servo motors that engage the control surfaces to accomplish the input set by the controller, for example, to catch a workpiece. The great number of applications based on this concept is based on the fact that this is the cleanest design because does not requires additional lines for hydraulic or pneumatic pressure supply.

Other types of fingered grippers may use pneumatic or hydraulic power to operate. Both options are based on the same working principle, where the documentation of the gripper is executed by a pressurized fluid, where the air is the fluid used in the pneumatic gripper. One of the benefits of this type of gripper is the velocity of moving the elements of the gripper and high force. However, these solutions require additional supply lines to provide the fluid flow. Different configurations are available on off-the-shelf fingered grippers as presented in Figure 5.21. However, based on the research, it was verified that these types of manipulators have been widely used for gripping small objects, and given the characteristics of the panel, mainly the dimensions of length and width, no suitable off-the-shelf solution was found for handling the components in question, and for this reason, it was not the chosen solution.

FIGURE 5.21 – Example of off-the-shelf fingered grippers, (ROBOTS, 2021).

The second philosophy of manipulator studied relies on suction to grab objects. This type of manipulator does not use mechanical force to catch the object but uses a vacuum to create a negative pressure that is able to hold the components through cups. One of the differentials from the previous philosophy defined is that industrial vacuum grippers can be designed to manipulate a variety of objects - from something small and delicate to a large and rigid element.

Nowadays there is a large variety of industrial vacuum-cup-based manipulators available on the market. Figure 5.22 shows some of the models studied for the application.

FIGURE 5.22 – Example of off-the-shelf vacuum grippers.

After researching different supplier's solutions it was found that the off-the-shelf vacuum grippers can meet the requirement of the minimum payload for the ceiling panels

application. However, was found a limitation regarding the dimensions of the contact area. Even in those models with an extension system, the maximum distance between handles is about 400 mm, as can be seen in Figure 5.23.

FIGURE 5.23 – Dimensions of an off-the-shelf vacuum gripper

The maximum dimension verified in the off-the-shelf manipulators was analyzed for the ceiling panels applications. The first aspect considered is that the Cobot will be responsible for the installation of the ceiling panel, wich has 3 fixings along the length, one in the center and two near the ends. Additionally, the ceiling panel is not a rigid object, presenting some flexibility that in this case, depending on the type and position of handling can compromise the correct assembly. For this reason, it was verified that the manipulator should be able to grab the ceiling panel near the points of attachment in the aircraft structure, requiring a longer distance between the handles than provided by the identified off-the-shelf vacuum grippers. As a result, it was not the chosen solution and consequently, it was verified the need for a customized manipulator to be developed for the application.

Considering that a customized vacuum effector was for the application, a conceptual design was identified. The main drivers considered were related to the dimensions to comply with the size of the panels and guarantee the feasibility of the assembly, positioning the vacuum cups near the attachment points of the panel. Also, the number and dimensions of the vacuum were preliminarily defined to assure the force capacity for holding the ceiling panels. As a result, Figure 5.24 shows the concept defined.

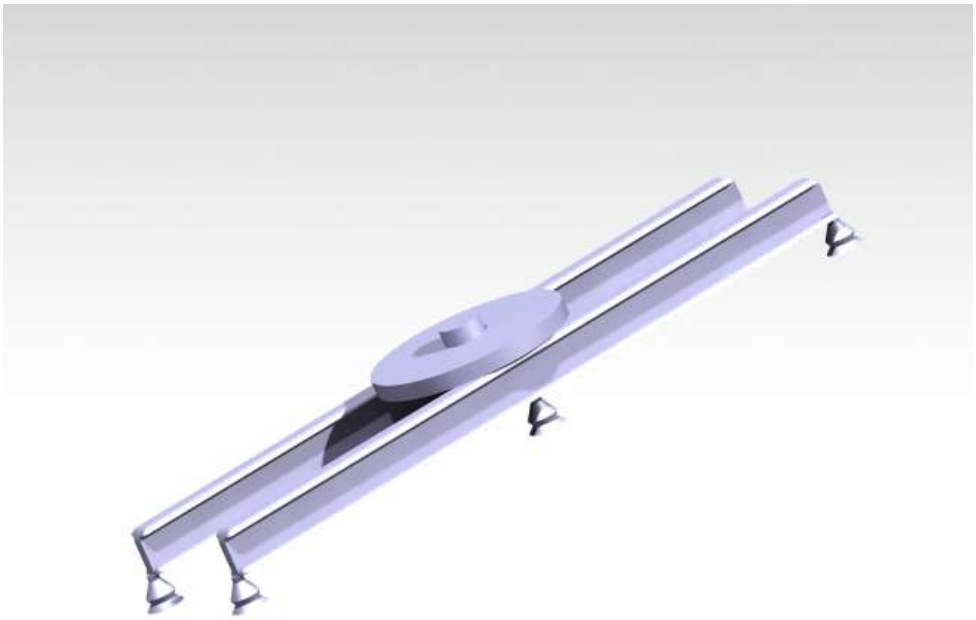


FIGURE 5.24 – COncept of vacuum cup end effector for ceiling panels installation.

Considering that the manipulator would be a customized solution, a solution using the fingered mechanical philosophy was also studied. The development of an effector with a handle that adheres to what is necessary given the dimensions of the ceiling panel would be possible and would enable the use of the vision system necessary for the effector. However, in this application, the imagined concept that would be viable would be similar to a vise, which would take the panel at its borders to assemble. The concept is presented in Figure 5.25.

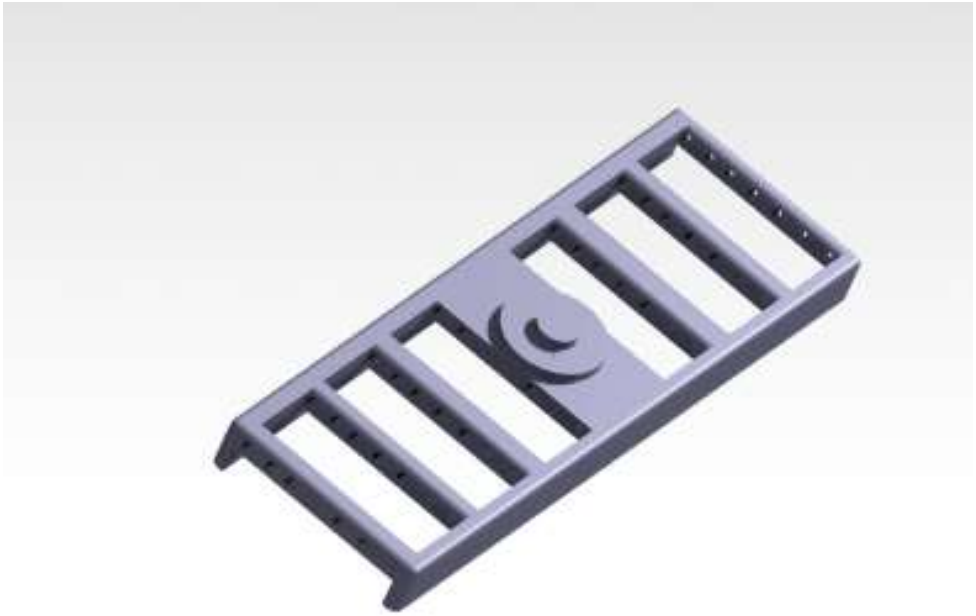


FIGURE 5.25 – Conceptual end effector, similar to a wise

In this sense, compared to the vacuum one, it appears that this solution would have a wider effector, which is not a positive aspect given the limited space inside the fuselage. Furthermore, it could come into contact with the roof side panels, and eventually damage them during the assembly procedure. Finally, ceiling panels are surfaces exposed to customers, and with this type of effector, there could be damage to the quality of the panels. For these reasons, this was not the solution used.

Considering the characteristics of the product, the use of an effector with a vacuum system makes a lot of sense. With a specific design, all requirements can be met, including a larger dimension to apply the force close to the installation position, integration with a vision system, and smaller dimensions than the custom gripper. For these reasons, it was the chosen solution.

5.5.2.2 Vision System

Considering the envisioned mission defined in the CONOPs additional functionalities were identified leading to the need for a vision system for the end-effector. The vision system is required for two main functionalities: the identification of the correct position to install the panel; and the inspection of the assembled panel to assure gap and step quality. Three options for the vision system were identified based on supplier catalogs, technical publications, and research works.

The first identified solution is the use of a system composed of a 2D camera and laser

sensors. This system is widely used in various applications such as sealant application, drilling, and riveting solutions. In this case, for the panel installation, the 2D camera is used to identify the installation points of the ceiling panel in the xy plane, while the laser sensors are used for the measurement of the distance and the angular components. These measurements are used for the fine controlling of the Cobot, feedbacking the controller, and allowing the correct installation. Additionally, laser sensors are used to scan the assembly quality. Figure 5.26 presents an example of the first solution application.

FIGURE 5.26 – Vision system example, (GMBH, 2022)

The main advantages of this solution are the high precision in positioning, an aspect very important for the installation procedure. However, once the system is required to perform quality inspection and verify gap and step alignment, the 2D camera with a laser sensor may not be enough.

Another option identified for the application consists of the use of a most sophisticated camera, with 3D capacity. In this case, the camera system does not require a laser system once the camera system is able to measure in three dimensions. Furthermore, the use of a 3D camera increases the capacity of the system to perform a quality inspection after the installation procedure, once the system would be able to capture images for gap step measurement.

This second option seems an interesting solution, once requires a lower number of components. However, for the assembly operation, the system relying only on image capture may lead to a loss of accuracy when compared to the system with laser sensors. This fact motivated the study of a third alternative, which relies on the application of a 3D camera with laser sensors.

The application of a 3D camera in conjunction with laser sensors composes a more robust solution to the application. In this case, the use of a 3D camera allows a better quality analysis of assembly quality, while the system maintains the accuracy level for the panel installation due to the application of the laser sensors. It is important to highlight that this recommendation is based on a conceptual level of the solution design, where based on benchmarking, suppliers' information, and research studies the selected

solution is the most robust and will comply with the identified requirements. However, with detailed studies, it may be possible to use the first or second options for the system aiming to reduce the system costs.

5.5.3 Mobile Base

5.5.3.1 Mobility System

One of the bases of the solution system is the capability of moving inside the aircraft cabin to perform the ceiling installation. As defined in the system's architecture, the mobile base is composed of different components in which the mobility system is the main element that connects all the others, being responsible for supporting the Cobot and for allowing the movement of the solution.

The implementation of mobile robots and in this case mobile collaborative robots in industrial applications is not a recent event, since a considerable number of current applications can be verified. Through a search of published works, supplier catalogs, and technical publications four main concepts were identified as potential for the mobility system: Off-the-shelf AGV, customized AGV, Customized Movable manual powered base, and Customized Movable motor-powered base.

The first option considered is the use of an off-the-shelf AGV as the mobile base solution. There are several models available in the market from low to high payload capacities and as can be verified in Figure 5.27, currently, there are options available that already are developed to work with Collaborative robots, the case of the KUKA KMR iiwa, for example.

FIGURE 5.27 – Off-the-shelf AGV already composed with Collaborative robots, (CORPORATION, 2023a), (KYASKAWA, 2023)

Considering this solution option, the main advantages verified are that the solution already contains all sensors and software to allow the correct movement and operation.

Also, the solution in most of the cases already has built-in interfaces for integration with the Cobot system. Finally, once is an off-the-shelf solution, no development costs are required for the solution.

However, because the solutions are already developed, and ready to be used in different applications the dimensions and capacities are standardized (not able to adapt according to specific application requirements such as space restrictions). Beyond that, once the solution is off-the-shelf, several built-in systems not required for the current application already make up part of the solution and will not be able to aggregate value. As a consequence, this solution presents a high acquisition cost that can compromise the feasibility of the application.

Considering these aspects presented, a possible second option was identified, being a customizable AGV. This solution was defined based on the scenario of customizing an existing AGV to comply with the characteristics of the application. In this case, the solution already contains all sensors and software to allow the correct movements, and the dimensions and capacities can be adapted to address specific application requirements, as the example presented in Figure 5.28. As the base of the solution is already existing, the acquisition of a customized AGV can have a shorter lead time compared to the Customized Moveable base solutions. It's important to highlight however that the customization process of one off-the-shelf AGV brings additional development costs, leading to the probably most expansive solution at all. Furthermore, this solution is dependent on the engagement of an AGV manufacturer as a project partner, interested in developing the customization required for the application.

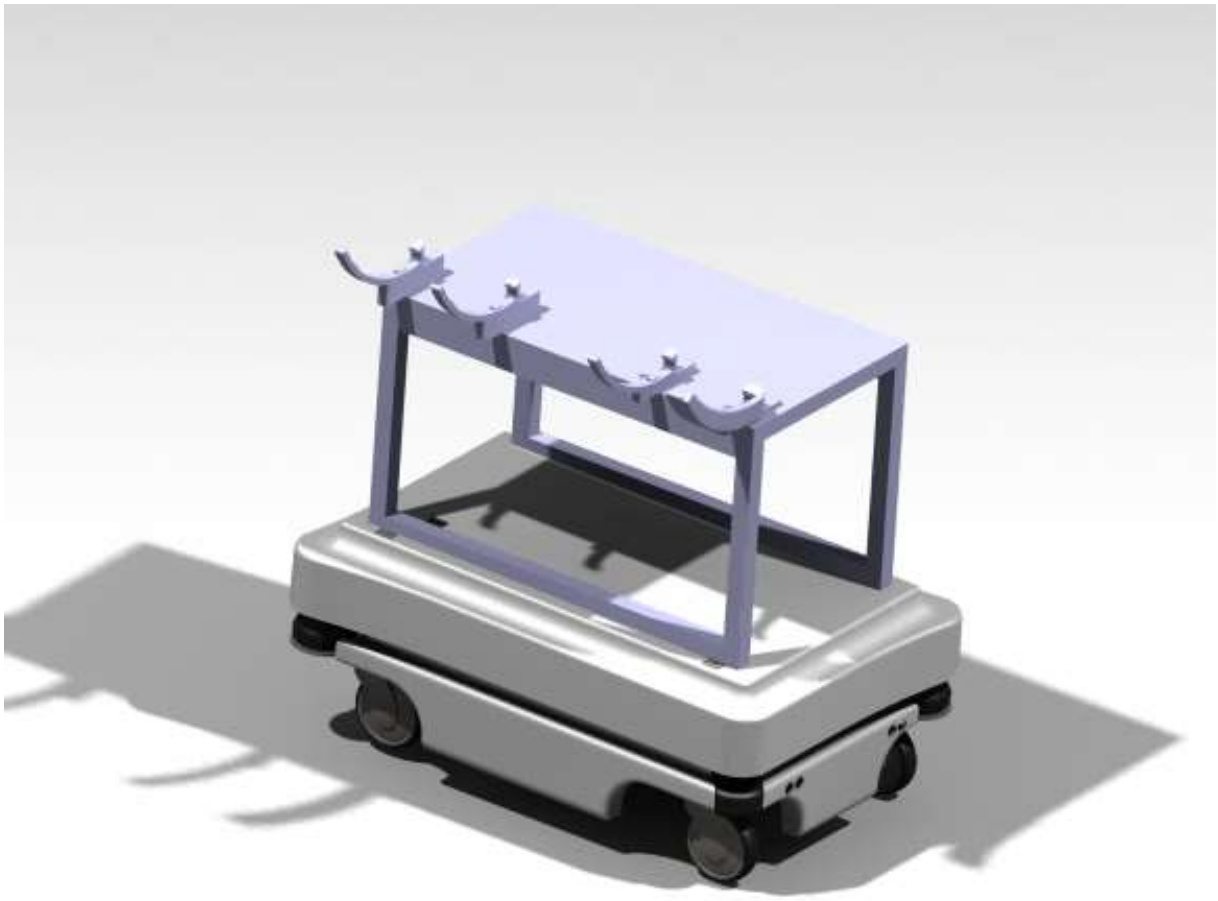


FIGURE 5.28 – Concept of customized AGV.

These characteristics of the previous solutions motivated the search for different alternatives for the mobile base. Additionally, some characteristics of the application encouraged the seek for simpler solutions. First, different from applications on the shopfloor such as logistics, machine feedings, etc., the ceiling panels installation is performed inside the aircraft cabin, which is a restricted space, and in this case the system movement is practically unidirectional along the length of the fuselage, and therefore, the mobile base does not need to present all the movement capabilities of an AGV. Second, the central part of the fuselage floor, where the solution will be moving to perform the installation, have seat tracks that could be used as a guide for the solution movement inside the aircraft.

Considering these characteristics two additional solution alternatives were defined, both based on the development of a customized mobile base simpler than an AGV. In this sense, the third alternative identified was a Customized Movable manual-powered base, which is basically a mobile base to support the installation of a Collaborative robot, that is moved and guided manually by the human operator, presented in Figure 5.29.

The main advantage identified for the solution is that being a customized mobile base,

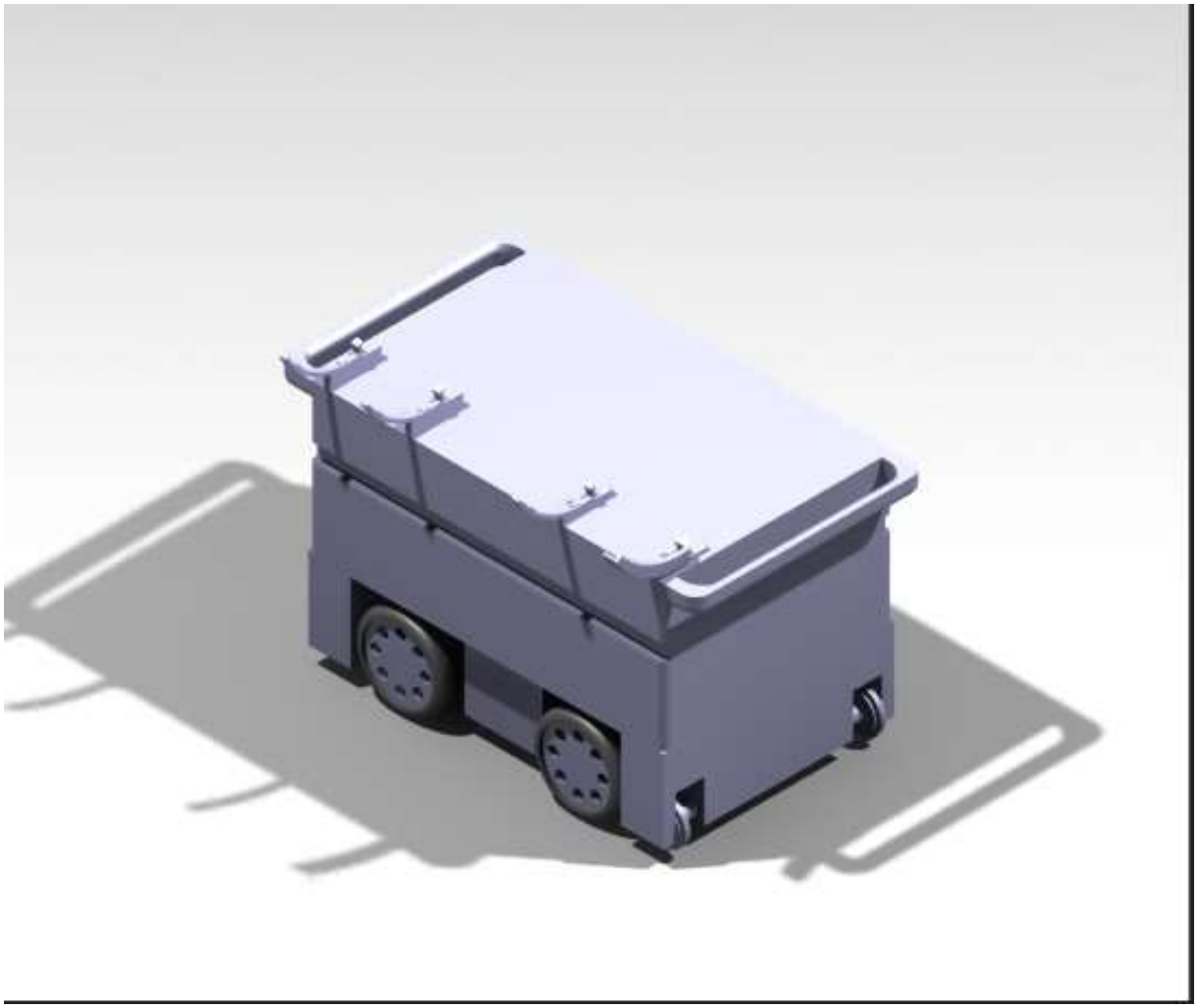


FIGURE 5.29 – Customized Movable manual-powered base

only the sensors, software, and elements necessary for the application would be required to be integrated into the solution. Also, the dimensions and capacities can be adapted to address specific application requirements. Finally, considering the application characteristics, the solution can be simpler than an AGV, consequently presenting considerably lower implementation costs, with less implementation complexity (set-up) and lower maintenance complexity and complexity.

This solution however brings some disadvantages. Once the solution is customizable, requires development costs as well as the development of this solution depends on the availability of intellectual resources to be performed. However, the main detractor of this solution is the fact that a hand-powered base can lead to non-ergonomic operations for humans due to the weight of the solution, a negative fact once one of the main motivations of the application of the reduction of the human workload. Additionally, manual handling of the base can lead to imprecise base positioning, precluding the completion of the operation. Once the human operator is responsible for the base handling, non-smooth movement can occur, leading to Cobot loss calibration issues.

Based on these negative aspects, the fourth solution was identified and is presented in Figure 5.30. This solution is an evolution of the third alternative however is based on motor power to move. Basically, the same benefits from the manually powered base can be verified in this solution, such as lower implementation costs, with less implementation complexity (set-up) and lower maintenance complexity and complexity compared to the options based on AGVs. However, the motor-powered base can solve non-ergonomic issues present in the hand-powered solution, as well as the motor-powered solution also allows more precise positioning, solving the potential issue verified in the hand-powered solution. As the main challenges regarding this concept, it can be described as the fact that as well as the third solution, this solution brings development costs, and the development of this solution depends on the availability of intellectual resources.

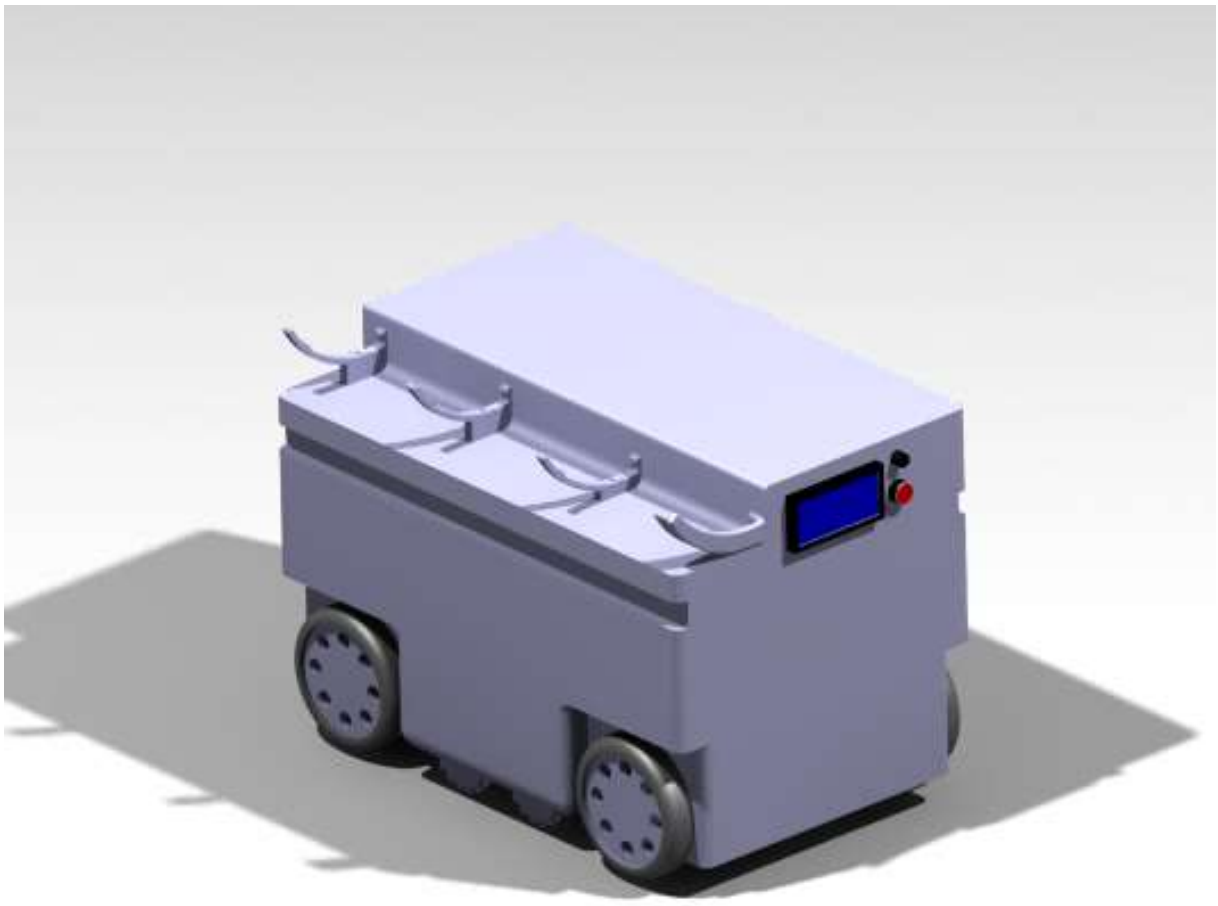


FIGURE 5.30 – Customized Movable motor-powered base

Considering the aspects discussed above, a comparative matrix was defined to help with the definition of the recommendation. For this analysis, five criteria were defined, and each solution was evaluated through the criteria with scores from 1 to 4. The criteria defined were:

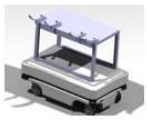
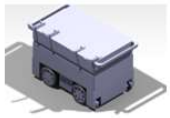
- Solution complexity: This criterion accounts for the level of complexity of the solution, encompassing the number of elements and systems, where the simplest is the

best (grade 4) and the most complex receives the lower value (grade 1).

- Alignment with operational needs: This criterion aims to evaluate if the solution architecture is aligned with the required characteristics of the application. The more the concept is aligned with the application characteristics, the higher the grade (tending to 4) while lower alignment receives a lower grade (tending to 1);
- Ergonomics: This aspect is focused on the evaluation of the ergonomics aspects influenced by the differences between concepts, where the lower values are for those solutions with worst ergonomics influences while the higher grades are for those that better ergonomics influences;
- Operational simplicity: This element was defined to capture the complexity of operating the different concepts. The higher complexity of operation receives lower grades, while solutions simpler to be operated receives higher grades (tending to 4);
- Solution costs: once the cost is an important aspect to be considered in the solution's design, this criterion was defined to allow the evaluation based on a qualitative analysis considering the characteristics of each solution. Additionally, to better capture the difference between the concepts, this element was broken into three sub-criteria, which are the acquisition, development, and Maintenance costs. For the evaluation, lower the costs higher the grades tending to 4.

Based on the characteristics exploited and discussed before, the solutions were evaluated, and the result is presented in Table 5.5. For the evaluation, the same weight was considered for the five criteria.

TABLE 5.5 – Mobility system trade-off.

					
		Off-the-shelf-AGV	Customized AGV	Customized Movable manual-powered base	Customized Movable motor-powered base
Solution complexity		1	2	4	3
Alignment with operational needs		4	4	3	4
Ergonomics		4	4	1	4
Operational simplicity		2	2	4	3
Solution Costs	Acquisition	1	2	4	3
	Development	4	2	3	2
	Maintenance	1	1	4	3
TOTAL		2,6	2,7	3,1	3,3

As can be verified the solution identified as the most indicated for the application is concept four, which is the Customized moveable motor-powered base. The result verified is compliant with the aspects discussed before and is most influenced by the possible

lower implementation costs when compared to AGVs while solving all the main problems verified in the manual-powered solution.

It is important to highlight that this recommendation is based on the CONOPs and use case defined for the ceiling panel installation and considers the data maturity with a conceptual phase level. Two main aspects must be refined in a further detailed phase for the confirmation of the recommended solution for the mobile base, which are the implementation costs and the availability of intellectual resources required for developing the detailed project of the solution.

5.5.3.2 Panel Support Structure

One important function of the mobile base also allows the positioning of the ceiling panels, allowing the Cobot to catch them and perform the installation operation. Two main design options were identified as possible for the solution, considering the mobility system defined and the other elements of the solution.

The first option considered is the panel support integrated with a mobile base. In this solution, the panels are positioned in the same structure that composes the mobile base, in a position that allows the Cobot to correctly catch the panel. However, in this solution, there is space for a maximum of two panels to be stored at the same time in the mobile base, requiring the human operator to be responsible for bringing the panels from the supply position in the back of the fuselage to the mobile base.

The second option evaluated is based on the use of an additional chart, connected to the mobile base, for panel support. In this configuration, there would be possible to eventually have a chart to store all the panels to be assembled, avoiding the need for these panels to be moved by the human operator. However, this option would require an additional device to be developed (the chart), increasing the costs of the solution. Additionally, as commented during the development of this work, the fuselage cabin is a restricted space, mainly in the stage of ceiling panel installations, where the overhead bins are already installed, and in this case, there would be an additional space being occupied by the chart.

It is important to highlight that, the scenario designed for the application in the CONOPS considers the human operator as responsible for carrying out one stage of the panel installation process (the panel's power supply cable connection). In this case, the utilization of the human operator to bring the panels to the mobile base makes sense once the activities performed for both Cobot and the human operator can be scheduled to occur synchronized, increasing the occupation of the human operator and consequently the efficiency of the workcell.

Based on the discussion above, the solution chosen for the application was the panel support structure integrated within the mobile base. The aspect that leads to this solution is no need for an additional element for the solution, reducing costs and the footprint inside the aircraft cabin, as can be verified in Figure 5.30.

5.5.3.3 Power Supply System

As part of the mobile base, the power supply system is responsible for providing all the energy required by the solution's operation. As defined in the end-effector section, the manipulator will require airflow for panel handling using a vacuum-based solution. For this reason, the two power sources for this operation are the electrical power and the power provided by the airflow.

Two solutions were identified as potential for this element of the solution design. The first is based on the use of batteries to provide electrical power, while in the second option, the electrical power is provided by power cables. For both, the airflow is also considered as a power source.

The system powered by batteries has the great advantage of not needing a connection via cables to carry out activities. However, the solution using batteries tends to be more expensive reducing the payback, and heavier, making it more difficult for the insertion in the fuselage cabin. The KUKA KMR iiwa platform, shown in Figure 5.27, has a weight of almost 400kg due to the batteries, (considering this weight is significant inside the aircraft, and also to lift given the height it has to be lifted).

As explained one of the advantages of the use of batteries is that no cable connections are required, which is an important aspect for restricted areas such as the aircraft cabin. However, considering the application characteristics where the Cobot motion is practically unidirectional inside the aircraft cabin, mobility would not be impaired using cables when compared to an application on a factory floor, for example.

Another important consideration is related to the autonomy of the batteries. As verified on existing platforms, the batteries have autonomy of between 4 and 8 hours, with about 1 to 2 hours for recharging. Depending on the production rate, this means that the autonomy is only 1 work shift, and to guarantee the operating capacity, there would be required more than one Collaborative solution to allow the required continuous assembly process, since the manufacturing cycle is higher than the battery autonomy.

Given the operating scenario, the use of power cables tends to be the simplest and cheapest solution that allows for a continuous operation of the system. Furthermore, once it is necessary to provide airflow as a power source, the use of power cables is aligned with this requirement once is equally provided by flexible elements that must be connected to

an external power source.

Despite the use of batteries for this type of system being a trend given these application conditions, the initial indication is to use power supply cables for electrical energy power, to allow the addressing of the continuous operation. Considering that the proposed mobile platform is simpler than an AGV, the autonomy of the system using batteries may be higher than the off-the-shelf mobile base solutions identified. In this context is important to perform a detailed cost and autonomy analysis for the project in a detailed design phase, to confirm the conceptual definition.

5.5.3.4 Localization System

An important part of the mobile base is the localization system, which is responsible for providing the information required for the correct positioning of the base inside the aircraft cabin. The positioning of the mobile base is essential to assure that the Cobot is able to perform the installation of the ceiling panel, mainly due to the Cobot's reach.

Different solutions were identified for this component, and the main are presented in Figure 5.31. Magnetic Localization, for example, is a technology that has been used since the first generations of AGVs (Automated Guided Vehicles), in which vehicles follow the path indicated by tapes or magnetic points installed on the floor. Beyond the fact that it is a technology that has already been tested and matured, is reliable and competitive prices. However, considering the scope of this application, is not understood to be an interesting technology since it involves the installation of components on the floor of the aircraft, to allow the movement of the floor leading to an additional laborious activity to both install and remove the magnetic elements. Furthermore, it would increase the risk of occurrences of FOD (Foreign object detection) and eventually quality escapes in the assembly process.

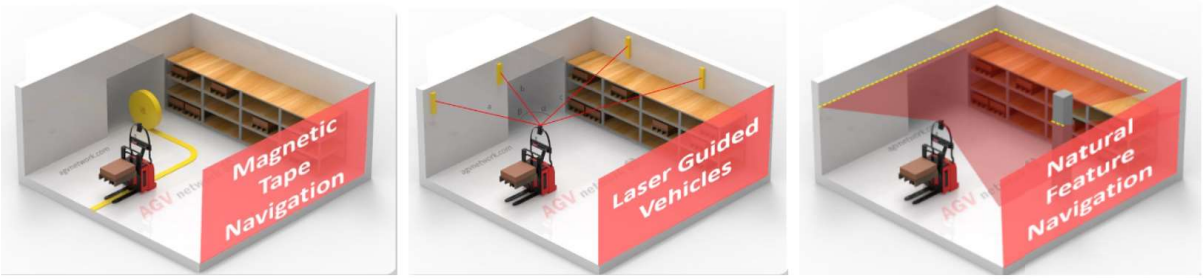


FIGURE 5.31 – Localization systems for AGVs and AMRs, (TELLA, 2023)

Another technology identified is laser localization, using laser reflectors. In this tech-

nology, the vehicle has a system that emits laser beams that impact reflectors installed in the appropriate places. Based on the reflection of the laser the AGV can locate itself. This technology is also used in various applications for AGVs, and is highly recommended when speed and accurate tolerance are required, and also for multiple destinations and tracks. However, Like the previous one, this technology involves adding elements inside the aircraft, which is not an interesting solution. In addition, the fuselage is a space with restricted dimensions and could have elements that would end up impacting the transmission of the signal. Due to this tense difficulty and the need for additional infrastructure, this was not the chosen solution selected.

The third identified technology is Free localization, the most common technology used nowadays in AMR (Autonomous Mobile Robots) vehicles. Different from previous solutions, these systems do not need any kind of hardware to be installed in the plant, or in the fuselage on the scope of this application. The system tracks the environment through different technologies such as contour, SLAM (Simultaneous Localization And Mapping), LiDAR (Light Detection and Ranging) and define their path thanks to the info received from the surrounding environment.

Most of the AGVs apply LiDAR navigation technology, in which the distance from targets is measured by the analysis of the illumination and reflection of laser lights. Basically, the system emits a laser that reflects on the objects, and the difference in laser return timing and wavelengths is used to create digital 2D or 3D representations of the space.

Although, currently, the base for free navigation relies on SLAM technology, which is based on calculations using complex algorithms that can translate the information collected by different sensors into localization information for the mobile vehicle. The process of localization using SLAM technology starts with Landmark Extraction, where the AGV navigates and maps the environment. Then, the data received is compared with digital information provided about the environment (e.g. CAD plant of the factory), and based on the information the AGV estimated where is located in the real world. This data is collected during the operation and feedbacked to the vehicle.

Free navigation is a very robust solution and the focus for further rent development of AGVs and AMRs. However, despite the robustness and the fact that this technology does not require the installation of additional equipment, it would not be necessary for the application, since space is confined within the fuselage and the movement of the mobile base is limited. Furthermore, as defined in the mobility system section, the mobile base will use the fuselage floor tack as a guide, leading to an almost unidirectional movement during the panel installation inside the fuselage. In this way, this technology presents much more than what is necessary for the application. And because it has all the features, it obviously has an additional acquisition cost.

Since the solution for the base was the motorized base, considering the previously discussed solutions, a simpler technology would be enough to guarantee the location for navigation. In this case, the use of a sensor that allows referencing the motor rotation with its displacement as an encoder would be enough to guarantee the location. From a reference point, base movement can be programmed to move as needed to the next ceiling panel mounting station, considering connecting the base to the seat rail provides precision movement. This solution is simpler and cheaper than the previous discussed, attending the requirements defined for the application and is aligned with the definition of the mobility system. For this reason is the solution selected for the localization system. meets the need to move the base.

5.5.3.5 Sensing System

A sensitive system is one of the most important safety systems used in mobile robots, once allows the detection of elements nearby the robot and consequently collision avoidance. Considering that the mobile base is also a mobile base, a sensing system was required to guarantee safe collaboration with humans inside the aircraft cabin (no collision with the base).

Based on research on supplier catalogs, technical publications, and research works, two main options were identified as potential to be used as sensing systems. The first is the application of a laser sensor system that is able to identify any obstacles in the proximity of the mobile base and send feedback to the controller to perform the correct action to assure no collision. This system is the most used of all types of AGVs and AMRs because is simple and allows correct identification and information processing. Figure 5.32 shows an illustration of the sensing system present in a commercial AGV.

The second identified possible solution is the application of a camera system to work as a safety system. In this case, a set of cameras positioned in adequate positions around the mobile base collects visual information that is processed and used as input for collision avoidance. Despite being able to perform the system function, this solution has a higher acquisition cost and requires more processing to report the necessary information when compared to the laser system. For these reasons and because it does not present any additional advantage verified in comparison to the laser system, the laser system was the solution chosen for this element of the system architecture.

FIGURE 5.32 – Example of laser sensors used in AGVs,(WORLD, 2019)

5.5.4 Control System

5.5.4.1 System Controller

One important part of the system is the system controller, which will be responsible for connecting all the other systems to work in a synchronized and safe way. The system's control architecture is not the scope of this work once composes the detailed phase of the development, however, based on the systems architecture, a general view of the control system unit and its connections is presented in a draft scheme in Figure 5.33.

Considering that the Collaborative robot already contains a native control system, the solutions control system is responsible for connecting the Cobot system control to the other elements of the solution. The Cobot, in turn, is responsible for the integration of the data with the end effector mainly related to the vacuum and vision system.

Furthermore, the control system is responsible for integrating the HMI and Safety system to execute the human command and assure the safety of the operation. Finally, the control system is responsible for the control of the mobile base, including the processing of the motion and localization system, as well as the sensing system.

The conceptual control system structure presented was developed considering the characteristics of the application, and based on different works such as Schlette *et al.* (2020), Bejarano *et al.* (2019), Wedin *et al.* (2020), Papanagiotou *et al.* (2021), and Dimitropoulos *et al.* (2021). Furthermore, as verified in most of these cited works, this control system could be developed considering the application of ROS (Robot Operating System) frame-

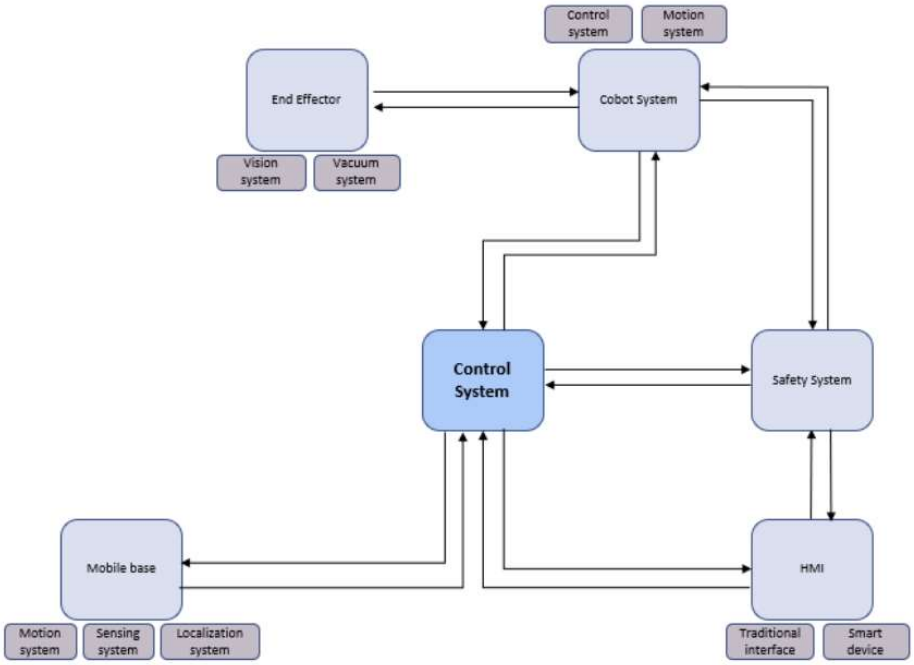


FIGURE 5.33 – Scheme of a conceptual control system architecture for the solution.

work.

As indicated before, a deeper discussion related to the control system is not part of this work. However, the schematic presented in Figure 5.33 serves as an initial guide for the system definition, including the main connections and information exchange required.

5.5.4.2 Collaborative Safety System

The most important concern in implementing collaborative solutions is safety. Despite collaborative robots being able to operate in collaboration, and sharing the workspace with humans, some applications require a deeper study of safety aspects to allow the operation.

The installation of ceiling panels inside the aircraft cabin involves the operation of the Cobot in a restricted space, with the end-effector operating at a height above the human operator's head. Furthermore, to allow the correct installation of the ceiling panel, the end effector defined previously has a considerable dimension that increases the workspace of the Cobot when operating. As a result, it can be verified that safety is an important aspect and will require special attention in this application and motivated the definition of this component in the systems architecture.

Based on the application characteristics detailed by the requirements and the operation described by the CONOPs it was defined that the solutions will operate according to the SSM (Speed and separation monitoring) operating mode. In this operating mode, the Cobot reduces or increases the speed when the human operator enters or leaves the

workspace.

To allow the operation of the solution in the SSM mode a system must be defined to measure and verify when the human operator enters the shared workspace. Also, this system must communicate with the Cobot controller to reduce or increase the operating speed according to the monitoring of the workspace.

Different options for the collaborative safety system were identified through research on technical publications, and research works. The first option identified is the use of Laser scanners, which is the simplest technology implemented for this purpose, which uses laser sensors to delimit collaborative spaces. Several applications consider the use of this technology, and Figure 5.34 presents two examples.

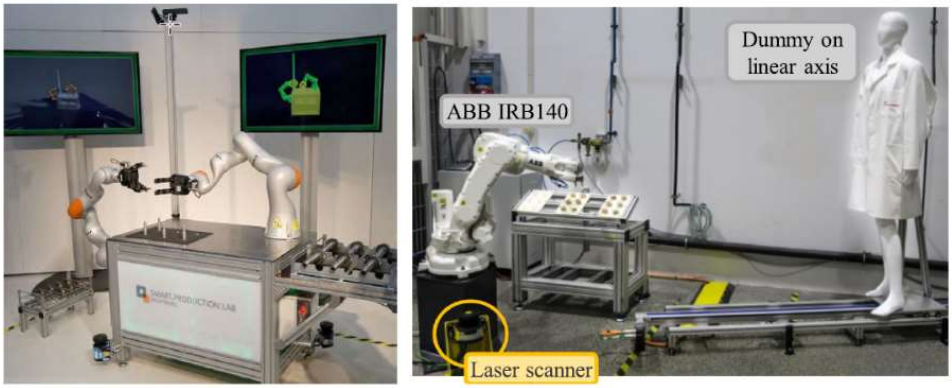


FIGURE 5.34 – Laser scanners safety systems. (FERNÁNDEZ *et al.*, 2017) (left), (BYNER *et al.*, 2019) (right).

One important point regarding this technology is that because of its simplicity, usually leads to a larger operating space whose Cobot speed is reduced. Therefore, the Cobot may be operating at reduced speed for longer times considering that the fuselage cabin is a restricted area, and the probability of the human operator entering the operating space is high. In this case, the productivity of the solution may be compromised.

Solutions that use camera systems are a trend for this type of application, due to the greater precision and ability to differentiate between moving objects and humans (for example between a human and an AGV). Due to the higher precision, in this system, the operating space can be reduced which is a positive aspect of the application in the study. In this case, the use of a 2D Thermographic camera plus a Depth camera or RGB-D camera is the second option identified. In this case, there is a need to use image processing which often requires the use of artificial intelligence and image fusion algorithms to allow the correct identification of the humans and objects in the workspace, increasing the implementation costs and complexity. Figure 5.35 presents an example of the application of this technology.

A third alternative identified is the use of a Stereo Camera with a "ToF" (Time-

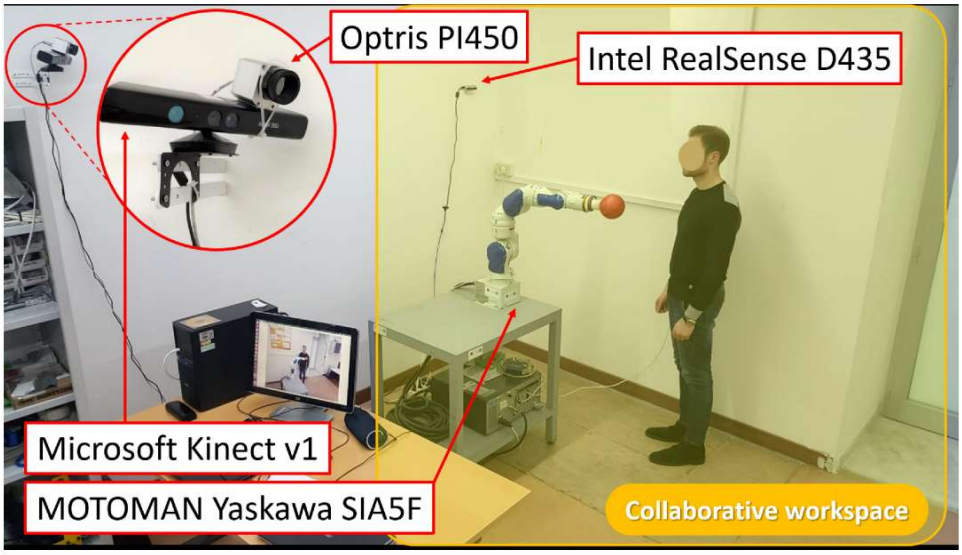


FIGURE 5.35 – Camera-based safety systems, (COSTANZO *et al.*, 2021)

of-Flight) Camera. This solution also uses cameras and shares the same robustness and positive aspects, being the subject of several studies and applications. Figure 5.36 presents an example of the application of this technology.

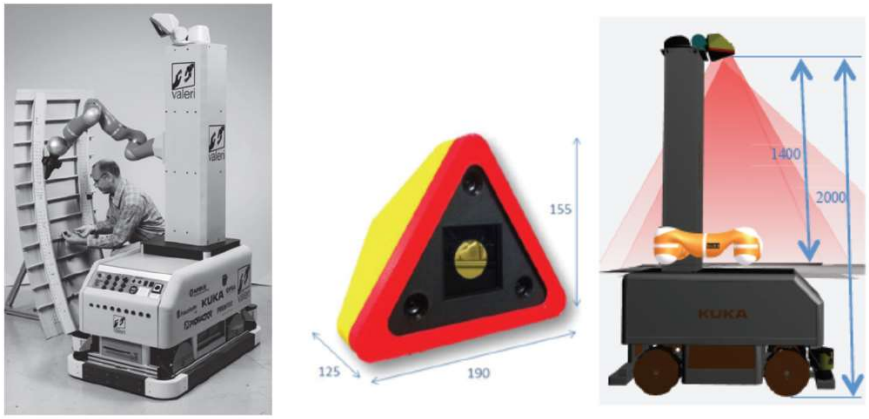


FIGURE 5.36 – Time-of-Flight and stereo cameras safety systems, (SAENZ *et al.*, 2017b)

However, similarly to the previous solution, the use of this application requires the installation of the cameras at a certain height to allow for capturing images and defining the operating space. It should be noted that this would require a structure attached to the base of the Cobot for positioning the cameras, limited by the height of the plane's entrance.

This structure should be installed in the mobile base to allow the use of this technology. However, in addition to hindering operator access, can limit Cobot movements, which is not attractive for the application.

Looking for a solution that does not compromise access or limit the movements, a

fourth option was found. This solution is based on the installation of "ToF" laser sensors mounted on the Cobot, as presented in Figure 5.37.

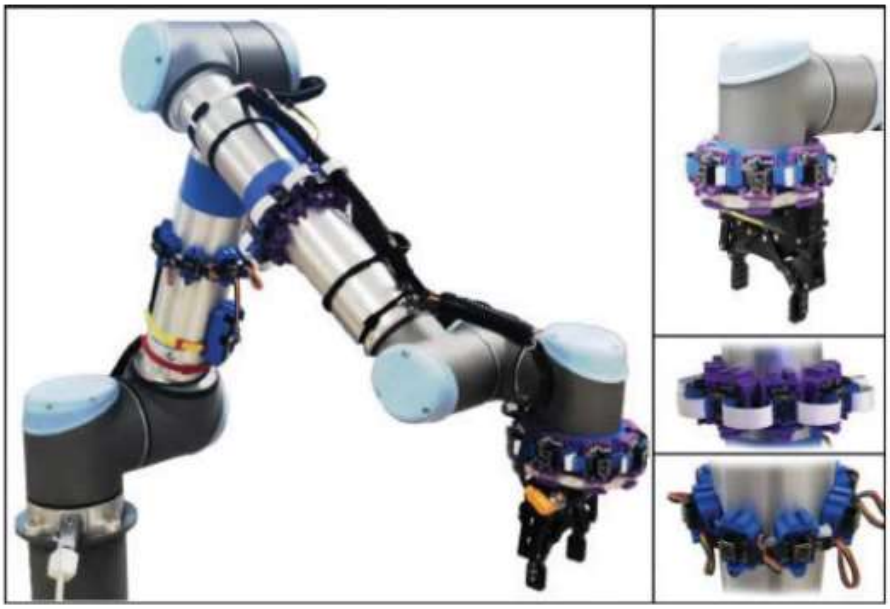


FIGURE 5.37 – UR10 robot with time-of-flight sensor arrays mounted on the robot link centers, used to perform SSM monitoring, (KUMAR *et al.*, 2019)

As can be seen, the great advantage of this solution is that the sensors are connected directly to the Cobot's links not requiring any additional infrastructure to be mounted. Furthermore, an even greater benefit is that it allows for a reduced operating zone where the Cobot must reduce the operating speed since the sensors are attached to the collaborative robot. Also, safety can be increased once all Cabot's links are monitored, allowing higher proximity between the robot and the human operator.

Considering the interior of the aircraft is a restricted space, this solution tends to greatly benefit the productivity of the application, and that is why it was the recommended solution. It is important to highlight that the development of the collaborative safety system is one of the most important elements of the Collaborative panel assembly system, and the final selection of the solution for this component will occur during a detailed phase of the project. However, the indicated solution presents great potential to be used for the scope of this project.

5.6 Conceptual Solution

Based on the discussions presented in the last sections, it is possible to identify in the morphological chart presented at the beginning of this chapter, the alternatives selected. Figure 5.38 retakes the design alternatives identified based on the system's architecture,

highlighting the components selected to compose the solution for the ceiling panels installation.

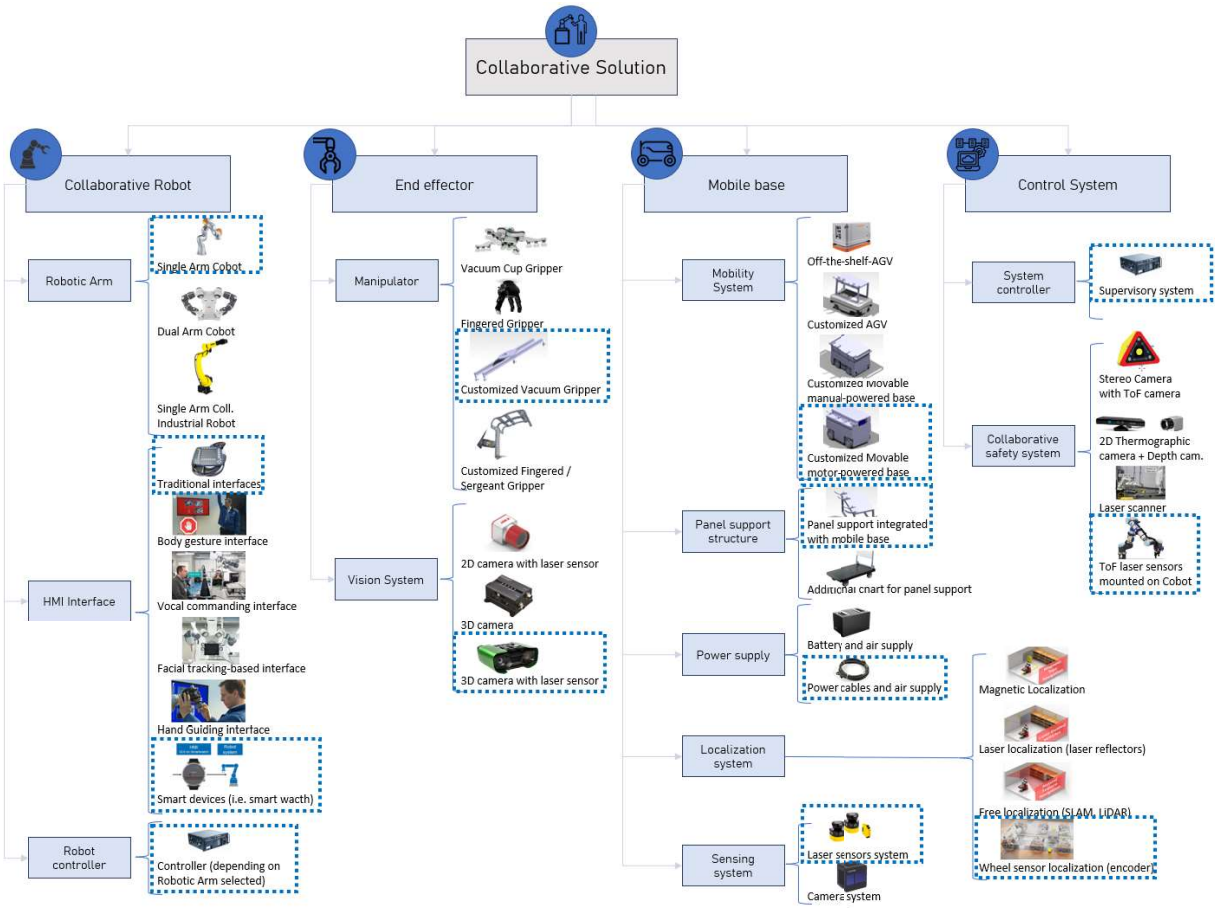


FIGURE 5.38 – Solution architecture with selected components.

Resuming, the defined solution concept is composed of a customized movable base, motor power-driven, to which the other elements of the solution are connected. Mounted on the top of the base, is the Collaborative robot which is responsible for the manipulation of the ceiling panels. Connected to the Cobot, there is the end effector which is a customized solution based on vacuum cup technology and cameras to allow the correct assembly process.

The mobile base is also the element on which the control system is mounted. Additionally, as defined, the mobile base design accounts for supporting the placement of the ceiling panels, where the Cobot catches it using the end effector.

To allow the definition of the conceptual solution, the components were modeled in the CATIA V5® software, in which the design of the components was updated according to the evolution of the studies and the verification of the requirements. Additionally, the solutions were simulated using the CopeliaSim® software, where the operation of the solution was simulated to verify aspects such as clearance, Cobot movement, and reach, and allowed the verification of compliance with some requirements, especially those related

to restrictions collected in the stakeholders' analysis phase. As a result, Figure 5.39 shows the final conceptual design for the solution described above.

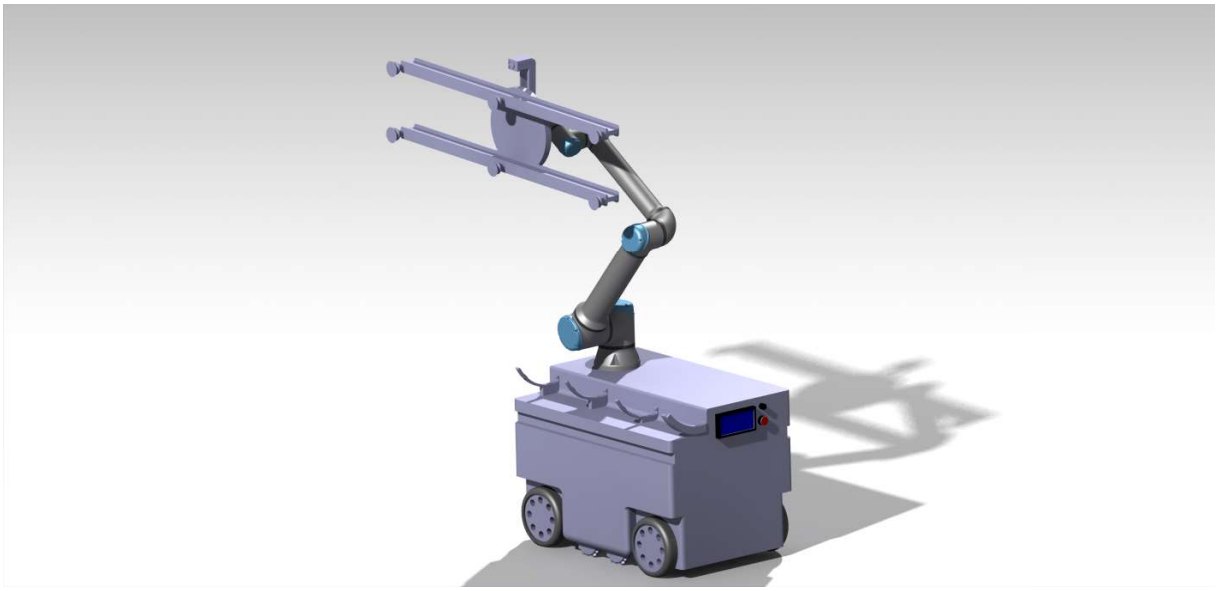


FIGURE 5.39 – Conceptual solution for the ceiling panel installation application.

Regarding the end-effector, section 5.5.2 presented the process of selecting the solution applied to the scope of the ceiling panel installation solution. As discussed, the concept defined was a Customized Vacuum Gripper considering all the aspects associated with the manipulation of the panels. To allow the evaluation of working spaces and verify compliance with the requirements, a conceptual design was proposed and is presented in Figure 5.40.

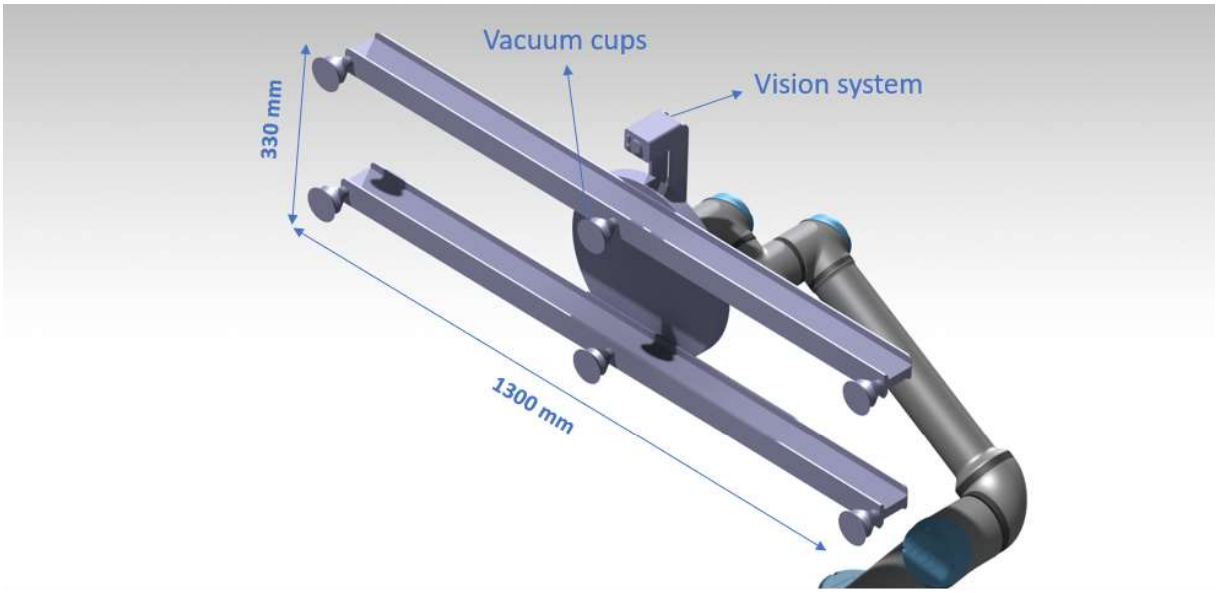


FIGURE 5.40 – Details of the conceptual solution for the end effector.

The proposed solution is based on a metallic base that is connected to the Cobot flange, in which longitudinal metallic (aluminum or other lightweight metal) bars are mounted.

These bar elements support the Vacuum cups that are responsible for gripping the ceiling panel.

The dimensions of the longitudinal elements were defined to allow the positioning of the vacuum cups near the connection points of the ceiling panel. The ceiling panel is fixed to the fuselage structure in 6 points, three on each side, and are distributed one in the longitudinal center and two near the edges, as presented in the schematic of Figure 5.41.

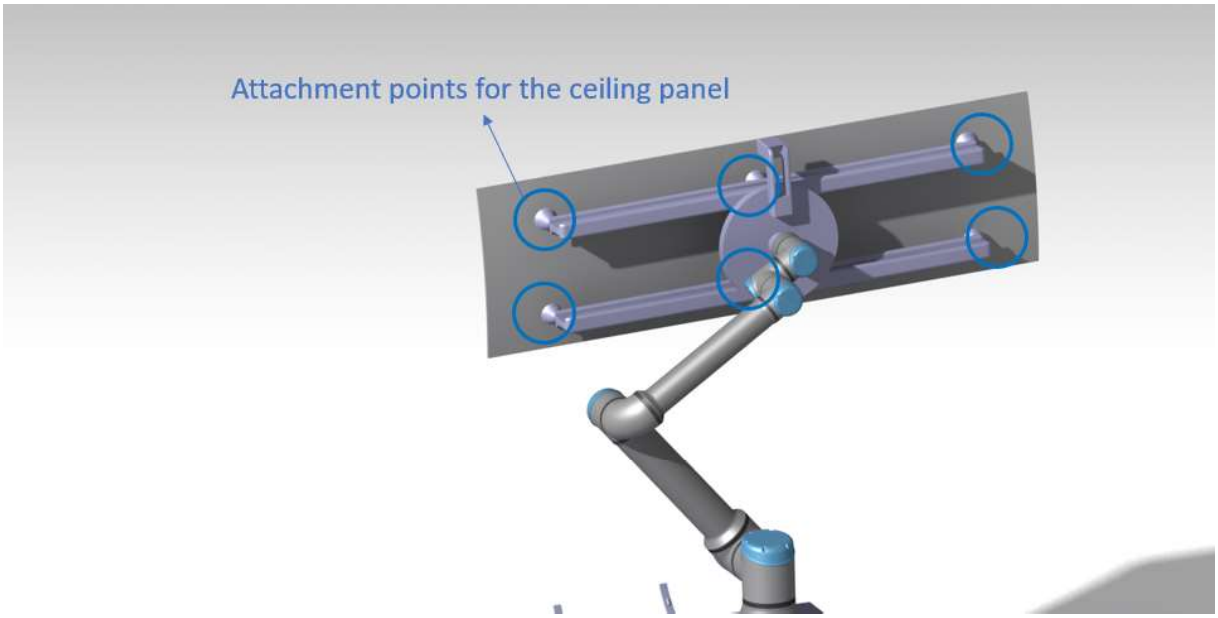


FIGURE 5.41 – Details of the ceiling panel attachment points.

Considering the dimensions of the ceiling panel and the position of the fixing points, the dimension of the end effector was defined. Additionally, the estimated number and dimensions of the vacuum cups were defined. Considering conventional feed pressures and air flows, it was verified that an arrangement of 6 cups with a diameter of 50 mm would be sufficient for handling up to 5 Kg of payload. These conclusions were based on the evaluation of three cases of handling, presented in Figure 5.42.

Considering also that the "Fc" must be supported by the vacuum force provided by the system, which can be defined as:

$$FP = \frac{A * P * n}{1000} \tag{5.1}$$

Where "A" is the area in mm^2 for each vacuum cup, P is the vacuum pressure in and n is the number of cups. Considering for the calculations, common vacuum cups with 22mm radius, 6 cups, and as recommended by (ROBOTIQ, 2021), 60,8 KPa of pressure:

$$FP = \frac{\pi * (22^2 * 60,8 * 6}{1000} = 554,7N. \tag{5.2}$$

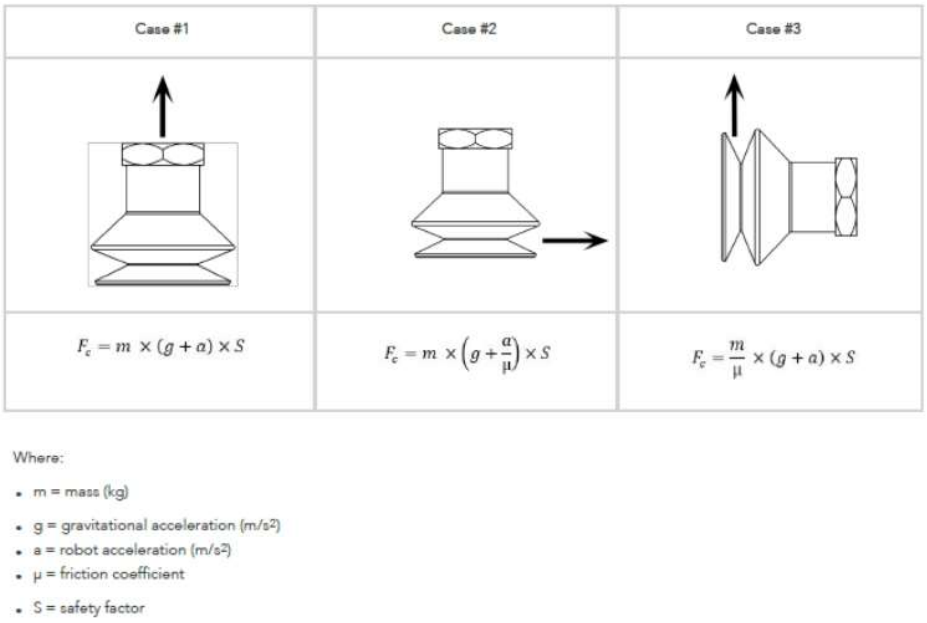


FIGURE 5.42 – Equations for the definition of the number of Cups required, (ROBOTIQ, 2021)

Considering yet for the three handling cases:

- g = 9,81 m/s²
- μ = 0,5
- a = 1,2 m/s²
- m = 5 kg

$$Fc1 = 5 * (9,81 + 1.2) = 55,1N. \tag{5.3}$$

$$Fc2 = 5 * (9,81 + \frac{1.2}{0.5}) = 61,1N. \tag{5.4}$$

$$Fc3 = \frac{5}{0.5} * (9.81 + 1.2) = 110,1N. \tag{5.5}$$

As can be verified, the most critical condition is the third, and compared with the provided force, it is possible to verify a safety factor higher than 5. This result is aligned with the recommended value of using a safety factor of 4 or more according to (ROBOTIQ, 2021).

$$S = \frac{FP}{Fc3} = \frac{554,7}{110,1} = 5,0. \tag{5.6}$$

It is important to highlight that this is an initial sizing, applicable for a conceptual stage of development. Further verification regarding the end effector capacity must be performed in the most detailed phases.

Another aspect considered in the concept of the end effector is the vision system, required for the execution of the assembly process. The dimensions and positioning of the system were based on the dimensions verified in the vision system suppliers' datasheet.

For the simulation of the conceptual development, the Universal Robot UR10e model was selected as presented in section 5.5.1. This Cobot presents a payload up to 12,5Kg and 1300 mm range.

The Cobot dimensions were also an input for the definition of the mobile base dimensions. As presented in Figure 5.43, the final estimated dimensions for the mobile base were 1000 mm long, 650 mm wide, and 700 mm high. The length of the mobile base was driven mainly by the panel support while the width was limited by the dimensions of the aircraft door (requirement ID 25). With these dimensions, it is also expected that all required components can be installed in the mobile base. The design of the mobile base concept considered no sharp edges to avoid any potential safety risk for human operators.

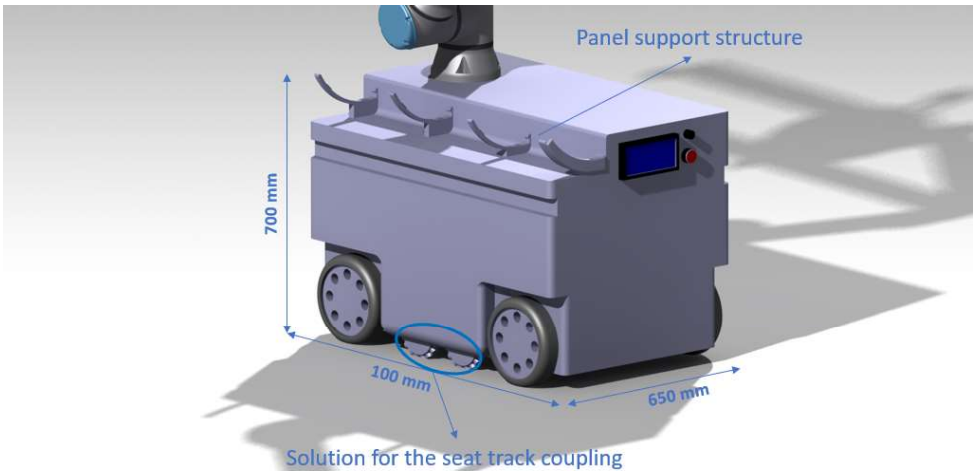


FIGURE 5.43 – Details of the mobile base concept

An important aspect considered in the mobile base was a poka-yoke panel support structure, to assure that the human operator will always position the panel in the same position. As shown in Figure 5.43, the support elements have a similar curvature to the ceiling panels, preventing the operator put the ceiling panel invented. Also, the support elements contain rollers and have different heights, which by the effect of gravity, guarantee that the panel will be in the same longitudinal position. In this case, the position of the panel for the Cobot picking is always assured.

Another relevant aspect of the mobile base is the mechanism defined to allow the use of the existing seat tracks on the fuselage floor. As shown in Figure 5.43, the concept predicts the existence of sprockets that are connected to the mobile base motor, which attaches

to the seat track to perform the mobile base movement. This solution was conceived to guarantee that the movement will be performed precisely and to assure that the mobile base is always aligned in the center of the fuselage.

As a result of the discussions, Figure 5.44 shows the final conceptual design of the solution, for the ceiling panels installation process.

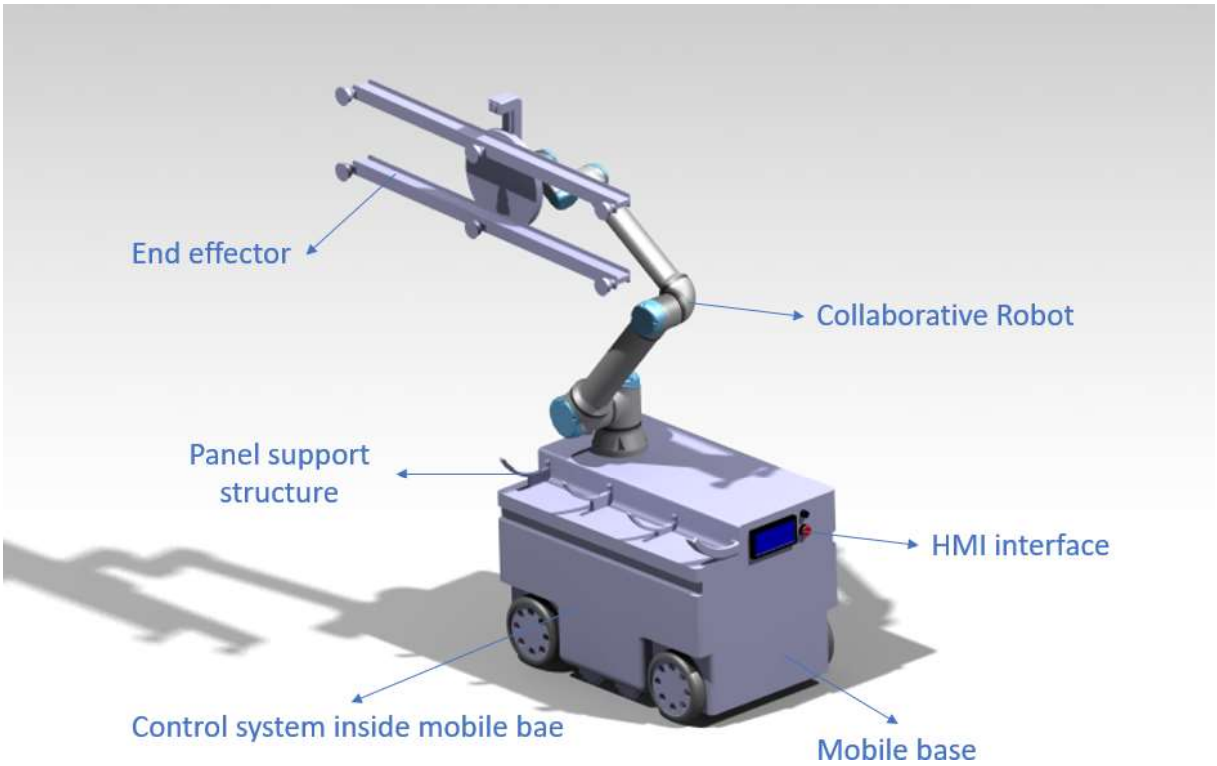


FIGURE 5.44 – Conceptual solution for the ceiling panel installation, with major components identified

5.7 Requirements Verification

The process for defining the conceptual solution, as presented in the beginning of Chapter 5, is recursive once requires the constant evaluation of the solution against the requirements defined. As presented in the last section, after several evaluations and studies, the conceptual solution was defined, and in this section, the verification of the requirements is presented.

As presented also in the last section, the components of the solution were modeled in CAD software (CATIA V5®) to allow the definition of the most indicated design for the ceiling panels installation process.

To help the definition of the solution, additional software was used to perform virtual simulations of the operation with the defined design solutions. The software used was CopeliaSim®, and the selection of this software to be used was based on the fact that the

software is recommended for factory automation simulations as well as fast prototyping and verification, has a free educational license version, and was already used in previous studies of the CCM (Centro de Competência em Manufatura – ITA) laboratory.

The simulations performed using the software were focused on evaluating the operational aspects of the design proposed, including robot movements, trajectory, and space restrictions. For this analysis, a representative fuselage section was modeled, considering the dimensions of the cross-section, length, and floor area of a commercial single-aisle jet. Furthermore, to allow the analysis of the requirements, the overhead bin, and the aircraft cabin doors were also modeled. Figure 5.45 shows a visualization of the scenario and components used in the simulations.

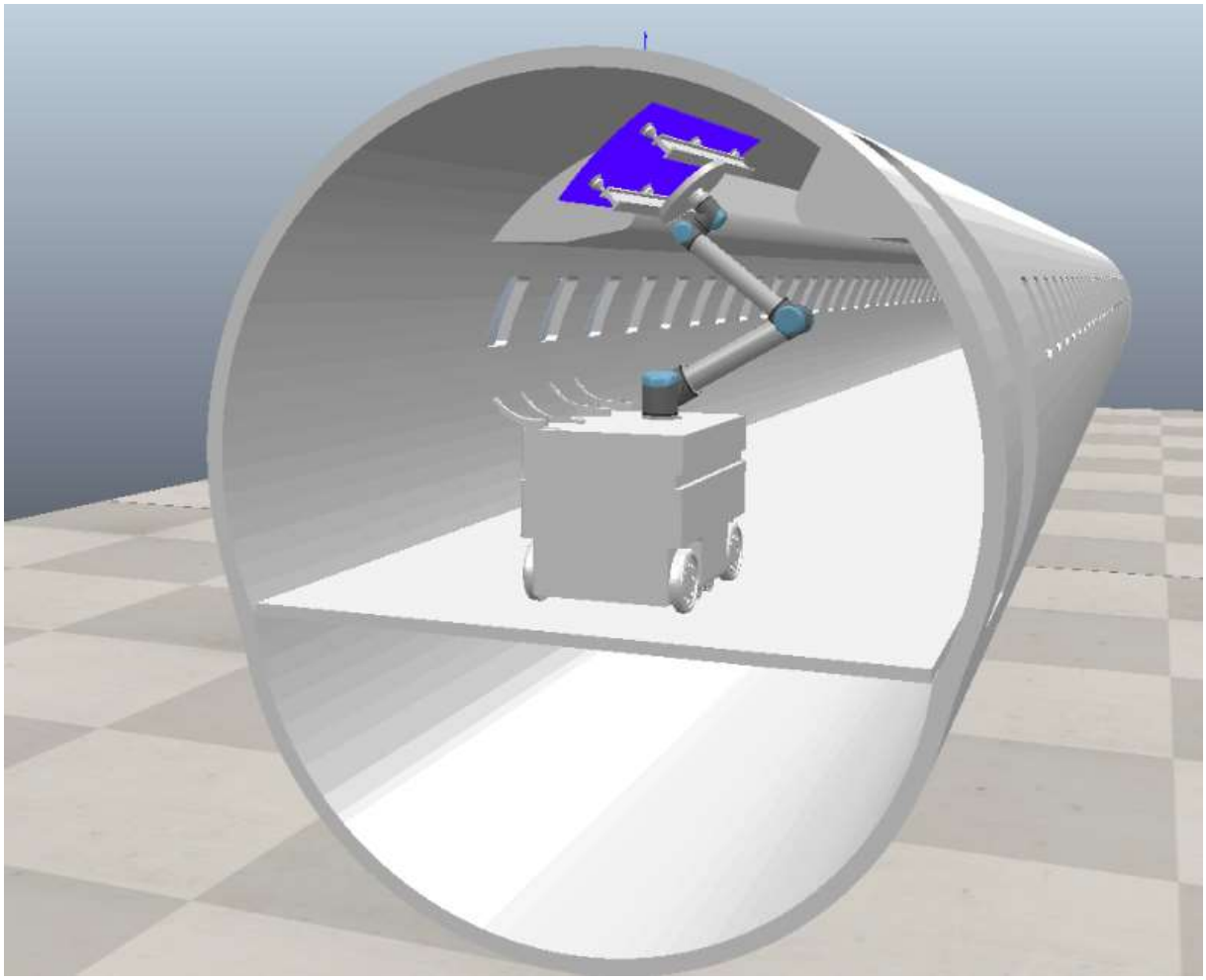


FIGURE 5.45 – Simulation environment implemented in the CopeliSim software.

Based on the simulations performed and the characteristics and components defined for the solution, compliance with the requirements could be evaluated for the final conceptual design presented in the last section. For the evaluation, the requirements list presented in section 5.3 is resumed below, in which the compliance is analyzed for each requirement, being the status supported by the evidence described. Table 5.6 presents the requirements verification result for the final conceptual design solution.

TABLE 5.6 – High-level requirements verification.

ID	Requirement	Verification	Mean of Compliance (MoC)
1	The Cobot System shall provide means to identify the target position to move during the panels assembly.	Compliant	The solution considers the use of motor power to move, and encoders to measure the wheels rotation. These information allows the controller to know the position and the path performed. (Section 5.5.3)
2	The Cobot System shall provide means to move inside the aircraft cabin during ceiling panel assembly.	Compliant	The solution contains a mobility system that moves using the seat track inside the fuselage. The motion is provided by motors, supplied by energy cables. (Section 5.5.3)
3	The Cobot System shall be able to avoid collisions with the cabin interior elements during its movement inside the aircraft cabin.	Compliant	The architecture is composed by a collaborative safety system which allows the tracking of the operating zone. Also, the mobile base contains sensing system to avoid colisions (Sections 5.5.3 and 5.5.4)
4	Solution must be able to identify the correct position of the panel to be taken.	Compliant	The proposed solution is based on a poka-yoke panel support strucuture that assures that the caeiling panel will be located at the same position. (Section 5.5.3.2)
5	The Cobot System shall provide means to identify the target position to move the panels for power suply connection.	Compliant	Once the Cobot will be moving at the center of the fuselage, the programming yet can assure the correct positioning of the ceiling panel for the power cables connection.
6	The Cobot system shall provide means to pick up the panels without damaging their surface.	Compliant	The end effector solution proposed is based on vacuum cups, allowing the handling without damaging the ceiling panel. (Section 5.5.2)
7	The Cobot system shall provide means to pick up panels with at least **W_p** kg of weight.	Compliant	The indicated Collaborative Robot have a payload compliant with the requirement. Also, the end effector was designed considering the payload required. (Sections 5.5.1, 5.5.2 and 5.6)
8	The Cobot system shall be able to move panels at least **H_F** m high from the ground and at least 0.95 m circumferentially from the Cobot center point at the ground level.	Compliant	The mobile base heght and the Collaborative robot range were defined/selected to allow the positioning of the ceiling panels and to comply with this requirement.
9	The Cobot System shall provide means to identify the target position to install the panels.	Compliant	The conceptual solution for the end effector considers the application of a vision system in conjuction with the controller. (Section 5.5.2.2).
10	The Cobot System shall be able to move the panel to the installation position.	Compliant	By the use of the collaborative robot and the end effector, considering the execution of the previous programmed routine, the sistem is able to move the panel to the installation position. Also, the vision system helps the fine tune position to installe the panels.
11	The Cobot System shall be able to apply the correct force to fit the panels in their position.	Compliant	The selection of a Collaborative robot as robotic arm allows the execution of the assembly with controlled force. (Section 5.5.1)
12	The Cobot System shall collect visual parameters related to the relative position of the installed panel.	Compliant	The vision system considered in the conceptual design is abble to collect informations that in conjunction with AI routines are able to verify the installation performed. (Section 5.5.2)
13	The Cobot System shall be able to process the collected visual information indicating the quality of the assembly.	Compliant	The vision system considered in the conceptual design is abble to collect informations that in conjunction with AI routines are able to verify the installation performed. (Section 5.5.2)
14	The Cobot system shall register the information of the assembly process.	Compliant	The vision system considered in the conceptual design is abble to collect informations that in conjunction with AI routines are able to verify the installation performed. (Section 5.5.2)
15	The Cobot system shall provide means to allow full operational control by the human operator.	Compliant	The solution considers SSM operating mode, also, traditional interfaces where the human operator can operate the solution. Additionally, it was considered the use of smart devices, to allow the human to operate the solution by distance. (Section 5.5.1)
16	The Cobot system shall provide indicative interfaces, status, and warning messages to the operator.	Compliant	The solution considered the use of traditional interfaces. Also, the mobile base considers the existence of a HMI interface. (Sections 5.5.1, 5.5.3)
17	Solution must reduce operators non ergonomics activitie	Compliant	The reduction of operators non ergonomics is the base of the solution, and described in Section 5.2.
18	The Cobot system shall be able to operate alongside humans operators in the same workspace without separation structures.	Compliant	The solution is comosed by Collaborative robot, which allows the operation alongside humans. Furthermore, considering the SSM operating mode and the collaborative safety system, this requirement is compliant. (Sections 5.5.1, 5.5.4)
19	The Cobot system shall be able to perform at least one different assembly activity inside the aircraft cabin	Compliant - Further	The design of the mobility system (Section 5.5.3) and the payload/range provided by the Cobot allows for the use of the solution for other application such as side panels positioning. The requirement is considered compliant but further analysis are required, out of the scope of this work.
20	The mean installation time for a panel should be lower than current mean installation time for panels without any non-conformity	Compliant - Further	Considering the simulations performed, the solution is compliant with the requirement. However, further studies are required in the detail phases.
21	The Cobot system must be able to perform the installation resulting in an acceptable level of Gap between panels.	Compliant - Further	The characteristics of the Cobot (repeatibility) with a mobile base that is well connected in the floor (connection eith the seat track), is expected that the installation will result in acceptable level of gap. However, this reuquirement must be evaluated in further phases of the project.
22	The Cobot system must be able to perform the installation resulting in an acceptable level of Step between panels.	Compliant - Further	The characteristics of the Cobot (repeatibility) with a mobile base that is well connected in the floor (connection eith the seat track), is expected that the installation will result in acceptable level of step. However, this reuquirement must be evaluated in further phases of the project.
23	The Cobot system must be able to perform the installation resulting in an acceptable level of alignment between panels.	Compliant - Further	The characteristics of the Cobot (repeatibility) with a mobile base that is well connected in the floor (connection eith the seat track), is expected that the installation will result in acceptable level of slignment. However, this reuquirement must be evaluated in further phases of the project.
24	The Cobot system must be able to be moved inside the aircraft by the entrance door, **H_D** m high from the ground.	Compliant	The solution is compliant with the requirement considering its dimensions and weight (strengthened by the use of power cables instead of batteries). (Section 5.5.3)
25	The Cobot system must be able to enter inside the aircraft by the pax door (**h_D** m height x **w_D** m width).	Compliant	The solution is compliant with the requirement considering the dimensions of the base and the Cobot. (Section 5.5.3 and 5.6)
26	The Cobot system must be able to operate inside the cabin (Circular with a **W_F** m floor width and **H_F** m height in the centerline).	Compliant	The solution is compliant with the requirement considering the dimensions of the base and the Cobot. (Section 5.5.3 and 5.6)

As can be verified, all the proposed requirements were compliant considering the conceptual phase of the development of the solution. The requirements indicated with the "green check" are those that the definitions performed at the conceptual level already assure their fulfillment. The requirements indicated with the "blue check" are the ones that the conceptual design already assures that they are addressed, however, the final verification will be possible only in further detailed phases of the project.

For each requirement defined, beyond the verification, the mean of compliance is presented in Table 5.6. In general terms, the requirements that indicated the need for any kind of functionality were addressed by the design of the solution, for example, ID 1, ID 2, ID 4, and others.

Considering some special cases, requirement ID 17, for example, is the core of the

solution. As presented in Chapters 2 and 3, ergonomics improvement is one of the pillars of the application of Cobots. In this case, for the ceiling panel installation, the solution operating mode and scenario presented in Section 5.2, were defined in the way of allowing the reduction of human workload and improving ergonomics conditions.

The requirement ID 19 indicates a need for the solution to be able to perform another operation inside the aircraft cabin. In this sense, since the selection of the ceiling panel installation as the application, the side panels positioning application was identified as a potential expansion of the use of the solution developed. Furthermore, by the design defined for the solution, including the mobility system and the Cobot payload range, it is possible for the solution to perform the side panels positioning activity. However, the status of the requirement was set as "Compliant-Further" once the evaluation of the required end effector, programming approach, and operating modes were out of the scope of this work, requiring further studies to assure full compliance.

Requirements ID 21, 22, and 23 are other cases of requirements in that the conceptual design considerations are aligned, however, only in further detailed phases it will be possible to assure the expected values of gap, step, and alignment resulting from the assembly process.

For requirement ID 20, the CopeliaSim software was used to allow a first evaluation of the operating cycle. The results showed that the requirement trend to be fulfilled, however, the verification must be performed in further detailed phases of the project, considering a deeper evaluation of the interaction between human and cobot, as well as the operating zone defined by the collaborative safety system, aspects that influence directly on the operating cycle.

Finally, Figure 5.46 shows images of the simulations performed in the CopeliaSim software, which allowed the verification of the space required for the Cobot to perform the installation of the panels, as well as verify if the Cobot reach was adequate to allow the execution of the activity, information used to refined the dimensions of the solution.

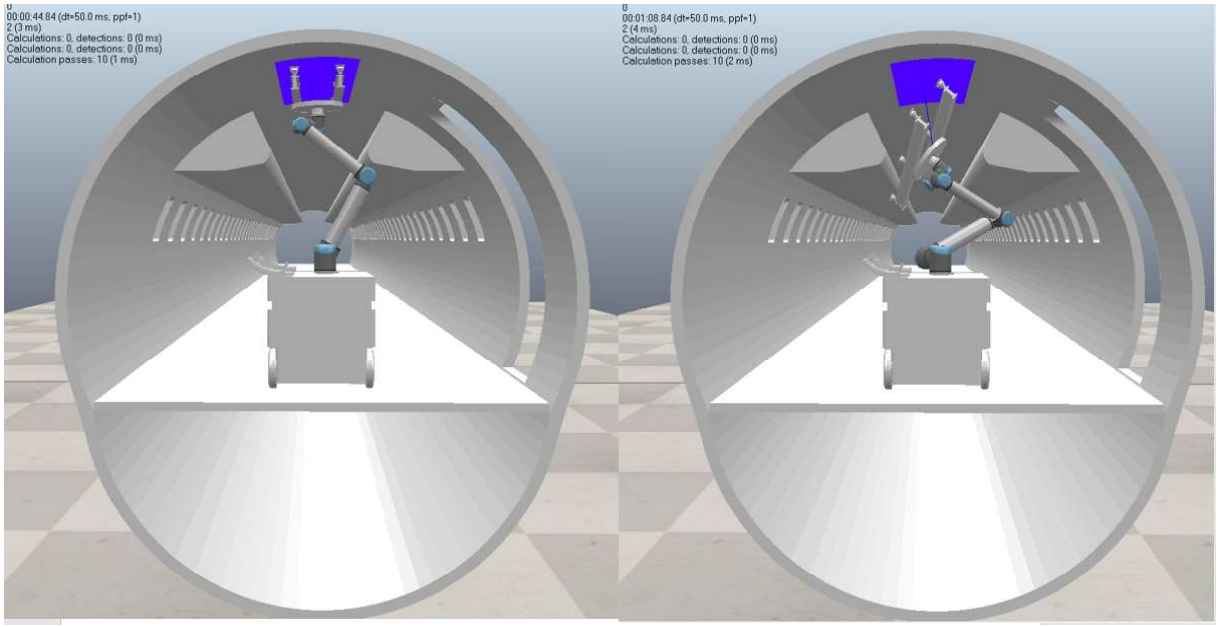


FIGURE 5.46 – Simulation environment implemented in the CopeliSim software.

6 Conclusion

The application of automation has played an important role for decades in aircraft manufacturing processes, with solutions for drilling, deburring, and riveting that allowed productivity, quality, and efficiency improvements in structural assembly stages. The aircraft assembly, however, is a very complex process with characteristics that have been limiting the increase of automation, such as a high number of different tasks and components, activities performed in confined spaces, and low production rates (compared to other industries). This is the case in the final assembly stage which is still highly dependent on a manual workforce.

The application of Collaborative robots has been seen as an opportunity to increase the level of automation in aircraft manufacturing stages where the insertion of industrial robots is not feasible, once they can be deployed to work side by side with human operators, combining the advantages of automation with the cognitive and flexibility of humans. The deployment of this technology, however, is still a distant reality in the aeronautical industry, with few published studies focusing on the aircraft's final assembly stage.

In this context, the objective of this work was to evaluate the application of Collaborative Robots in aircraft's final assembly process, to increase the level of automation in this manufacturing stage. The study was focused on the final assembly process of a typical single-aisle commercial jet with up to 150 pax capacity.

The first step of work was to map the macro process and activities related to the final assembly process, to allow the identification of potential applications for Collaborative robots. As presented in Chapter 2, a comprehensive bibliographical review was performed, identifying current applications and studies for Collaborative robots in the aeronautical, aerospace, automotive, electrical/electronic, and metal mechanics sectors. Furthermore, the studies presented allowed the identification of the reference final assembly process for aircraft. Additionally, in Chapter 4, the real final assembly process was mapped, in which 36 macro activities were identified in two macro stages of the final assembly process, the systems equipping and the final equipping phase.

Through the execution of interviews with specialists from different technologies such as sealing, interiors, hydraulics, AMS, and mechanical systems, 8 applications were iden-

tified as potential for the application of Collaborative robots. The potential applications identified were focused on activities performed inside the fuselage cabin and the cargo compartment, including insulation blankets installation, side and ceiling panels installation, overhead bin positioning, floor fixing, cargo floor fixing, and cargo ceiling positioning. These results allowed the fulfillment of the first specific objective defined for this work.

The potential applications identified were then evaluated through 8 parameters to compose the score for the expected benefit and the development complexity of each potential application. This evaluation was performed through the extensive use of the AHP multi-criteria method, which allowed compare the importance of the different criteria defined and evaluate the potential applications. As a result of the process, an evaluation matrix was obtained which allowed the selection of the ceiling panels installation as the application to be studied in more detail in this work, and the second specific objective was achieved.

Based on the selected application, the requirements, restrictions, and main functions required for the application were defined. Following the systems design process methodology, a deeper characterization of the current operational scenario was performed, including the identification and evaluation of the 7 main stakeholders, the identification of 20 stakeholders' needs, 8 constraints as well as the mapping of the operator's journey. These evaluations allowed the discussion for the envisioned automated scenario with the application of the Cobot. For this evaluation 5 different use cases were identified, and for the selected one, the design reference mission was established, characterizing the operational flow, task sharing between human and robot, as well as the definition of the SSM as the indicated operating mode for the Cobot. This information composed the CONOPs, which allowed the functional analysis to be performed and the definition of the requirements for the conceptual solution development, the third objective of the study.

Based on the CONOPS and the requirements defined, a system decomposition was performed allowing the definition of the solution architecture. For each component of the architecture, 36 different solutions were identified for all the subsystems required in the solution. As presented in Chapter 5, these identified solutions were evaluated through trade-off studies allowing the definition of the final conceptual solution for the ceiling panel installation application. The development of the conceptual solution also counted with the use of CAD tools for modeling the solutions and the use of virtual simulation performed in the CopeliaSim software, to verify operational aspects such as Cobot movement and reach, as well as test identified restrictions.

Finally, the defined conceptual solution for the ceiling panel installation was confronted with the previously defined requirements, concluding the fulfillment of the fourth and fifth specific objectives of the work.

It should be noted that, in addition to meeting all previously defined objectives, this work has made a notable contribution to the development of a methodology used to evaluate identified potential applications for Collaborative robots. Such methodology presented in Chapter 3 was defined based on different methods and frameworks studied, whose best practices were combined to allow the definition of a methodology that allowed a quick response and with a secure technical basis of what should be the potential applications to be prioritized.

6.1 Future Work

Based on the results provided in this study, alternatives for future works were identified as follows:

- First, to continue the study developed in this work, it is suggested the execution of the further stages of the project of the solution, including the detailed design phase, the virtual commissioning, manufacturing, and testing of the solution in a representative environment. With these steps, it will be possible to mature the solution for a future application in the real final assembly environment. A special point to be further detailed regarding the solution is the evaluation of safety aspects, including simulations and tests for the operation modes, sensors, and the whole interaction during the operation. Another opportunity for the detailed project is the evaluation of the other use cases identified for the application, mainly those where the operation of ceiling panel installation is performed by both human and robot using hand-guiding operation modes. In this case, the objective is to compare this scenario with the baseline use case in terms of the ergonomics impact on the operator versus the solution design simplification.

- Another suggested future work is to perform a detailed financial analysis regarding the application, including the evaluation of recurring and non-reoccurring costs, and financial savings due to the productivity, quality, and ergonomics improvements. With these values, it will be possible the definition of the financial indicators of the project, such as return on investment, net present value, and payback, which are important aspects required to approve these types of implementation in the companies.

- Finally, it is suggested the development of a detailed study regarding the application of DFMA guidelines in the concept of ceiling panels. The main objective is to define product changes that are relevant to assure the feasibility of automation by the application of Collaborative robots.

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Appendix A -

A.1 Questionnaire - Development Complexity Evaluation

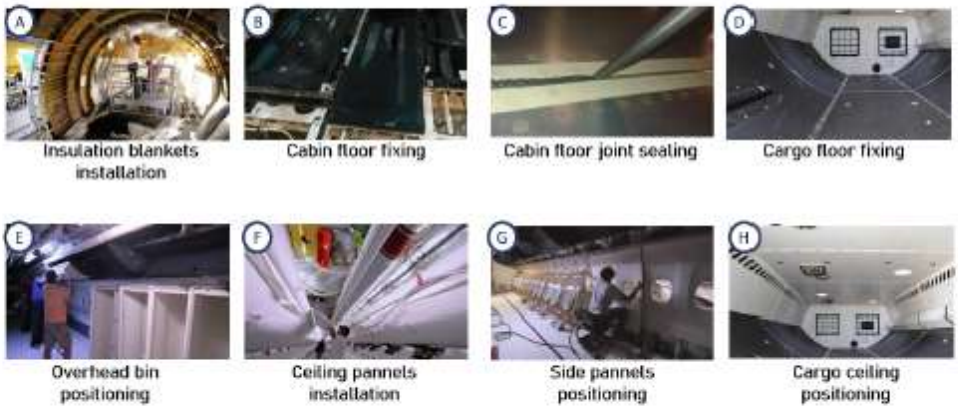
COBOTs Application Evaluation on Aircraft Final Assembly Processes

This questionnaire aims to collect the perception of the difficulty in developing collaborative robotics solutions for the identified possible applications.

The result will help the work of the master's student Felipe Miotto, who aims to carry out a conceptual project of a collaborative robotics application in the aircraft's final assembly.

*Obrigatório

Identified Possible Applications:



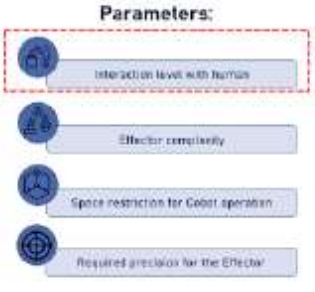
1. 1. Name (optional):

2. 2. E-mail (optional):

3. 3. Occupation Area (ex: Product development) *

4. How do you classify the difficulty in developing the solution, for each proposed application, with respect to the 1st parameter? *

The level of interaction brings different complexity to the application: If humans and robots work in separated workspaces and are detached from each other (independent), the solution is easier than a Simultaneous or supportive level, where human and robot need to interact and work together to complete the activity.



More difficult — Easier

Marcar apenas uma oval por linha.

	1°	2°	3°	4°	5°	6°	7°	8°
A	◯	◯	◯	◯	◯	◯	◯	◯
B	◯	◯	◯	◯	◯	◯	◯	◯
C	◯	◯	◯	◯	◯	◯	◯	◯
D	◯	◯	◯	◯	◯	◯	◯	◯
E	◯	◯	◯	◯	◯	◯	◯	◯
F	◯	◯	◯	◯	◯	◯	◯	◯
G	◯	◯	◯	◯	◯	◯	◯	◯
H	◯	◯	◯	◯	◯	◯	◯	◯

FIGURE A.2 – Questionnaire page 2.

5. How do you classify the difficulty in developing the solution, for each proposed application, with respect to the 2nd parameter: *

Depends on the characteristics of the manipulated object: (Fragile, delicate, flexible, large or small, very heavy or very light.), it also depends on the type of operation performed by the robot: (Pick and place, Screwing, tightening, holding) and mainly on the number of features required for the Cobot.

Applications:

Parameters:

More difficult — **Easier**

Marcar apenas uma oval por linha.

	1°	2°	3°	4°	5°	6°	7°	8°
A	◯	◯	◯	◯	◯	◯	◯	◯
B	◯	◯	◯	◯	◯	◯	◯	◯
C	◯	◯	◯	◯	◯	◯	◯	◯
D	◯	◯	◯	◯	◯	◯	◯	◯
E	◯	◯	◯	◯	◯	◯	◯	◯
F	◯	◯	◯	◯	◯	◯	◯	◯
G	◯	◯	◯	◯	◯	◯	◯	◯
H	◯	◯	◯	◯	◯	◯	◯	◯

FIGURE A.3 – Questionnaire page 3.

6. How do you classify the difficulty in developing the solution, for each proposed application, with respect to the 3rd parameter: *

It depends on whether there is a height restriction or a lateral space restriction for the cobot operation: The greater the space restriction, the greater the complexity in the solution development.

More difficult — Easier

Marcar apenas uma oval por linha.

	1°	2°	3°	4°	5°	6°	7°	8°
A	◯	◯	◯	◯	◯	◯	◯	◯
B	◯	◯	◯	◯	◯	◯	◯	◯
C	◯	◯	◯	◯	◯	◯	◯	◯
D	◯	◯	◯	◯	◯	◯	◯	◯
E	◯	◯	◯	◯	◯	◯	◯	◯
F	◯	◯	◯	◯	◯	◯	◯	◯
G	◯	◯	◯	◯	◯	◯	◯	◯
H	◯	◯	◯	◯	◯	◯	◯	◯

FIGURE A.4 – Questionnaire page 4.

7. How do you classify the difficulty in developing the solution, for each proposed application, with respect to the 4th parameter: *

It basically depends on: a) Required positioning accuracy; b) Precision required in the execution of the task (process parameters) and its tolerances. The greater the precision required, the more complex the solution.

More difficult — Easier

Marcar apenas uma oval por linha.

	1°	2°	3°	4°	5°	6°	7°	8°
A	◯	◯	◯	◯	◯	◯	◯	◯
B	◯	◯	◯	◯	◯	◯	◯	◯
C	◯	◯	◯	◯	◯	◯	◯	◯
D	◯	◯	◯	◯	◯	◯	◯	◯
E	◯	◯	◯	◯	◯	◯	◯	◯
F	◯	◯	◯	◯	◯	◯	◯	◯
G	◯	◯	◯	◯	◯	◯	◯	◯
H	◯	◯	◯	◯	◯	◯	◯	◯

FIGURE A.5 – Questionnaire page 5.

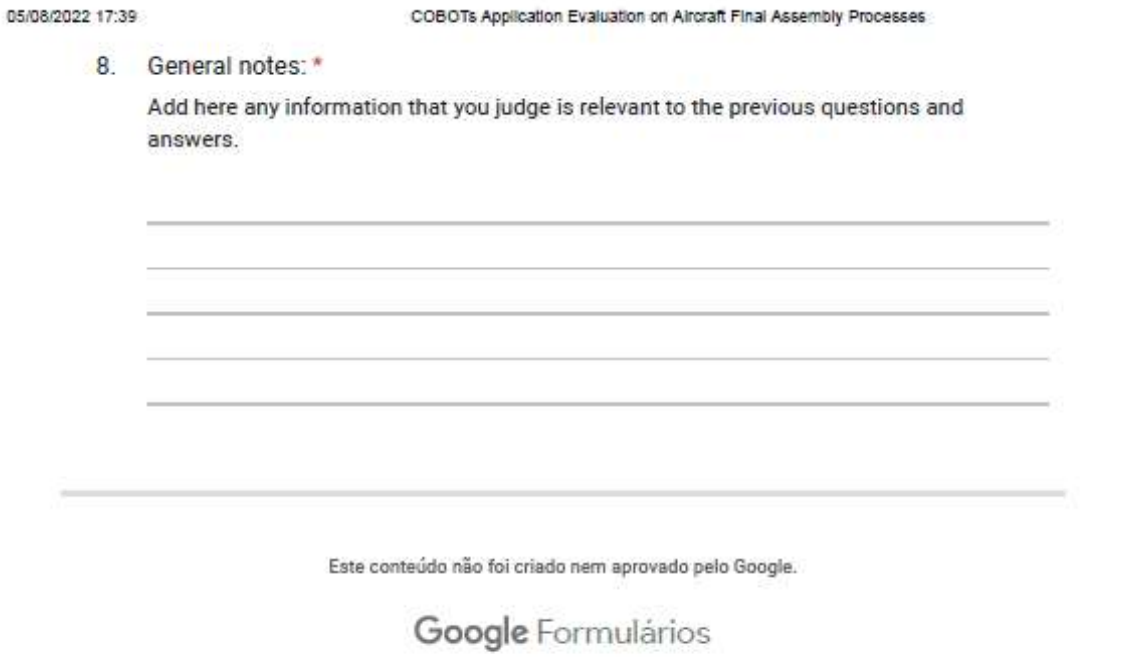


FIGURE A.6 – Questionnaire page 6.

Results obtained through the questionnaire application:

TABLE A.1 – Results obtained from the questionnaire, for the first criteria - Interaction Level With Humans.

	Answer 1	Answer 2	Answer 3	Answer 4	Answer 5	Answer 6	Answer 7	Answer 8	MEAN
Insulation blankets installation	4	5	2	1	5	7	5	2	3,9
Cabin floor fixing	7	7	5	7	8	5	5	3	5,9
Cabin floor joint sealing	6	7	7	8	8	8	5	6	6,9
Cargo floor fixing	7	7	1	7	8	6	5	5	5,8
Overhead bin positioning	2	1	5	1	1	1	5	5	2,6
Ceiling pannels installation	5	3	6	5	8	4	5	8	5,5
Side pannels positioning	3	5	7	3	8	2	5	6	4,9
Cargo ceiling positioning	1	2	1	4	8	3	4	8	3,9

TABLE A.2 – Results obtained from the questionnaire, for the second criteria - Effector complexity.

	Answer 1	Answer 2	Answer 3	Answer 4	Answer 5	Answer 6	Answer 7	Answer 8	MEAN
Insulation blankets installation	1	3	1	1	1	5	5	7	3,0
Cabin floor fixing	4	5	7	6	4	7	5	7	5,6
Cabin floor joint sealing	3	4	8	4	6	8	3	8	5,5
Cargo floor fixing	4	4	3	5	4	7	5	7	4,9
Overhead bin positioning	5	1	5	1	8	2	5	7	4,3
Ceiling pannels installation	2	2	4	2	8	1	5	6	3,8
Side pannels positioning	6	3	6	3	8	4	5	5	5,0
Cargo ceiling positioning	7	1	3	3	8	3	3	5	4,1

TABLE A.3 – Results obtained from the questionnaire, for the third criteria - Space Restriction for Cobot Operation.

	Answer 1	Answer 2	Answer 3	Answer 4	Answer 5	Answer 6	Answer 7	Answer 8	MEAN
Insulation blankets installation	5	6	8	4	8	6	5	5	5,9
Cabin floor fixing	5	6	8	7	8	8	5	4	6,4
Cabin floor joint sealing	5	7	8	7	8	7	2	6	6,3
Cargo floor fixing	2	4	1	3	1	2	4	7	3,0
Overhead bin positioning	3	5	8	1	8	4	4	7	5,0
Ceiling pannels installation	4	6	6	2	8	3	5	7	5,1
Side pannels positioning	5	6	6	5	8	5	5	5	5,6
Cargo ceiling positioning	1	2	1	4	1	1	3	5	2,3

TABLE A.4 – Results obtained from the questionnaire, for the fourth criteria -Required Precision for the Effector .

	Answer 1	Answer 2	Answer 3	Answer 4	Answer 5	Answer 6	Answer 7	Answer 8	MEAN
Insulation blankets installation	1	5	2	4	1	4	4	5	3,3
Cabin floor fixing	3	6	3	3	3	1	4	5	3,5
Cabin floor joint sealing	2	7	5	4	6	2	2	8	4,5
Cargo floor fixing	3	6	2	3	3	1	4	6	3,5
Overhead bin positioning	5	1	5	1	8	3	4	7	4,3
Ceiling pannels installation	4	2	6	2	6	2	4	7	4,1
Side pannels positioning	5	3	7	3	8	3	4	7	5,0
Cargo ceiling positioning	5	2	2	2	8	3	4	7	4,1

FOLHA DE REGISTRO DO DOCUMENTO

1. CLASSIFICAÇÃO/TIPO	2. DATA	3. DOCUMENTO Nº	4. Nº DE PÁGINAS
DP	09 de agosto de 2023	DCTA/ITA/DP-005/2023	180

5. TÍTULO E SUBTÍTULO:

Evaluation of collaborative robots application in aircraft final assembly

6. AUTOR(ES):

Felipe Julian Miotto

7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES):

Instituto Tecnológico de Aeronáutica – ITA

8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR:

Cobot; Final Assembly; Aircraft Manufacturing

9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO:

Dinâmica de robôs; Planejamento de tarefas (robótica); Linhas de montagem; Fabricação de aeronaves; Tomada de decisão; Robótica; Indústria aeronáutica; Computação.

10. APRESENTAÇÃO: (**X**) **Nacional** ( ) **Internacional**

ITA, São José dos Campos. Curso de Mestrado Profissional em Engenharia Aeronáutica e Mecânica. Orientadora: Profa. Dra. Emília Villani; coorientador: Eng. MEng. João Marcos Gomes de Mello. Defesa em 02/08/2023. Publicada em 2023.

11. RESUMO:

This work presents studies for the application of Collaborative robots in the final assembly stage of aircraft manufacturing. Initially, a comprehensive bibliographical analysis was carried out in order to understand the processes and activities involved in this stage of the aircraft production process. In addition, the current scenario of Collaborative robotics was characterized, through the identification of applications and studies for the implementation of Cobots in assembly processes, including cases from the aeronautical industry and from other sectors such as automotive, electrical / electronics, and metal mechanics. Different frameworks for evaluating the application of collaborative robots were identified in the literature, which allowed the development of a methodology that would allow the mapping, identification, and selection of the most appropriate application. This methodology was defined in order to allow decision-making taking into account the balance between the expected benefits and the complexity in the development of the solution. Based on this methodology, an in-depth mapping of the final assembly processes of a single-aisle commercial aircraft was carried out, allowing the identification of eight potential applications for Collaborative robots. Such applications were then evaluated according to the criteria defined in the methodology, allowing the identification of the installation of fuselage ceiling panels as the most suitable application to be implemented. From the definition of the application, studies were carried out for the development of a conceptual project of the robotic solution to be used, following the methodology of the system design process, part of the concept of systems engineering. Starting from the mapping and assessment of stakeholders, and needs and restrictions identification, it was possible to map the current scenario and define the future scenario within the scope of CONOPS, leading to the definition of application requirements. Finally, from the mapped requirements, a systems architecture was defined, and different solutions were identified for each component. Such solutions were evaluated through trade-off studies and simulations in a computational tool, allowing the definition of the most adequate conceptual solution for the studied application.

12. GRAU DE SIGILO:

( ) **OSTENSIVO** (**X**) **RESERVADO** ( ) **SECRETO**